Guided Group Featurizers (GGF): Improving Block-Sparse Featurizers with Object-Guided Learning

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https://github.com/MechInterpreter/Guided-Group-Featurizers

Abstract

Block-sparse featurizers (BSFs) use groups of learned directions to capture the multidimensional structure of visual concepts. However, grouping alone does not ensure that information about a concept is consolidated within a single group. We introduce Guided Group Featurizers (GGF), a novel object-guided extension that uses object annotations for initialization and training while preserving the BSF architecture. We assess whether each group's strongest response regions share an object type, part or material across images. Blinded GPT reviews give GGF groups an average semantic coherence score of $93.4 \pm 1.6\%$*, versus* $83.4 \pm 2.5\%$ *for SLiCS and* $67.9 \pm 3.1\%$ *for supervised Vanilla BSF. GGF also leads in a separate blinded Gemini review. GGF's consolidation advantage extends to unguided categories: one selected group per category covers* $66.1 \pm 1.9\%$ *of target-containing crops, versus* $59.2 \pm 1.8\%$ *for supervised Vanilla BSF, at approximately 5% false-positive rates for both. Across COCO and VOC, GGF also improves single-group object Capture and localization over Vanilla BSFs, SLiCS and Concept-SAE. Its groups produce more focused and reproducible object responses while supporting recognition and retaining associations under rotation, scale and brightness changes. GGF also uses fewer learned parameters than Concept-SAE and shorter recorded VOC training loops than both supervised alternatives. These findings support object guidance for organizing more interpretable BSF groups and show that response concentration extends to categories excluded from guidance.*

1. Introduction

Sparse autoencoders (SAEs) reconstruct neural activations as sparse combinations of inspectable learned directions [16]. Accurate reconstruction, however, does not explain how those components organize meaningful concepts.

This distinction matters when a concept has multidimensional structure. Its activations may trace a curve or surface, forming a *concept manifold*. SAEs can *tile* these manifolds [5]: different features cover local regions (shattering), or changing, overlapping subsets provide redundant representations (dilution). Compact capture instead uses a small shared set of decoder directions, with coefficients that vary across manifold points. Reconstruction can succeed in each case, while the organization across examples differs (Figure 1).

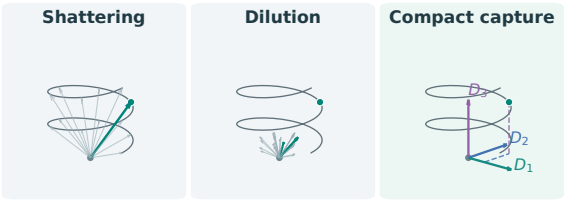


Figure 1. **Shattering, dilution, and compact capture.** The helix represents a concept manifold; arrows denote dictionary directions. Shattering uses local directions, dilution uses changing, overlapping subsets, and compact capture reuses a small shared set across the manifold. Schematic adapted from Bhalla et al. [5, Fig. 3].

Block-sparse featurizers (BSFs) make this richer structure an architectural possibility by replacing individual directions with groups of directions [14]. Sparsity selects groups together, while input-dependent coefficients describe positions within each active group's subspace. This represents multidimensional structure as one unit, improving manifold recovery and activation description length.

Providing that capacity does not ensure that training consolidates a concept within it. In their analysis of Grassmannian BSFs, Jerpelea and Ananthram [21] find that synthetic factors can still split across blocks, including pairs that capture complementary parts of the same factor. This gap between representational capacity and learned organization motivates asking whether explicit guidance can help groups capture more of a concept's information.

Object annotations provide a practical way to investigate this question. They connect annotated regions to groups during training. Objects serve as concept proxies: a model's concepts can also describe properties shared across object categories. We ask: *can object guidance help groups consolidate concepts, or annotated objects used as concept proxies, more effectively?* Here, consolidation means making substantial object-related information accessible within an

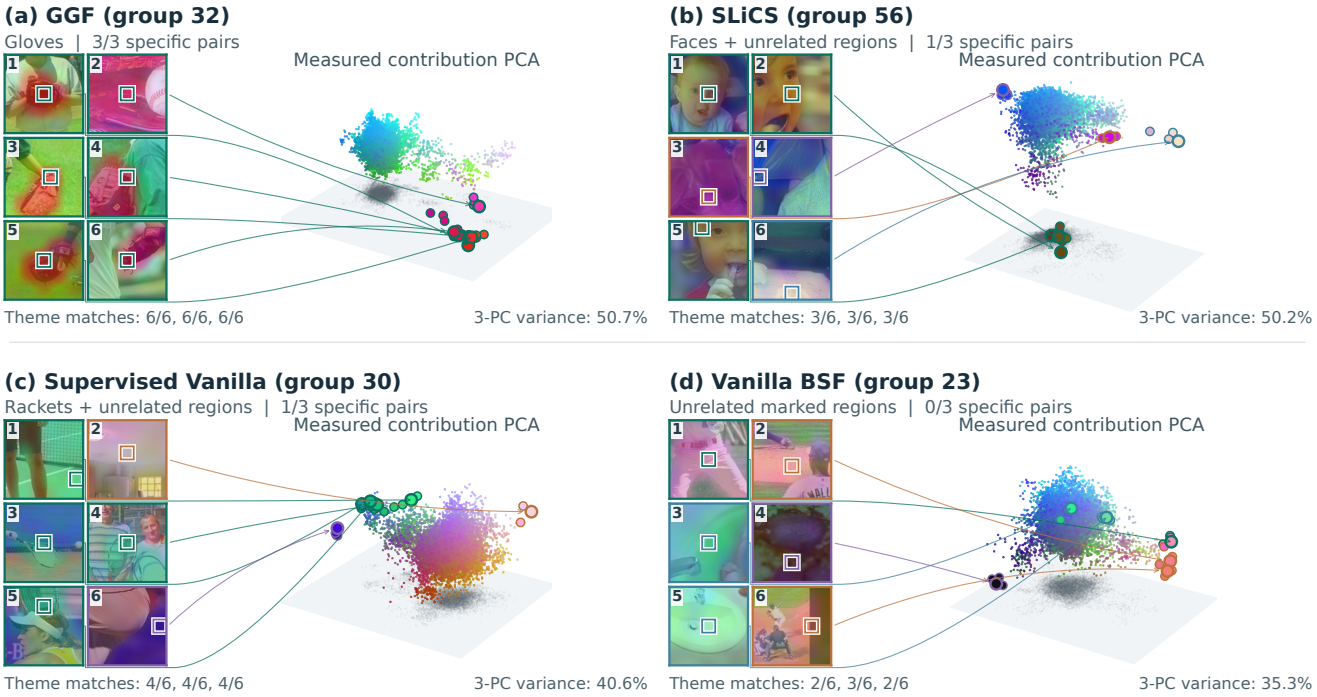


Figure 2. **Coherent and mixed responses in selected native groups.** Numbered crops show the strongest marked patches from six images for one native group per 75-group, seed-21 model. *Specific pairs* counts majority-confirmed pairs sharing an object type, part, material or other specific meaning, out of three fixed pairs. *Theme matches* gives the view count (out of six) fitting each of three blinded AI reviewers' themes. Clouds project decoded patch contributions using principal component analysis (PCA); *3-PC variance* is reference variance retained by the first three components. RGB encodes PCA coordinates; borders, arrows and outlines mark visible content. Arrows locate shown patches; outlined points include all four marks per image. PCA bases are fitted separately (Appendix F.1). These selected illustrations complement aggregate comparisons of every group (Table 1).

individual group. We separately test reconstructed regional information and the concentration of category responses within small group subsets.

Figure 2 illustrates the semantic coherence of a group's responses across images. The selected GGF group responds to baseball gloves across six images; all three fixed pairs confirm that specific meaning. The selected SLiCS, supervised Vanilla and Vanilla groups mix unrelated marked regions, confirming only one, one and zero pairs, respectively. The exhaustive 75-group comparison gives GGF the highest aggregate specific-pair coherence among the four matched methods (Section 5.1, Table 1).

Contributions. Our contributions are as follows:

- **Object guidance for more interpretable BSF groups.** We introduce GGF, a novel object-guided extension that makes object-related responses more semantically coherent and information more accessible within individual BSF groups. Object-informed *Initialization* and adaptive *Guided Learning* preserve the native architecture and additive activation reconstruction.
- **Evidence of coherent and concentrated object responses.** We provide exhaustive coherence and concentration comparisons of matched 75-group models and show concentration benefits for categories excluded from guidance. COCO and VOC comparisons, recognition tests, response consistency across images and appearance changes, and visualizations establish the scope of these interpretability gains.
- **Explaining and validating consolidation.** We explain which training components improve single-group reconstruction of object-versus-background feature contrast, measured by Capture, and which improve localization. Tests of group subsets and contribution removal assess information accessibility and effects on predictions. Geometric diagnostics show how groups use multiple directions, while recovery of known synthetic factors directly tests consolidation.
- **Practical capacity and resource tradeoffs.** We quantify tradeoffs among object alignment, reconstruction, group capacity, activity, training duration and resources for training and inference. These comparisons make GGF's practi-

cal costs explicit, including fewer learned parameters than Concept-SAE and shorter recorded VOC training loops than SLiCS and Concept-SAE on the tested hardware.

2. Related Work

From sparse directions to concept geometry. Manifold analyses examine whether sparse learned directions [16] capture concepts compactly or tile local regions [5]. BSFs address this mismatch through structured group sparsity, connecting concept geometry to reconstruction and description length [14, 18]. Yet related information can remain distributed across blocks, as Jerpelea and Ananthram [21] show for Grassmannian BSFs. This motivates studying how training organizes the capacity that groups provide.

Supervision for concept representations. Annotations offer another way to influence that organization. Task-driven dictionary learning establishes supervision as part of sparse representation learning [25], while SpLiCE connects representations to word-derived concept directions [4]. For multi-dimensional concepts, SLiCS learns concept-specific groups of atoms with nonnegative, input-dependent coefficients [23]. Concept-SAE instead learns representations supervised for concept existence and location, combining concept tokens with free tokens to reconstruct activations [8]. These methods connect supervision to the content of a concept unit. Other approaches supervise class-aligned scalar features [28], centralize concepts for diffusion unlearning [6], and anchor directions or subspaces [15]. GGF instead guides consolidation within the native BSF architecture.

From spatial alignment to group organization. Group evaluation connects spatial responses to represented information. Network Dissection measures alignment with annotated regions [3], and LUCID-SAE evaluates the spatial grounding of sparse concepts [20]. Foreground/background prototype pooling [30] and support decontamination [31] further motivate comparing object information with surrounding evidence. We connect spatial alignment to the object-related activation contrast reconstructed by a group. Measuring recovery through group subsets and removing selected contributions from fixed readouts then reveals how concentrated that information is and how the readouts use it. Profiles of object responses and individual direction contributions show whether these associations repeat across images and agree within groups.

3. Methodology

GGF uses Vanilla BSF's encoder, group gate, decoder and fixed groups [14, 18]. We choose it over Grassmannian and Group Lasso for its initial COCO Capture lead, supported by synthetic recovery and reconstruction (Appendix C.7). Figure 3 follows two stages: Initialization establishes a reconstruction model, then Guided Learning adds object guidance. The supplementary code and evidence bundle accompanies this manuscript.

3.1. Features and object evidence

We use DINOv3 ViT-B/16 [27] as our frozen backbone, following Fel et al. [14], who trained BSFs on DINOv3's final-layer patch activations. Each image supplies 196 vectors of dimension $d = 768$. We resize the shorter side to 256 and center-crop to 224×224, applying the same geometry to masks. Training-derived positional means remove average spatial patterns, and a shared scale normalizes magnitude (Appendix A.1). All methods share these cached inputs. Because attention combines information across the image, regional features can also reflect context beyond their own pixels.

Write $x_{ip} \in \mathbb{R}^d$ for the normalized vector at patch p of image i. Let $\mathcal{C}$ be the supervised category set; $\mathbf{1}[\cdot]$ is an indicator and $\|\cdot\|_2$ the Euclidean norm. For object instance o of category c in image i, let f_{op} be its coverage of patch p, q_{icp} the category-union coverage, and u_{ip} the crowd-mask coverage. Write $v_{ip} = \mathbf{1}[u_{ip} \le 0.1]$ for usable patches and $b_{icp} = v_{ip}\mathbf{1}[q_{icp} \le 0.1]$ for comparison patches. Foreground/background prototypes motivate regional pooling [30]. Here we define an activation contrast by subtracting the comparison mean from the coverage-weighted object mean:

$$w_{op} = \frac{f_{op}v_{ip}}{\sum_{p'} f_{op'}v_{ip'}} - \frac{b_{icp}}{\sum_{p'} b_{icp'}}, \qquad \delta x_o = \sum_p w_{op}x_{ip}. \tag{1}$$

A contrast requires at least two usable foreground patch-equivalents and four comparison patches. Since comparison regions can contain other objects, subtraction reduces shared components without isolating a pure concept or removing all context. Images without eligible instances still contribute to reconstruction.

3.2. Initialization

Object patches and starting axes. Initialization places strong foreground directions together so guidance starts from object-associated groups (Figure 3). A dictionary contains G groups of r directions, with decoder $D_g \in \mathbb{R}^{d\times r}$, encoder $E_g \in \mathbb{R}^{r\times d}$, and bias $b_g \in \mathbb{R}^r$ for group g. Initialization uses training patches with at least 50% category coverage and no more than 10% crowd coverage. For each retained category, the leading r eigenvectors of the uncentered foreground second moment initialize one decoder group, retaining high-energy foreground structure. The encoder starts as the decoder transpose with zero bias, while random reserve groups can model other activation structure. Keeping a shared origin lets the axes reflect both category means and

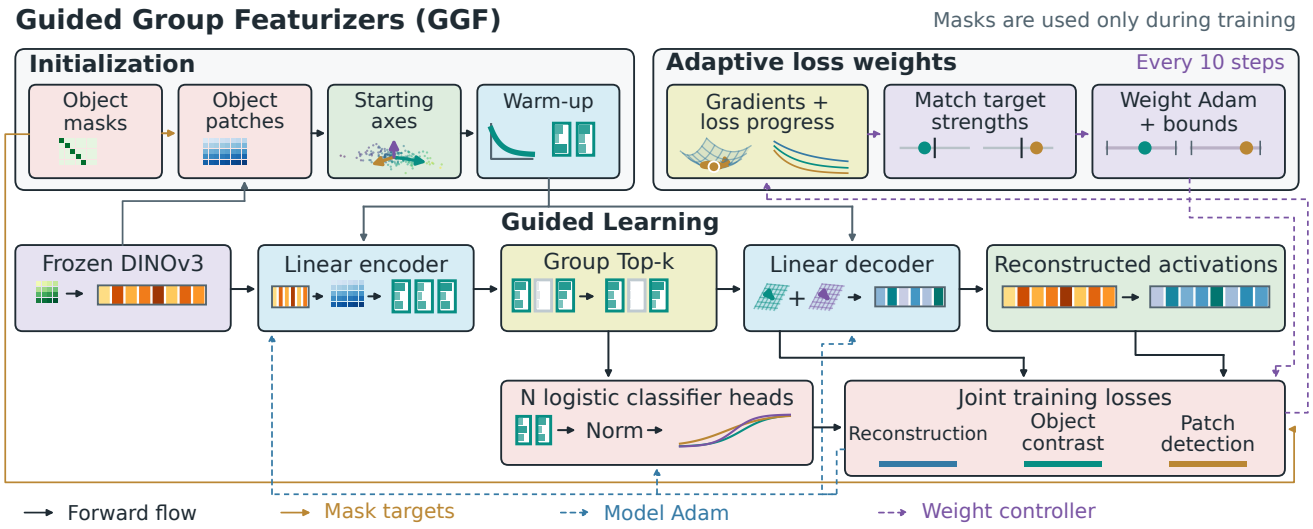


Figure 3. GGF training workflow. Upper left: masks initialize axes before reconstruction warm-up. Guided Learning encodes, selects and decodes groups from left to right. For N supervised categories, N logistic heads map group norms to detection loss, alongside reconstruction and object contrast. Model Adam updates the representation and heads; the upper-right controller adjusts loss weights.

variation. Training can then change a group's orientation or the categories it represents (Appendix A.2).

Warm-up. Reconstruction-only warm-up adapts the starting directions to the full activation distribution before applying object losses. The resulting encoder and decoder initialize *Guided Learning*, where selected groups receive object-specific guidance.

3.3. Guided Learning

Forward pass. For a normalized patch vector x, the encoder produces coefficients $a_g(x)$. Group Top-k retains k groups with largest coefficient norms; $K(x)$ is their index set, $z_g(x)$ their gated codes, and $r_g(x)$ their decoded contributions. The reconstruction is $\widehat{x}$ (Figure 3):

$$\begin{aligned} a_g(x) &= E_g x + b_g, \quad K(x) = \mathrm{TopK}_k\{\|a_h(x)\|_2\}_{h=1}^{G}, \\ z_g(x) &= a_g(x)\mathbf{1}[g \in K(x)], \\ r_g(x) &= D_g z_g(x), \qquad \widehat{x} = \sum_g r_g(x), \end{aligned} \tag{2}$$

Top-k retains at most kr scalar coefficients per patch.

Fixed readouts and logistic classifiers. After warm-up, training data select each category's contrast readout g_c and detector readout g_c^{det} independently because the tasks can favor different groups (Section 4). Readouts may share groups or differ from initialized slots. Fixing these assignments gives guidance stable targets without evaluation-set selection.

For $N = |\mathcal{C}|$ supervised categories, each head maps its group norm $h_{icp} = \|z_{g_c^{\mathrm{det}}}(x_{ip})\|_2$ to logit ℓ_{icp} and probability π_{icp}. It learns slope parameter α_c and intercept β_c, with fixed training-derived center m_c and scale $t_c \geq 10^{-3}$:

$$\begin{aligned} \ell_{icp} &= \mathrm{softplus}(\alpha_c)\frac{h_{icp} - m_c}{t_c} + \beta_c, \\ \pi_{icp} &= \mathrm{sigmoid}(\ell_{icp}), \end{aligned} \tag{3}$$

The positive slope calibrates activity while preserving norm ordering. The heads add $2N$ parameters (150 for COCO, 38 for VOC). Evaluation uses the raw norms, so the heads affect training but do not enter reconstruction or Top-k selection.

Reconstruction, contrast, and detection. With B a clean patch batch and ϵ standard normal noise, denoising [29] trains tolerance to feature perturbations using $\widetilde{x} = x + 0.1\,\mathrm{std}(B)\epsilon$. At update u, weights $\lambda_C(u)$ and $\lambda_D(u)$ scale contrast loss $\mathcal{L}_{\mathrm{contrast}}$ and detection loss $\mathcal{L}_{\mathrm{det}}$; $\mathcal{L}_{\mathrm{rec}}$ is reconstruction loss and $\mathcal{L}$ the total. The expectation averages training patches and noise:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\mathrm{rec}} + \lambda_C(u)\mathcal{L}_{\mathrm{contrast}} + \lambda_D(u)\mathcal{L}_{\mathrm{det}}, \\ \mathcal{L}_{\mathrm{rec}} &= \mathbb{E}\|x - \widehat{x}(\widetilde{x})\|_2^2/d. \end{aligned} \tag{4}$$

This noise is used only for reconstruction; object contrast and patch detection are computed from clean inputs.

Preserving regional contrast. Using clean inputs, the chosen group's regional contribution is $\delta r_{og} = \sum_p w_{op} r_g(x_{ip})$. For batch categories $\mathcal{C}_B$ with eligible instances $\mathcal{O}_{Bc}$, the

object loss is

$$\mathcal{L}_{\text{contrast}} = \frac{1}{|\mathcal{C}_B|} \sum_{c \in \mathcal{C}_B} \frac{1}{|\mathcal{O}_{Bc}|} \sum_{o \in \mathcal{O}_{Bc}} \frac{\|\delta x_o - \delta r_{og_c}\|_2^2}{\max(\|\delta x_o\|_2^2, 10^{-6})}. \tag{5}$$

Normalizing each target and averaging categories equally prevents larger objects or frequent categories from dominating this objective.

Localizing category evidence. A group can reconstruct a regional contrast without responding selectively at the object location. Detection therefore uses the classifier probabilities to reward activity on object patches. Usable patches with coverage at least 0.5 are positive and those with coverage at most 0.1 are negative. Intermediate patches are ignored. Let $\mathcal{C}_B^{\text{det}}$ contain batch categories with usable detection patches. Write $\mathcal{A}_{c,t}$ for their patch pool with sign t ($+1$ foreground, -1 background), and T_c for its nonempty signs:

$$\mathcal{L}_{\text{det}} = \frac{1}{|\mathcal{C}_B^{\text{det}}|} \sum_{c \in \mathcal{C}_B^{\text{det}}} \frac{1}{|T_c|} \sum_{t \in T_c} \frac{1}{|\mathcal{A}_{c,t}|} \cdot \sum_{(i,p) \in \mathcal{A}_{c,t}} \text{softplus}(-t\ell_{icp}). \tag{6}$$

Averaging the positive and negative pools separately prevents the more numerous background patches from dominating. A task contributes zero when its eligible pool is empty.

Model Adam and adaptive loss weights. The three losses differ in scale and learning speed, so the model and its loss weights are updated separately. Adam [22] updates the encoder, decoder and heads, followed by unit normalization of decoder directions (Figure 3). A separate controller compares encoder-gradient norms and relative loss progress. Every ten steps, its own Adam optimizer adjusts the two bounded object weights while keeping the reconstruction weight at one. Gradient norms and targets are detached, so these updates require no second-order differentiation and take effect at the next model step. This GradNorm-inspired refinement [7] and its exact bounds appear in Appendix A.4. Constant-weight controls distinguish object guidance from the controller's smaller contribution. Algorithm 1 summarizes the full workflow.

4. Experimental design

Data and evaluation. We use COCO train2017 [24], retaining 75 categories. The shared allocation has 2,000 training images and three 400-image cohorts for selection, confirmation and readout calibration. The 75-group study calibrates on confirmation images and scores 1,500 reserved audit images (Appendix B.1). For cross-dataset evaluation, we fit and develop models on VOC2012 training images and test on official validation images [9, 10]. VOC normalization, initialization and readouts use its own training data (Appendix C.4).

Group count and category splits. We use 75 groups per method for interpretability: SLiCS requires one dictionary per category, and matching that count makes exhaustive review practical for 75 categories. The methods can still differ in parameters, activity and learned meanings. Other counts answer complementary questions: 256 groups support the BSF screen, while 1,024 and 4,096 groups support reconstruction, capacity and training-component comparisons. These BSF control arms match group count within each comparison (Appendices C.7 and C.5).

To test transfer beyond guidance, two complementary folds guide training on 37 or 38 categories while retaining 75 groups. Excluded annotations enter neither initialization, guided training nor model selection, although their objects may occur in training images. Their readouts are selected after fitting on separate calibration data. Shared Vanilla counts once per run. Native SLiCS with 37 or 38 dictionaries is capacity-unmatched; Concept-SAE with 111 tokens remains exploratory (Appendix G.6).

Comparisons and controls. Supervised Vanilla shares GGF's object losses and classifier heads, testing what initialization and adaptive weighting add (Appendix C.1). Mask-initialized and fixed-guidance arms separate these components (Appendix D.1). We adapt SLiCS to patch-level training and retain Concept-SAE's native training stages [8, 23]. Both use COCO-selected learning rates on VOC (Appendices C.2, C.3). Further studies vary capacity, activity, loss weights and continuation (Appendices D, E), while Manifold Zoo tests recovery of known generating factors (Appendix F.3).

Primary interpretability measures. We assess coherence [2, 13, 17] using six strongest-response views for every group and three fixed pairs. Reviewers judge specific shared meaning (e.g., object type, part or material) and visual structure separately for each pair. A whole-group theme must fit at least five views, with specificity rated separately. We score majority votes across three fresh blinded GPT-6.1 Sol contexts at high reasoning effort [26]. All planned pairs and groups count toward their respective totals, including those with unclear judgments. Separately, Gemini 3.5 Flash-Lite [19] applies the same criteria to the same blinded cards once per task and group (Appendix G.6).

Response concentration asks how broadly a small subset detects an object category at a measured false-positive rate. Calibration ranks all 75 groups by balanced accuracy for each category and sets thresholds at the 95th percentile of target-negative responses. We freeze the top 1, 2, 4 or 8

groups before audit scoring. A crop is detected if any selected group exceeds its threshold. Coverage is the fraction of known target-present crops detected; the false-positive rate is the fraction of known target-absent crops detected [12]. We average these rates equally over the 73 supported categories. Unknown presence labels are excluded, with support retained for all 75 targets. Mathematical definitions are in Appendix G.6.

Supporting alignment and reconstruction measures. Object contrast Capture measures how much regional object-versus-background information one group reconstructs, adapting block recovery [14] through foreground/background pooling [30]. Patch AP measures how well native responses rank object patches above comparison patches [11]; reconstruction VE is one minus normalized MSE relative to the evaluated split's global mean [16]. Definitions, readout conventions and the 32-patch benchmark versus full-grid audit distinction appear in Appendix B.2.

Complementary organization tests. We also examine whether object associations repeat across image sets, remain stable under appearance changes and depend on selected groups. These tests use category-response profiles, training-selected subsets, contribution-PCA maps [14] and fixed linear-probe removals [1, 16]; their protocols are given in Appendix G.

Uncertainty and cost. The 75-group study uses seven training orders (seeds 21–27) for coherence, concentration, Capture, Patch AP and applicable reconstruction metrics. The original COCO GGF–SLiCS benchmark reports five of the seven orders (21–25) using its 32-patch AP and joint-reconstruction protocol. Most other comparisons, including paired capacity and continuation, use three runs. Screens and pilots use one or two (Appendices B–E). For n orders, we report sample SD ($n-1$ denominator) with shared initialization and images, training-run Student-t intervals and conditional image-bootstrap intervals [11]. Appendix H details fitting time, latency, memory and partial-FLOP coverage.

5. Results

We begin with semantic coherence and response concentration, then examine the object information these groups retain.

5.1. Coherent meanings at a common group count

With all category labels, the GPT majority gives GGF $93.4 \pm 1.6\%$ specific-pair coherence, versus $83.4 \pm 2.5\%$ for SLiCS, $67.9 \pm 3.1\%$ for supervised Vanilla and $59.9 \pm 4.6\%$ for Vanilla (Figure 4(a), Table 1). The whole-group test gives $99.4 \pm 1.5\%$ for GGF and $97.1 \pm 1.4\%$ for SLiCS,

Table 1. Blinded coherence at 75 groups: mean ± sample SD (%) across seven training orders. Pair meaning, visual structure and five-of-six whole-group themes are separate AI-majority endpoints. Every planned pair/group is retained.

Method	Meaning	Visual	Whole
All category labels			
GGF	93.4 ± 1.6	86.0 ± 5.2	99.4 ± 1.5
SLiCS	83.4 ± 2.5	80.4 ± 4.3	97.1 ± 1.4
Supervised Vanilla	67.9 ± 3.1	62.9 ± 2.4	70.7 ± 5.3
Vanilla BSF	59.9 ± 4.6	54.7 ± 1.8	66.1 ± 4.4
Category-split guidance			
GGF	79.9 ± 1.7	74.0 ± 4.3	82.8 ± 2.4
Supervised Vanilla	72.8 ± 3.1	68.0 ± 5.0	75.4 ± 4.1
Vanilla BSF	60.4 ± 4.4	57.2 ± 3.5	64.4 ± 6.5

placing both near ceiling on the broader theme criterion. Parts, textures and cross-category motifs are valid meanings, so intended category names do not determine the judgment. Figure 2 shows specific-pair successes and failures.

With category-split guidance, GGF's specific-pair score is $79.9 \pm 1.7\%$, versus $72.8 \pm 3.1\%$ for supervised Vanilla, while their whole-group scores are $82.8 \pm 2.4\%$ and $75.4 \pm 4.1\%$ (Figure 4(b)). These averages cover all groups in each fit. Some folds tie or reverse.

Gemini scores are lower overall, but GGF still leads on all three coherence measures in both settings. It leads on specific-pair coherence in all seven orders, averaging 9.5 percentage points above SLiCS with all labels and 4.7 above supervised Vanilla with split guidance (Appendix G.6).

5.2. Concentration beyond the guidance categories

The strongest-response examples show what a group responds to, but not how broadly or selectively it does so. We therefore select small group subsets using 400 calibration crops, freeze them and evaluate category presence on 1,500 audit crops (Appendix G.6). In the full-category condition, one GGF group covers $90.62 \pm 0.10\%$ of target-present crops, compared with $79.60 \pm 0.79\%$ for SLiCS. Their false-positive rates are $5.05 \pm 0.07\%$ and $4.66 \pm 0.11\%$, respectively, so SLiCS has slightly fewer false positives. Adding groups increases both coverage and false positives (Figure 4(c,d)).

This coverage advantage extends to categories whose annotations never entered initialization or guided training. One GGF group covers $66.13 \pm 1.87\%$ of their target-present crops, versus $59.20 \pm 1.76\%$ for supervised Vanilla and $57.15 \pm 1.99\%$ for Vanilla. The corresponding false-positive rates are $5.09 \pm 0.14\%$, $5.27 \pm 0.23\%$ and $5.22 \pm 0.10\%$. Excluded objects can still occur in training images, and their labels calibrate readouts after fitting. The coverage advantage thus reflects calibrated category transfer within the existing image pool.

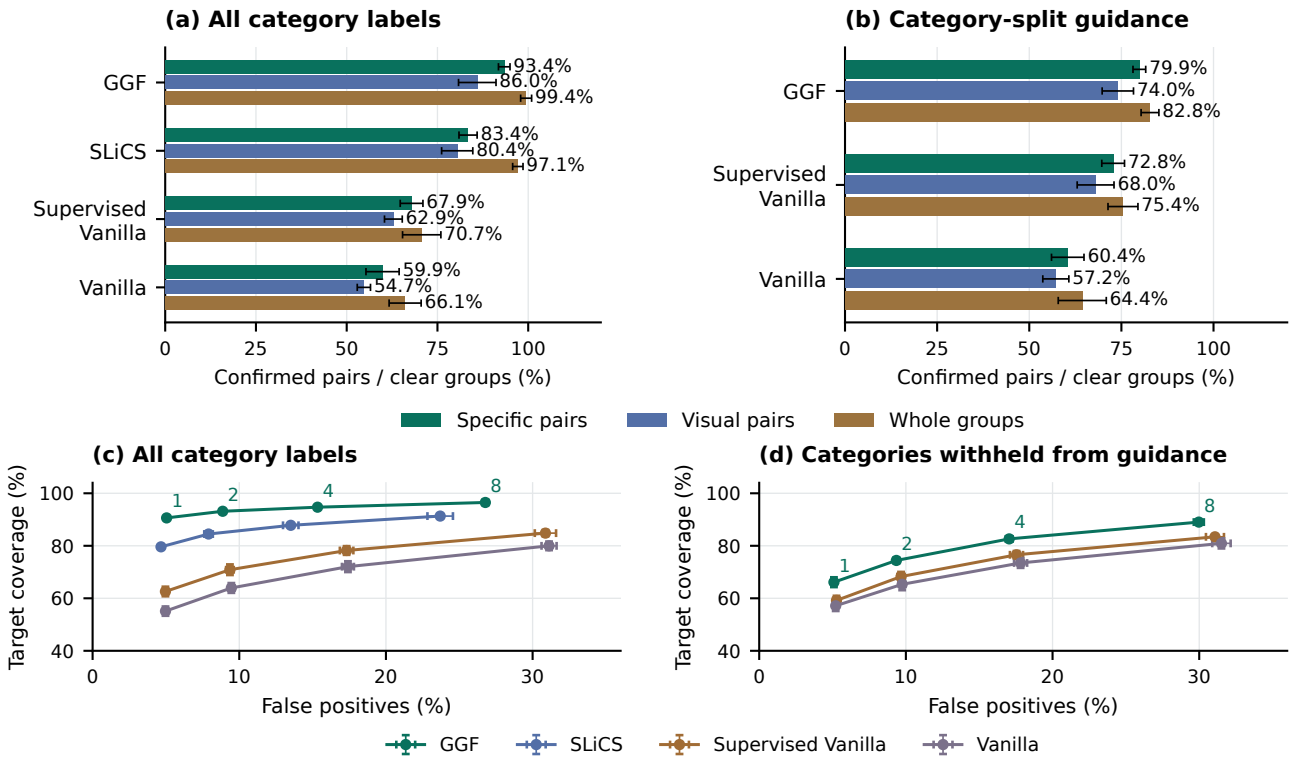


Figure 4. **Coherence and response concentration at 75 groups.** (a,b) Three blinded AI reviews assess specific-pair meaning, visual structure and whole-group themes ($\geq 5/6$ views). Category-split scores include all groups. (c,d) Calibrated subsets detect category presence across 73 supported targets; labels give GGF's subset sizes. Panel (d) withholds categories from guidance. Means and sample SD (both axes for curves) summarize variation across seven training orders with shared initialization and images. Protocols are in Appendix G.6.

5.3. Object alignment and reconstruction

These more interpretable responses also retain object-region information. At 4,096 groups, GGF achieves $42.77 \pm 0.10\%$ Capture and $74.76 \pm 0.14\%$ Patch AP, versus $21.37 \pm 0.84\%$ and $51.66 \pm 1.55\%$ for Vanilla. Capture improves for all 75 categories in every run, and mean Patch AP improves for 72 categories. Paired image bootstrap intervals support gains in both metrics. The improvement has a reconstruction cost: GGF explains $76.94 \pm 0.27\%$ of activation variance, versus $80.35 \pm 0.39\%$ for Vanilla (Appendix C.5). Extending guided training at 1,024 groups retains this alignment–reconstruction tradeoff (Appendix E.1).

The supervised comparisons place these gains alongside alternative representations. On COCO and VOC, GGF exceeds SLiCS in Capture, Patch AP and reconstruction VE in every run. The VOC gains are 21.31, 16.33 and 22.32 percentage points, respectively (Figure S2; Appendix C.2). GGF also exceeds Concept-SAE on all three metrics, with VOC gains of 42.69, 27.31 and 14.80 points (Appendices C.3 and C.4).

On withheld categories in the 75-group study, GGF gains 2.63 Capture points over supervised Vanilla. Full-grid Patch AP is $22.92 \pm 1.17\%$ for GGF and $22.19 \pm 0.69\%$ for supervised Vanilla, with GGF ahead in 5 of seven orders. Vanilla BSF has the highest reconstruction VE in both guidance settings (Tables S40 and S41).

Separate analyses compare the 1,024-group BSFs with native supervised alternatives. GGF's intended category accounts for $77.1 \pm 0.4\%$ of positive category-response profile weight, versus $63.9 \pm 1.6\%$ for supervised Vanilla. Its profiles are the most reproducible across two image halves (Appendix G.5). Across six categories, controlled rotation, scale and brightness changes give GGF $80.92 \pm 0.33\%$ worst-condition recognition AP, close to supervised Vanilla's $80.52 \pm 1.22\%$. Recognition is clearly higher than Concept-SAE's. Each transformation also gives GGF higher regional Capture than both BSF controls (Appendix G.4). Both reuse existing images in configurations separate from the 75-group study.

5.4. What initialization and guidance contribute

Supervised Vanilla shares GGF's object objectives and classifier heads, allowing us to distinguish their benefit from that of initialization and adaptive weights. At 1,024 groups on COCO, these shared components recover most of the Capture gain. GGF adds 0.73 points, with a 95% interval of

Table 2. Component ablation: mean ± sample SD (%). All arms use 4,096 groups, 12 directions, Top-32 and 10,640 updates. Fixed guidance uses training-derived weights.

Arm	Capture	Patch AP	VE
Vanilla	21.37 ± 0.84	51.66 ± 1.55	80.35 ± 0.39
Mask + rec.	23.56 ± 0.05	68.55 ± 0.17	79.56 ± 0.23
Fixed guidance	42.25 ± 0.12	74.63 ± 0.19	77.33 ± 0.20
GGF	42.77 ± 0.10	74.76 ± 0.14	76.94 ± 0.27

$[-0.05, 1.52]$, and 10.91 Patch AP points, with an interval of $[10.70, 11.11]$, while reconstruction VE decreases by 1.35 points. On VOC, GGF gains 1.72 Capture and 1.36 Patch AP points at a 0.37-point VE cost. Both object metrics improve in every run, although the intervals support an additional gain only for Capture (Appendices C.1, C.4).

The component comparison in Table 2 explains this pattern further. Initialization primarily improves localization, while adding object losses supplies most of the Capture gain. Fixed guidance retains 97.6% of the full Capture improvement over Vanilla, and adaptive weighting adds a clear improvement only for Capture (Appendix D.1).

5.5. Capacity and known-factor recovery

Increasing capacity offers an additional way to improve recovery. With equal additional updates, expanding groups from eight to twelve directions adds 0.82 Capture and 1.60 VE points, at the cost of 50% more parameters and active coefficients (Appendix E.1).

We also test consolidation where the generating factors are known. On Manifold Zoo, GGF improves recovery and reconstruction over the archived BSFs, and every assigned group has factor-recovery score $R^2 > 0.90$ in every run (Appendix F.3, Table S25).

6. Discussion

BSFs give concepts room to occupy several directions within one group. GGF uses that capacity to improve interpretability by making responses more semantically coherent and concentrating object-related information within individual groups.

The selected bird and doughnut readouts make the semantic benefit concrete (Figures S16 and S17; Appendix I). GGF repeatedly marks feathers or doughnut surfaces, while SLiCS combines hair with feathers or different topped foods. The Vanilla variants also include fingers, sky or unrelated regions. GGF has three confirmed specific pairs in both cases, compared with one for each alternative. Its advantage is a description that remains specific across varied examples, allowing the same group to retain a consistent meaning (Appendix G.6).

The consolidation tests quantify a complementary benefit: information available within individual groups. GGF's higher Capture indicates more regional information. Removing its contributions also weakens fixed readouts (Appendices G.2, G.3). The 75-group coverage curves show that small GGF subsets detect objects in more crops at the measured false-positive cost (Appendix G.6). Within the existing image pool, this advantage extends to categories excluded from guidance even where Patch AP is close to supervised Vanilla (Table S40). Useful information can still remain in other groups.

That organization should also persist as an object's appearance changes. In the six-category controlled 2D study, GGF and supervised Vanilla recognize objects similarly under rotation, resizing and brightness changes, but GGF retains greater mean regional Capture (Figure S11(b,c); Appendix G.4). GGF's category preferences also reproduce more consistently across image halves in the guided-category development study (Table S30). Jointly rotating a BSF's encoder and decoder can change direction profiles while preserving group norms and contributions [14, Sec. 3.5]. Whole-group focus and reproducibility therefore assess object association more directly (Appendix G.5).

The component studies trace most Capture gains to object losses. Initialization mainly improves localization, and adaptive weighting adds a smaller Capture benefit (Appendix D.1). Recovery of known synthetic factors supplies a direct consolidation check (Appendix F.3). Capacity studies place these gains alongside reconstruction, size and activity costs (Appendices E.1, E.2).

These improvements remain efficient to train. At 1,024 groups, GGF uses $9.3\times$ fewer learned parameters on COCO and $4.9\times$ fewer on VOC than Concept-SAE, excluding the frozen backbone and probes (Appendix C.3; Table S43). On VOC, the 19-group comparison gives a $18.33\times$ mean training-loop speedup over SLiCS on A100. At 1,024 groups on T4, GGF is $1.66\times$ faster than Concept-SAE, with 3.4% overhead over supervised Vanilla (Appendices H.1, H.2). This makes coherent, concentrated groups practical within the same BSF architecture.

7. Conclusion

GGF shows how object guidance can make block-sparse visual representations easier to interpret. Its groups more often retain a recognizable meaning across images, and responses to annotated categories are accessible through fewer groups. This concentration advantage also extends to categories excluded from guidance, suggesting that the resulting organization can be useful beyond the supervision vocabulary. Reconstruction, localization and appearance analyses provide complementary evidence about the visual information these groups retain. GGF makes object guidance a practical means of learning more coherent, interpretable groups that consolidate object-related information.

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Supplementary material

Appendix contents

A. Implementation and training 10
B. Data and evaluation protocol 12
C. Model comparisons 15
D. Components and loss-weight studies 22
E. Capacity and continuation studies 25
F. Response geometry and synthetic recovery 26
G. Group organization and functional diagnostics 33
H. Fitting costs, execution profiles and FLOP coverage 46
I. Qualitative comparisons 49

A. Implementation and training

A.1. Feature construction and objective details

Normalization and masks. For training-image indices $\mathcal{I}_{\mathrm{tr}}$, let $N_{\mathrm{tr}} = |\mathcal{I}_{\mathrm{tr}}|$ and let $a_{ip} \in \mathbb{R}^d$ be DINOv3's patch vector at position $p \in \{1, \dots, 196\}$, with $d = 768$. The training positional mean is μ_p. The positive scale s_{norm} makes average squared training coordinates approximately one and is unrelated to random seeds. With $\|\cdot\|_2$ denoting Euclidean length:

$$\mu_p = \frac{1}{N_{\mathrm{tr}}} \sum_{i \in \mathcal{I}_{\mathrm{tr}}} a_{ip}, \quad x_{ip} = s_{\mathrm{norm}}(a_{ip} - \mu_p),$$
$$s_{\mathrm{norm}} = \sqrt{\frac{d}{\frac{1}{196 N_{\mathrm{tr}}} \sum_{i \in \mathcal{I}_{\mathrm{tr}}} \sum_{p=1}^{196} \|a_{ip} - \mu_p\|_2^2}}. \tag{S1}$$

All splits reuse training-only means and scale, without coordinate rescaling or whitening. The frozen backbone is `facebook/dinov3-vitb16-pretrain-lvd1689m`. Its final patch tokens retain attention context. We discard the CLS token and four registers. Images use bicubic resizing and masks use nearest-neighbor resizing with identical crop geometry. Mask thresholds and object contrasts follow Section 3.1. Appendix B.1 specifies stored mask precision and split allocation.

Loss averaging. The reconstruction loss $\mathcal{L}_{\mathrm{rec}}$ compares predictions from noisy inputs with clean targets. Let B be a batch of clean d-dimensional patch vectors and $|B|$ its number of patches. For each $x \in B$, let $\widetilde{x}$ be the noisy input, $\widehat{x}(\widetilde{x})$ its reconstruction, and ϵ a vector of independent standard normal draws. The scalar $\mathrm{std}(B)$ is the standard deviation over all batch entries:

$$\widetilde{x} = x + 0.1\,\mathrm{std}(B)\epsilon, \qquad \mathcal{L}_{\mathrm{rec}} = \frac{1}{|B|d} \sum_{x \in B} \|x - \widehat{x}(\widetilde{x})\|_2^2. \tag{S2}$$

The averaging gives patches and coordinates equal weight. Reconstruction and contrast update both encoder and decoder. Detection updates the encoder and the two-parameter heads, with no decoder gradient. Object losses use clean inputs. Their averaging, head initialization and fixed training normalization are specified in Equations 3, 5, and 6. Training uses all 196 patches. Detection audits rank raw group norms on the deterministic sample described in Appendix B.1.

Protocol choices and empirical support. Mask/support cutoffs, normalization, noise and warm-up length are study conventions, not literature-prescribed constants. Component/loss controls (Appendix D), capacity/continuation studies (Appendix E) and mask-threshold sensitivity (Appendix B.2) test selected choices, not every constant.

A.2. Mask-derived starting directions

The fresh configuration uses the leading $r = 12$ directions. The original three-direction study uses $r = 3$. For category c, let $\mathcal{F}_c$ contain normalized training patch vectors covered at least 0.5 by the union of that category's masks and at most 0.1 by crowd masks. Crowd annotations mark regions without separate instance masks. For $x \in \mathcal{F}_c$, $x^\top$ denotes transpose. We compute the second-moment matrix $S_c = \sum_{x \in \mathcal{F}_c} x x^\top$ in float64. Its r leading unit eigenvectors retain maximal foreground energy under rank-r orthogonal projection and initialize one decoder group. In the original three-direction runs, we obtain these directions from the saved singular value decomposition (SVD) table, without subtracting a category mean after shared normalization. This is the sense in which the calculation is uncentered. Each direction is oriented so positive projections contain at least as much squared energy as negative ones. GGF coefficients remain signed. We reuse the verified float32 table to avoid differences from recomputing the SVD.

In the original 1,024-group study, groups 0–74 receive the category directions in order. Groups 75–1023 retain same-seed random directions. The encoder starts as the decoder transpose, with zero bias. Across seeds, the 75 SVD groups are identical. The 949 random groups, noise, and data order vary. The runs therefore share one SVD dictionary rather than independently estimating it. After warm-up, we search

all 1,024 groups for each category's contrast and detector readouts, the groups used to score it. Initial category slots are not enforced: categories may share groups, and contrast and detection may choose different groups.

A.3. Main GGF training algorithm

Let X denote normalized training patch activations, Y their aligned masks, $\mathcal{V}$ the development split, and s a random seed. The dictionary has G groups of width r, with k retained per patch. T stores fixed training-selected category readouts. w_C, w_D are the bounded object-loss weights. Development data record scores and eligibility. They do not supply the controller's gradients or targets.

Algorithm 1: GGF: Initialization and Guided Learning

Input: $X, Y, \mathcal{V}, s, G, r, k$ and the fixed training schedule.

1: Initialize one category group from each category's TRAIN foreground second moment and randomly initialize reserve groups.
2: Run reconstruction warm-up and choose T using TRAIN masks and activations.
3: Initialize logistic classifier heads and bounded loss weights w_C, w_D.
4: **for** each scheduled guided update **do**
5: Sample training images and compute reconstruction on noisy inputs and object contrast/detection on clean inputs.
6: Retain the task weights for this model update. Every ten updates, use encoder gradient norms and loss ratios to adjust only the two log weights with detached targets.
7: Update encoder, decoder and logistic classifier heads with the retained weights and apply controller changes to the next step.
8: Unit-normalize decoder directions and periodically save model, optimizer, controller and sampling states.
9: **end for**
10: Freeze the scheduled endpoint and fixed readouts before evaluation.
11: **return** GGF and its training records. Group membership never changes.

A.4. Bounded gradient-based weight adjustment

The controller is inspired by GradNorm [7], with a fixed reconstruction anchor and bounded object weights rather than canonical GradNorm's jointly normalized task weights. Let $j \in \{R, C, D\}$ index reconstruction, contrast and detection. Let $L_j(u)$ be the corresponding unweighted loss at update u, and L_j^0 its mean over the first 20 batches whose three losses are finite and greater than 10^{-12}. Let θ_E collect the shared encoder matrix and bias, and $n_j(u)$ denote the Euclidean norm of that task's gradient on θ_E. Write $w_R = 1$, $w_C = \lambda_C$ and $w_D = \lambda_D$. Define the normalized loss ratios ρ_j, weighted gradient strengths s_j, and target strengths a_j:

$$n_j = \|\nabla_{\theta_E} L_j\|_2, \qquad \rho_j = \frac{L_j / L_j^0}{\frac{1}{3}\sum_{h\in\{R,C,D\}} L_h / L_h^0},$$
$$s_j = w_j n_j, \qquad a_j = \left(\frac{1}{3}\sum_{h\in\{R,C,D\}} s_h\right)\rho_j^{1.5}. \tag{S3}$$

A larger ρ_j means slower relative loss reduction and gives that task a larger target strength. Every ten model updates, separate Adam at learning rate 0.01 adjusts $\log w_C$ and $\log w_D$ to reduce $|s_C - a_C| + |s_D - a_D|$. Gradient norms and target strengths are detached: the controller differentiates only through its two weights, without second-order model derivatives. It skips adjustment if the reference is unavailable or any shared gradient norm is at most 10^{-12}.

Weights start at 0.05 and 0.01. After each controller step, the log weights are projected to contrast bounds $[0.0125, a]$ and detection bounds $[0.0025, a]$. The cap a is 0.20 in the first width/weight study, 0.40 in subsequent capacity studies and 0.60 in the fresh 4,096-group comparison. New weights apply to the next model update. Reconstruction remains at one. Only training batches drive the controller. The empirical cap comparison is in Appendix D.2.

A.5. Released baseline implementations

The pinned Goodfire classes `VanillaBSF`, `GrassmannianBSF`, and `GroupLassoBSF` are unchanged. Source identities accompany the paper. Our sampling, fitting, and audits are separate, so this is not full replication.

Vanilla. Decoder directions start at unit length. The encoder starts as their transpose with zero bias, then learns independently. Top-k retains the k largest group norms with signed coefficients. Training minimizes reconstruction MSE, with sparsity supplied by gating alone.

Grassmannian. QR factorization keeps each decoder group mutually perpendicular and unit length whenever accessed. Its tied encoder uses those directions with one unconstrained learned multiplier per group. Top-k selects the requested k active groups by scaled coefficients. The loss is reconstruction error.

Released Group Lasso. For group g, let $\theta_g \geq 0$ be its learned activation threshold, a_g its encoder coefficient vector, and z_g its coefficients after gating. The implementation retains the entire signed vector when its length exceeds the threshold: $z_g = a_g \mathbf{1}[\|a_g\| > \theta_g]$, where the indicator is one when its condition holds and zero otherwise. It does not

shrink retained coefficients. Thresholds start at a quantile of observed norms. During backpropagation, a normal-density kernel approximates the threshold's derivative, with bandwidth tracking the norm scale. Let L_0 count active groups, $\mathbb{E}[L_0]$ be their mean count, and λ be the activity-penalty multiplier. Reconstruction loss plus $\lambda\mathbb{E}[L_0]$ targets mean activity 16 by updating λ separately. This hard gating differs from textbook group lasso, which shrinks coefficients through soft thresholding.

A.6. Numerical implementation and saved state

Numerics. We resize and crop images explicitly while retaining the processor's pixel rescaling and normalization. Cached half-precision features are loaded as float32. Positional means accumulate in float64 and are saved as float32. The shared scale uses float64 energy accumulation over float32-centered features. Verified memory maps stream normalized float32 arrays under Equation S1, without coordinate-wise standardization, PCA, or whitening.

Records and environment. Logs record loss values and weights, gradient norms and cosines, batch support, preservation checks, selected and final states, group activity, inactive groups, and shared readouts. Checkpoints retain the model, optimizer, heads, and random-number states, allowing recovery from local or verified Drive saves. Work since the last save may repeat. Batch hashes verify pairing rather than bitwise-identical optimization.

The scaled run records T4, PyTorch 2.11.0+cu130, NumPy 2.1.3, SciPy 1.16.3, scikit-learn 1.6.1, Transformers 4.57.3, pycocotools 2.0.10 and einops 0.8.2. Final extraction uses Transformers 5.18.0 with a verified state-key rename and unchanged tensor values. The VOC SLiCS fits and execution profiles use A100-SXM4-40GB, CUDA 13.0, Python 3.13.15 and the same PyTorch/NumPy/SciPy versions, with TF32 disabled. Run-specific versions and identities are retained. Matched updates do not imply equal FLOPs or time.

Saved identifiers. GGF denotes mask-derived Initialization followed by Guided Learning with fixed group membership. Adaptive refers to the task-loss weights. Saved `agf_g`, `GGF` and guided display labels map to GGF in `evidence/terminology.csv`. Run IDs, model keys and source hashes identify the saved records. Evidence paths in this Appendix are relative to the companion archive's `code/` directory.

A.7. Data stewardship and author responsibility

COCO can depict identifiable people. Object-response scores neither establish demographic fairness nor justify sensitive-attribute inference. Data, pretrained weights, and third-party code remain subject to their respective terms. Code, model identities, and evaluation records support reproduction. Availability of any redistributed assets must be verified before release. An AI assistant supported coding, analysis, figures, and writing. The authors remain responsible for the results and manuscript.

B. Data and evaluation protocol

B.1. Mask construction and split allocation

Data roles. We split COCO train2017 by image ID, so no image belongs to two splits. Training data update parameters and supervised assignments; development selects checkpoints. Some audits select readouts on calibration data, whereas the matched 1,024-group baseline comparison fixes them from training data (Appendix C.6). Test data only supply scores. Coverage checks use annotations, before model outcomes, to retain 75 categories. Sampling enough examples of scarce categories makes this a coverage-enriched benchmark, not a sample with natural COCO frequencies.

The shared data contain 2,000 training images and three 400-image sets for selection, confirmation, and readout calibration. In the 75-group study, the confirmation cohort serves post-fit calibration; the separate readout-calibration cohort is unused (Appendix G.6). The initial baseline screen uses 800 test images. Scaled guided learning uses a separate 1,500-image test. The matched baseline comparison uses another 1,500 images, excluding all recorded split IDs and requiring at least 20 usable images per category. It reuses the two development sets but freezes all runs before extracting the final test features. The following subsection details coverage requirements for contrast and detection.

Training schedules differ across experiments and are recorded with their results. The fixed-weight loss ablations select eligible development checkpoints. The matched-comparison studies report scheduled final endpoints. Experiments use seeds 21, 22 and 23 unless specified otherwise.

Geometry and storage. Let W and H be the original image width and height in pixels. The resize factor is $t = 256/\min(W, H)$, giving dimensions $(\lfloor tW \rfloor, \lfloor tH \rfloor)$, where the floor symbol rounds down to an integer. We center-crop to 224 pixels by halving the excess width and height and rounding as in the saved implementation. COCO's `annToMask` converts polygon outlines or compressed pixel runs into binary masks. Non-crowd instances stay separate for object contrast, while detection uses the union of all instances of a category. Different categories may overlap. Crowd masks form a separate union used to exclude uncertain regions.

Each patch's category-union coverage is rounded to an 8-bit unsigned integer and divided by 255 when loaded. Instance and crowd masks retain fractional coverage. All thresholds apply to these stored values. An eligible contrast

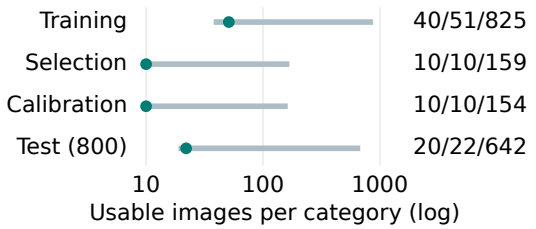


Figure S1. Usable-image coverage in the COCO split allocation. Lines show category minima to maxima. Dots show medians. Labels give minimum/median/maximum. Counts require both audit detection evidence and an eligible object contrast.

requires known foreground coverage totaling at least two full patches and at least four category-negative patches. Its foreground average uses all known foreground coverage, including patches below the detection-positive threshold.

Coverage rule for sampled-patch evaluation. For sampled-patch evaluation, a usable image must provide both a positive and a negative patch in its fixed 32-patch audit sample, as well as an eligible contrast computed from all patches. If imageID is the photograph's integer COCO identifier, the sampling seed is $61{,}002 + \text{imageID}$. Coverage checks, calibration, and test use this same sample. A category's presence somewhere in the original image is insufficient if cropping removes usable evidence. The allocator first supports scarce categories, then fills shared prevalence targets. Quotas do not equalize prevalence because one image can support several categories. These pool and usability rules leave 75 categories, excluding skis, sports ball, baseball bat, toaster, and hair drier.

Scaled fresh-test allocation. Before fitting, a constrained solver found that 800 images could not meet the declared quotas within the available pool. The revised allocation uses 1,500 images, the same 75 categories, and minimum quotas of 20 usable images per category. Annotation-only scans add at most 500 candidate images for each undercovered category. A binary covering solver selects a core set meeting the quotas. Prevalence-based filling completes the split. We verify the saved image IDs and coverage. Preparation stops if quotas cannot be met, rather than relaxing them. The matched baseline comparison uses separate quotas for contrast, positive detection and negative detection evidence.

Identity checks. We verify image IDs, feature/mask shards and crop geometry. The scaled fresh test excludes the recorded confirmation and test cohorts; paired models share images, normalization and category denominators. Readout and checkpoint-selection roles are specified above.

B.2. Additional audit conventions

Where the metrics are defined. Each metric has one canonical mathematical definition. Coherence votes, presence-based concentration and prediction overlap are defined in Appendix G.6; the supporting alignment and reconstruction scores are given below. Synthetic factor recovery is defined in Appendix F.3. Fixed-probe removal effects are in Appendix G.2, worst-condition recognition in Appendix G.4, and response fingerprints, focus, reproducibility and direction agreement in Appendix G.5. Resource ratios and per-image operation counts are defined in Appendix H.

Supporting alignment and reconstruction scores. In evaluation formulas, $\mathcal{C}$ denotes the evaluated category set. Let $\mathcal{O}_c$ contain its eligible objects of category c, and g_c be that category's fixed contrast readout. Regional Capture is

$$C_c = 1 - \frac{\sum_{o\in\mathcal{O}_c} \|\delta x_o - \delta r_{og_c}\|_2^2}{\sum_{o\in\mathcal{O}_c} \|\delta x_o\|_2^2}, \quad \overline{C} = \frac{1}{|\mathcal{C}|}\sum_c C_c. \tag{S4}$$

Errors and target energies are pooled within each category before categories receive equal weight. Concept-SAE uses its native-token calculation (Appendix C.3). BSF patch scores are gated norms, while SLiCS uses independent NNLS projections. The sampled benchmark uses 32 patches per image; the 75-group audit uses all 196. Patch AP measures ranking, rather than COCO segmentation AP. Reconstruction VE is one minus normalized MSE relative to the evaluated split's global mean [16].

Contrast uses all patches and eligible non-crowd instances. The sampled-patch benchmark evaluates 32 patches per image. Unlike the instance-normalized training loss, Equation S4 weights larger contrasts more heavily. Without across-instance centering, it is not regression R^2.

Because Capture and detection can use different readouts, their scores need not demonstrate both properties in one group. AP treats tied scores together, excludes ambiguous and crowd-invalid patches, and sums precision-weighted recall increments, measuring ranking rather than a fixed detection threshold or bounding-box overlap.

Patch ranking and reconstruction. Let $\mathcal{C}_{\mathrm{AP}}$ contain categories with valid positive and negative patches. Let $P_{c\ell}$ and $R_{c\ell}$ be precision and recall after including all patches at distinct score level ℓ in descending order, with $R_{c0} = 0$. The non-interpolated average precision and category-macro score are

$$\mathrm{AP}_c = \sum_\ell (R_{c\ell} - R_{c,\ell-1})P_{c\ell}, \quad \overline{\mathrm{AP}} = \frac{1}{|\mathcal{C}_{\mathrm{AP}}|}\sum_c \mathrm{AP}_c. \tag{S5}$$

Tied scores enter together; ambiguous and crowd-invalid patches are excluded. Categories must contain valid positives

and negatives. For N evaluated d-dimensional vectors x_i and their reconstructions $\widehat{x}_i$, let $\mu_{\mathcal{E}}$ be the evaluated split's global mean. Define squared error $E = \sum_i \|x_i - \widehat{x}_i\|_2^2$ and total variation $V = \sum_i \|x_i - \mu_{\mathcal{E}}\|_2^2$. Then

$$\mathrm{MSE} = \frac{E}{Nd}, \qquad \mathrm{VE} = 1 - \frac{E}{V}. \tag{S6}$$

MSE uses the training-normalized feature coordinates. Natural-image VE uses the evaluated split's global mean $\mu_{\mathcal{E}}$; synthetic VE instead uses the training mean, as specified in Appendix F.3. Thus MSE and $1 - \mathrm{VE}$ have different denominators. No reconstruction score is imputed for independent SLiCS projections.

Why measure object contrast Capture? Single-block recovery in Fel et al. [14] tests whether a block reconstructs a known synthetic factor's contribution. Natural images do not provide that latent contribution. We instead construct an observable regional target: the difference between mask-weighted object features and category-negative features in the same image. PANet [30] provides precedent for separately pooling foreground and background features. Their difference cancels a spatially constant feature offset. It does not remove every contextual influence. Our choice tests whether one group carries the measured object-versus-comparison contrast, rather than claiming access to a ground-truth concept vector.

Normalized reconstruction error is also used to evaluate sparse autoencoders [16]. Our denominator differs: it is the energy of the observed contrast, making zero contrast the reference prediction. For a nonzero target d and prediction r, the score is $1 - \|d - r\|^2/\|d\|^2 = (2d^\top r - \|r\|^2)/\|d\|^2$. Thus exact reconstruction scores one, no reconstructed contrast scores zero, and worse predictions score below zero. Increasing response magnitude alone cannot improve the score indefinitely: $r = ad$ scores $2a - a^2$, maximized at $a = 1$. A common rescaling of target and prediction leaves Capture unchanged. Pooling errors and target energies within a category avoids averaging unstable ratios for individual weak contrasts. Macro-averaging then gives categories equal weight. This construction is our adaptation, not the centered factor R^2 of the synthetic benchmark.

Capture complements AP: a group may rank object patches correctly while reconstructing little of their contrast, or reconstruct a contrast while responding elsewhere. Same-group AP, additional-group Capture and matched-control removal diagnostics therefore qualify consolidation claims. Neither Capture nor those diagnostics establish semantic exclusivity or recovery of every factor of variation.

Additional reconstruction and detection diagnostics. Calibration maximizes F1, the harmonic mean of precision and recall. With TP correctly detected positives, FP falsely detected negatives, and FN missed positives, test reports precision, recall, F1 and intersection over union:

$$\mathrm{F1} = \frac{2TP}{2TP + FP + FN}, \qquad \mathrm{IoU} = \frac{TP}{TP + FP + FN}. \tag{S7}$$

The sampled-patch diagnostic returns zero for a zero denominator; the presence audit retains undefined precision separately. Recall and false-positive rates use Equation (S26). Mask sensitivity changes foreground thresholds with fixed readouts. It does not optimize the two/four support requirements or validate every threshold. A calibration-greedy audit adds up to four groups while contrast error decreases, without refitting coefficients or directions. All-group Capture uses full reconstruction. These diagnostics measure reconstruction by selected groups.

A separate diagnostic fits a balanced logistic classifier to responses from up to four groups ranked on calibration data. Its response normalization also uses calibration data. The inverse regularization-strength parameter is $C = 1$, and the optimizer allows 500 iterations. Its extra flexibility differs from both single-group AP and the training heads. Reports retain category prevalence/support, mask eligibility, group IDs, thresholds, and negative outcomes. Appendix F.2 discloses visual selection.

Uncertainty. For n run-level scores x_s, indexed by $s = 1, \ldots, n$, let $\overline{x}$ be their mean and s_x their sample SD:

$$\overline{x} = \frac{1}{n}\sum_s x_s, \qquad s_x = \sqrt{\frac{\sum_s (x_s - \overline{x})^2}{n-1}}. \tag{S8}$$

For paired scores x_s, y_s, let $d_s = x_s - y_s$ have mean $\overline{d}$ and sample SD s_d. With $t_{0.975,n-1}$ the Student-t quantile for $n-1$ degrees of freedom, the reported paired 95% interval is

$$\overline{d} \pm t_{0.975,n-1}\, s_d/\sqrt{n}. \tag{S9}$$

These intervals concern the declared run variation. The 75-group primary figures report sample SD, not confidence bands. Percentile image-bootstrap intervals use the 0.025 and 0.975 quantiles of the paired resampled differences described below.

Paired bootstraps resample images with the same instance/patch multiplicities for both models. Means retain all 75 categories. Unsupported full-cohort draws are counted as invalid. The scaled comparison has 100 valid and zero invalid draws. Matched baseline intervals use 200 valid draws. Unless stated otherwise, intervals are percentile intervals conditional on fitted models and readouts. They omit training-seed, tuning, assignment and researcher-selection uncertainty and receive no multiplicity correction. Image-bootstrap VE differences remain point estimates. Training-run VE intervals are reported where available. The continuation and capacity

experiments reuse the inspected 1,500-image benchmark, so their comparisons are not independent confirmation. The 4,096-group audit also reports Student-t intervals across training starts. The VOC SLiCS comparison uses paired training-run intervals rather than image-bootstrap intervals (Appendix C.2). This limited replication does not support a well-powered population-level test.

Replication across comparisons. Most comparisons use three training runs per setting, including paired capacity and continuation studies. The COCO GGF–SLiCS comparison uses five training orders sharing one initializer. The baseline, loss-weight and width/activity screens, Vanilla group-count check, cap pilot, supervised Vanilla calibration and random-start diagnostic each use one run per setting. The auxiliary 1,024-group pilot uses two runs.

C. Model comparisons

We begin with the matched supervised Vanilla control, which distinguishes fixed object-supervised training from GGF's initialization and controller. The SLiCS and Concept-SAE comparisons then place these improvements alongside alternative supervised representations. VOC2012 tests the training recipe on another dataset. Capacity and baseline studies are reported separately because they use different cohorts, budgets and readouts. Component controls are collected in Appendix D.1. Group-organization diagnostics are collected in Appendix G.

C.1. Matched object-supervised Vanilla comparison

Run `ef2f97f3dc53` compares reconstruction-only Vanilla, fixed-weight supervised Vanilla and GGF at 1,024 groups, twelve directions and Top-16. All endpoints receive 7,640 reconstruction warm-up and 3,000 continuation updates at Adam learning rate 0.0003. The random-initialized arms share their warm-up. GGF uses mask-derived initialization and cap 0.40. Training-selected contrast/detection readouts remain fixed. The same normalization, image cohorts and within-run sampling rules apply. Reused checkpoints are verified by source and weight identity.

On COCO, fixed-weight calibration compares $(0.05, 0.01)$, $(0.20, 0.20)$ and $(0.40, 0.40)$ at the full continuation budget on the selection start. Development Capture $+0.2$ AP under the original warm-up MSE allowance selects $(0.20, 0.20)$, which satisfies that allowance. The chosen pair applies to the remaining starts. Reconstruction weight is one, and supervised Vanilla uses the same logistic classifier heads and both object objectives as GGF. Each category has one head with a learned slope and intercept on the selected group norm. It uses neither mask initialization nor adaptive weighting. Relative to Vanilla, this control adds annotation targets, contrast/detection objectives and trainable heads together. Their individual effects are not identified by this comparison alone.

Table S1. Matched 1,024-group COCO comparison on the inspected audit cohort: mean ± sample SD (%). Capacity, activity and updates match.

Method	Capture	Patch AP	VE
Vanilla	23.29 ± 0.84	58.72 ± 0.32	73.37 ± 0.16
Supervised	41.80 ± 0.24	65.47 ± 0.22	72.49 ± 0.10
GGF	42.54 ± 0.21	76.38 ± 0.24	71.13 ± 0.03

GGF-minus-supervised-Vanilla gains are 0.73 ± 0.32 Capture, 10.91 ± 0.08 AP and -1.35 ± 0.11 VE points, with paired 95% training-start intervals $[-0.05, 1.52]$, $[10.70, 11.11]$ and $[-1.62, -1.08]$. Compared with reconstruction-only Vanilla, gains are 19.25 ± 1.02, 17.66 ± 0.11 and -2.24 ± 0.17 points, respectively. The supervised control isolates GGF's initialization/controller combination under this calibration rule. The hyperparameter searches have different costs. All per-model, category and paired-image records, frozen identities and verified archive hashes are retained in `evidence/current_control/`.

Table S2. Matched BSF control design.

Arm	Initialization	Continuation
Vanilla BSF	Random	Reconstruction only
Supervised Vanilla	Random	All losses, fixed weights
GGF	Mask-derived	All losses, adaptive weights

C.2. Supervised comparison with SLiCS

We compare GGF and SLiCS on both COCO and VOC, retaining one dictionary per eligible category. The COCO study uses 75 groups and the VOC study uses 19. Both use twelve directions per group and 10,640 training updates. The dataset-specific protocols follow.

COCO data and frozen settings. SLiCS [23] applies alternating nonnegative-code/dictionary updates to DINOv3 patches, adapting the published image-level inputs to patch-level evidence. Both methods share 2,000 training and 400 development images, normalization, masks and 75 category slots. The 1,500-image test excludes prior study splits. Settings, hashes and readouts freeze before extraction. Test labels select no rate, checkpoint or group. Run `a9c638e1ec0b` retains the complete audit.

Settings and their rationale. SLiCS assigns each concept a dictionary [23, Sec. 3.1], giving 75 groups for our eligible categories. Their Sec. 4.2 relates width to capacity and concept overlap. Twelve atoms match GGF's decoder, rather than a published optimum. The 10,640 updates match our budget, replacing their ten-iteration image-level schedule (Sec. 4.4). Spatial masks provide patch-level label support.

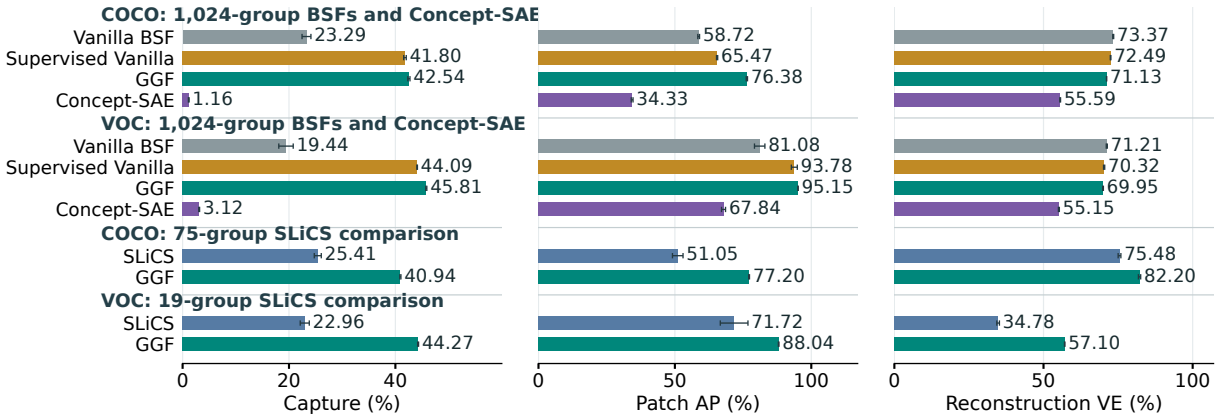


Figure S2. Mean ± sample SD. Upper blocks compare 1,024-group BSFs and Concept-SAE. Lower blocks compare GGF and SLiCS at 75 COCO and 19 VOC groups. Each block retains its own protocol and cohort (Appendices C.1–C.4).

Matched quantities and remaining differences. Both methods share 900 decoder atoms and training-mask SVD initialization [23, Sec. 3.3], with SLiCS signs oriented by positive projection energy. Training orders vary conditional on this initializer and fixed data. GGF uses 7,640 reconstruction plus 3,000 guided updates, Top-32 and cap 0.60. Development-selected learning rate 10^{-3} comes from $\{10^{-4}, 3\times10^{-4}, 10^{-3}\}$ and freezes for orders 24–25. SLiCS receives 10,640 alternating updates on matching image schedules. Mask support constrains training, not inference. The order budget freezes before test access.

GGF has 1,383,452 learned parameters, including its untied encoder, 150 head parameters and two controller weights. SLiCS has 691,200 dictionary parameters. GGF activates 32 groups per patch, versus approximately 75 for joint SLiCS reconstruction. Thus decoder capacity and updates match, while total parameters, sparsity and computation differ.

Object readouts and full reconstruction. GGF uses its gated encoder codes. SLiCS object scores use an independent nonnegative least-squares (NNLS) projection onto each category's twelve directions, preserving its partial-concept readout [23, Eq. (3)]. Both methods share contrast definitions, 32-patch detection samples, masks and category denominators. Reconstruction VE instead uses a joint NNLS solution over all 900 SLiCS directions: independent concept projections are never summed as if they formed one reconstruction. All 294,000 test patches are evaluated per model.

Solver verification and execution. A per-patch float64 projected-gradient KKT check enforces the 10^{-5} threshold against the full objective. Batched GPU FISTA warm starts remove certified rows before cached float64 active-set refinement. Excluded atoms can re-enter, with full SciPy NNLS up to 20,000 iterations as fallback. An eight-patch probe checks objectives against independent full NNLS before each dictionary's work, without retraining or selection. Evaluation resumes verified CPU/GPU prefixes. Records retain precision recovery and mixed execution. Nonunique codes may change activity at equal reconstruction. Audit-call timers exclude deferred NNLS and are not end-to-end inference times.

COCO training results. Table S3 reports all final endpoints. GGF improves Capture, AP and VE in every order. Training-order means are 40.94%, 77.20%, 82.20% for GGF and 25.41%, 51.05%, 75.48% for SLiCS. Paired Student-t intervals for GGF-minus-SLiCS are $[14.78, 16.28]$, $[23.63, 28.66]$ and $[6.19, 7.24]$ points, respectively. These intervals condition on fixed data and initialization. Separately, 1,000 whole-image bootstrap draws per order retain all 75 categories. Every Capture and AP interval excludes zero, conditional on the fitted model rather than combining training-order and image uncertainty.

Development-selected contrast. The original co-primary contrast selects checkpoints by development Capture $+0.2$ AP, including update zero. All GGF selectors retain the final endpoint, while all SLiCS selectors retain the common SVD initializer: 37.19% Capture, 65.76% AP and 67.87% VE. Its zero between-order SD reflects one identical checkpoint, not independent initializations. GGF gains 3.75 Capture and 11.44 AP points, with paired training-order intervals $[3.62, 3.88]$ and $[11.25, 11.63]$. Holm-adjusted two-sided p values are below 1.5×10^{-7} for the two co-primary object metrics. Every order's image interval also favors GGF. Those tests retain their original primary designation. Final-versus-final comparisons are separately reported secondary contrasts, not relabeled primary tests.

SLiCS training raises VE from 67.87% to 75.48% while

Table S3. Values with ± show mean and sample SD. Completed COCO training on the separate supervised test cohort (%). All models have 75 groups of 12 directions and 10,640 updates.

Seed	Method	Capture	Patch AP	VE
21	GGF	40.93	77.06	81.61
21	SLiCS	26.25	52.26	75.33
22	GGF	40.85	77.39	82.52
22	SLiCS	24.51	49.69	75.49
23	GGF	40.93	77.19	82.10
23	SLiCS	25.26	53.56	75.40
24	GGF	41.11	77.03	82.20
24	SLiCS	25.82	51.02	74.96
25	GGF	40.86	77.30	82.55
25	SLiCS	25.22	48.72	76.22
Mean	GGF	40.94 ± 0.10	77.20 ± 0.15	82.20 ± 0.38
Mean	SLiCS	25.41 ± 0.66	51.05 ± 1.94	75.48 ± 0.46

reducing its partial-projection object scores. Thus the larger final-endpoint object gap partly reflects this trajectory. GGF's improvement over the selected initializer is the more conservative object comparison. The result applies to the stated patch-input implementation, not a claim about every supervised method or a reproduction of published image-level retrieval results.

Capacity anchor and evidence. On this COCO cohort, saved 4,096-group GGF averages 42.69% Capture, 78.45% AP and 76.94% VE. These images differ from the 1,500-image pilot cohort that gives 42.77%/74.76%/76.94% in Appendix C.5. This larger anchor is excluded from the 75-group tests. Frozen identities, unrounded seed/category records, intervals and provenance are retained in `evidence/supervised_five_seed/`.

VOC transfer protocol. VOC uses the splits in Appendix C.4, fitting normalization and mask-SVD initialization on training data. Its 19 category dictionaries preserve SLiCS's design, giving 228 decoder directions per method. GGF uses Top-8, cap 0.60 and COCO-selected learning rate 10^{-3} with the same warm-up/continuation budget. Top-8 rounds the COCO active-group fraction $19(32/75)$ before fitting. Both methods retain one fixed group per category and a common training-mask SVD initializer, with SLiCS signs oriented by positive projection energy. Unlike the denoising 1,024-group study, this matched comparison uses clean reconstruction inputs and excludes patches with crowd/void coverage above 0.1. SLiCS receives 10,640 alternating updates. Seeds are 21–23 without a new rate search. Final endpoints are the prespecified primary comparison. The development-selected checkpoints remain separate diagnostic records. GGF has 350,476 learned parameters versus SLiCS's 175,104. SLiCS retains independent category NNLS readouts and joint full-dictionary reconstruction, with KKT tolerance 10^{-5}, float64 recovery and a 20,000-iteration fallback. Eight-patch checks for each SLiCS endpoint certify every row against full CPU NNLS at the 10^{-5} KKT tolerance, with largest scaled squared-error difference 3.48×10^{-10}. These category-matched models are separate from the 1,024-group study. VOC resource measurements use one device and remain separate from COCO timing records (Appendix H).

VOC results. The completed 19-group comparison gives GGF $44.27 \pm 0.09\%$ Capture, $88.04 \pm 0.11\%$ Patch AP and $57.10 \pm 0.07\%$ VE, versus $22.96 \pm 0.85\%$, $71.72 \pm 5.07\%$ and $34.78 \pm 0.44\%$ for SLiCS (Table S4). GGF improves every metric in every paired run. Mean gains are 21.31, 16.33 and 22.32 points. Their paired 95% Student-t intervals are $[19.12, 23.50]$, $[3.91, 28.74]$ and $[21.26, 23.38]$ points (Table S5). These intervals describe variation across paired fits, without pooling models or images from the COCO comparison. All 1,449 VOC validation images are evaluated, including joint full-dictionary reconstruction for SLiCS. The frozen run identity is `544561a40caa4025b92a`, with source lineage, per-seed scores, solver checks and the completion receipt in `evidence/voc_slics_final/`.

Table S4. VOC category-matched comparison (%). Both methods use 19 twelve-direction groups and 10,640 updates. Aggregates show mean ± sample SD.

Seed	Method	Capture	Patch AP	VE
21	GGF	44.21	88.09	57.08
21	SLiCS	22.41	77.18	34.27
22	GGF	44.23	87.92	57.04
22	SLiCS	23.94	67.15	35.04
23	GGF	44.37	88.12	57.18
23	SLiCS	22.54	70.82	35.02
Mean	GGF	44.27 ± 0.09	88.04 ± 0.11	57.10 ± 0.07
Mean	SLiCS	22.96 ± 0.85	71.72 ± 5.07	34.78 ± 0.44

C.3. Supervised comparison with Concept-SAE

Concept-SAE settings and their rationale. Concept-SAE retains presence/mask supervision and residual free modules

Table S5. Paired VOC GGF-minus-SLiCS gains in percentage points. SD and Student-t intervals describe paired training-seed differences.

Metric	Gain $\pm$ SD	95% interval
Capture	21.31 ± 0.88	[19.12, 23.50]
Patch AP	16.33 ± 5.00	[3.91, 28.74]
VE	22.32 ± 0.43	[21.26, 23.38]

[8, Secs. 3.2–3.4], substituting DINOv3 features and dataset masks. One token per eligible category and 36 free tokens follow Sec. 4.2. The 14×14 grid matches DINOv3. BSF capacity differs.

Section 4.2 specifies Adam, batch size 64 and 30/50/30 epochs, with tenfold rate decays after 20 tokenizer epochs and 30 aggregator epochs. Aggregation freezes the tokenizer. Free-module training freezes concept modules. Aggregator/free rates are 10^{-3}. Tokenizer selection includes the published 10^{-3} on COCO development, then freezes for other starts/VOC. Empty-mask MSE and constant-AP checks are additional diagnostics.

The pinned Concept-SAE release (`956d19b`) supplies linear score/mask outputs, merge-L_1 weight 0.01, aggregator KL weight 0.01/temperature 0.02 and aggregation-weight L_1 weight one. The paper instead uses a sigmoid for the presence score and a merge penalty of 0.1. Its mask output is linear. Training follows its longer schedule, not the release's ten epochs.

Completed comparisons. Completed Concept-SAE achieves $1.16 \pm 0.07\%$ Capture, $34.33 \pm 0.33\%$ AP and $55.59 \pm 0.12\%$ VE, versus GGF's $42.54 \pm 0.21\%$, $76.38 \pm 0.24\%$ and $71.13 \pm 0.03\%$ on the common COCO cohort (mean $\pm$ sample SD). This tests native-token outputs under the stated adaptation, not Concept-SAE's original benchmark. VOC results appear in Table S6. All Concept-SAE fits complete the 30/50/30-epoch schedule and pass the empty-mask/constant-output checks. On VOC, tokenizer and aggregator selection retains their last epochs. Schedule completion and these checks establish a functioning fit, not proof of convergence.

Parameter efficiency. GGF has 18,886,808 learned parameters (18,886,656 featurizer, 150 heads, two controller), versus Concept-SAE's 175,125,423. Excluding the frozen DINOv3 backbone and evaluation probes but counting all learned stages, GGF uses 10.8% as many parameters ($9.3\times$ fewer) while improving all COCO scores. On VOC, 19 categories reduce GGF's heads to 38 parameters: the total is 18,886,696 including two controller parameters. Concept-SAE has 92,000,983 parameters. GGF uses 20.5% as many ($4.9\times$ fewer). The concept/free-token architecture is retained, so this is measured parameter efficiency under the stated protocols, not a capacity-matched architecture comparison.

Concept-SAE Capture calculation. Let P_c be category c's biased projector. For image i, predicted mask $\mu_{ic} \in \mathbb{R}^{196}$ and presence $s_{ic} \in \mathbb{R}$ yield $v_{ic} = P_c(s_{ic}\mu_{ic}) \in \mathbb{R}^{768}$. Let r^{CS}_{ipc} denote its contribution at patch p and r^{free}_{ip} the summed free-token contribution. The bias-free aggregator $A \in \mathbb{R}^{196\times|\mathcal{C}|}$ gives

$$r^{\mathrm{CS}}_{ipc} = A_{pc} v_{ic}, \qquad \widehat{x}_{ip} = \sum_c r^{\mathrm{CS}}_{ipc} + r^{\mathrm{free}}_{ip}. \tag{S10}$$

For object o in image $i(o)$, the same weights w_{op} from Eq. (1) give

$$\delta r^{\mathrm{CS}}_{oc} = \sum_p w_{op} r^{\mathrm{CS}}_{i(o)pc} = \left(\sum_p w_{op} A_{pc}\right) v_{i(o)c}. \tag{S11}$$

Capture pools errors over eligible objects before averaging categories:

$$\begin{aligned} C^{\mathrm{CS}}_c &= 1 - \frac{\sum_{o\in\mathcal{O}_c} \|\delta x_o - \delta r^{\mathrm{CS}}_{oc}\|_2^2}{\sum_{o\in\mathcal{O}_c} \|\delta x_o\|_2^2}, \\ \overline{C}^{\mathrm{CS}} &= \frac{1}{|\mathcal{C}|}\sum_c C^{\mathrm{CS}}_c. \end{aligned} \tag{S12}$$

Category-token assignment is fixed, with no probe/decoder refit, rescaling or clipping. Free tokens enter full VE, not category Capture. Patch AP ranks native mask-times-presence scores $s_{ic}\mu_{icp}$.

C.4. PASCAL VOC2012 comparisons

The 1,024-group VOC comparison trains featurizers with COCO-selected GGF defaults and VOC-only normalization, initialization, readouts and evaluation probes. The separate 19-group SLiCS comparison appears in Appendix C.2. The official segmentation training split supplies 1,208 fit and 256 development images. All 1,449 official validation images supply held-out evaluation. Annotation-only eligibility retains 19 object categories. Flickr metadata are checked against COCO and exact decoded-image/Flickr duplicates across VOC splits. This is not an exhaustive perceptual near-duplicate audit. No GGF setting is selected on VOC validation. As on COCO, supervised Vanilla's fixed loss pair is selected on VOC development under the original warm-up MSE rule and frozen thereafter. This tests transfer of the training recipe to a new dataset, not zero-shot transfer of COCO-trained weights or evaluation on the withheld official test server.

Against completed Concept-SAE, GGF gains 42.69 Capture, 27.31 AP and 14.80 VE points. Paired training-run 95% intervals are $[42.57, 42.81]$, $[25.43, 29.19]$ and $[14.58, 15.02]$ points. All metrics favor GGF in each paired run. The methods retain different native output interfaces. This comparison does not equate Concept-SAE tokens with linear BSF groups.

Table S6. VOC2012 validation scores (%, mean ± sample SD). Readouts are fixed on VOC training data. All models use the same images and categories.

Method	Capture	Patch AP	VE
Vanilla BSF	19.44 ± 1.36	81.08 ± 1.91	71.21 ± 0.18
Supervised Vanilla	44.09 ± 0.05	93.78 ± 1.09	70.32 ± 0.20
GGF	45.81 ± 0.10	95.15 ± 0.05	69.95 ± 0.12
Concept-SAE	3.12 ± 0.07	67.84 ± 0.73	55.15 ± 0.16

Table S6 compares reconstruction-only training with the added object losses and classifier heads, then with GGF's initialization and adaptive weighting. The supervised training package raises Vanilla Capture from 19.44% to 44.09%. GGF reaches 45.81%. Against supervised Vanilla, GGF gains 1.72 Capture and 1.36 AP points at a 0.37-point VE cost. Paired training-run 95% intervals are $[1.36, 2.08]$, $[-1.41, 4.14]$ and $[-0.78, 0.04]$ points. Both object metrics improve in every paired run, but only Capture's interval excludes zero. Conditional image-bootstrap AP intervals favor GGF for the first two fits and include zero for the last. These do not replace training-run uncertainty. Category means favor GGF on Capture for 18 of 19 categories and AP for 11 of 19. Couch has lower Capture. One GGF endpoint exceeds the common development-MSE allowance and remains reported without relaxing or filtering by that rule.

Shared implementation and cross-dataset AP. Both datasets use the same Supervised Vanilla training code, BSF architecture, contrast and detection objectives, logistic classifier heads, Adam learning rate 0.0003 and 7,640/3,000-update warm-up/continuation budget. Supervised Vanilla's contrast/detection loss weights are selected separately on each dataset's development split. Both independent searches select $(0.20, 0.20)$. VOC fits fresh weights, normalization and readouts from its own training images. Its 19 categories require 38 head parameters, compared with 150 for COCO's 75 categories. Equal update counts therefore do not imply equal passes through the differently sized training sets.

Supervised Vanilla's 93.78% VOC Patch AP and 65.47% COCO Patch AP average over different categories, images and positive/negative patch populations. The mask formats and annotation coverage also differ. Even unsupervised Vanilla rises from 58.72% on COCO to 81.08% on VOC, consistent with a broader dataset/evaluation difference rather than a change to the supervised method. Category composition, object scale, context and patch prevalence can affect AP, but these comparisons do not isolate their contributions. Having fewer categories does not itself increase per-category binary AP.

Group-subset curves, removal controls and response spectra for these same models are analyzed in Appendix G.3.

C.5. Fresh 4,096-group comparison

This pilot uses newly initialized 4,096-group dictionaries with twelve directions per group, Top-32 and adaptive cap 0.60. The first 75 GGF groups use the twelve leading uncentered training-foreground directions. All remaining groups start randomly. Vanilla is reconstruction-only and has no semantic initialization. All six endpoints receive 7,640 reconstruction warm-up and 3,000 additional updates. Constant-rate Adam at 0.0003 differs from the OneCycle warm-up schedule. Paired branches share training streams. The original feature cache, masks, normalization and 75 categories are reused. The endpoints are fresh 12D fits, not expansions of trained 8D dictionaries.

Table S7. Completed pilot: development selection-set scores (%). Seed standard deviations are descriptive, not confidence intervals.

Seed	Model	Capture	AP	VE
21	GGF	42.48	80.46	77.29
21	Vanilla	20.82	51.03	80.66
22	GGF	42.28	80.54	76.69
22	Vanilla	20.48	52.71	79.82
23	GGF	42.24	80.33	76.49
23	Vanilla	21.90	55.24	79.77

Mean GGF scores are 42.33% Capture, 80.44% AP and 76.82% VE. Vanilla scores 21.07%, 52.99% and 80.08%. GGF seed standard deviations are 0.13, 0.11 and 0.42 points. Vanilla's are 0.74, 2.12 and 0.50 points. Every seed favors guidance on object metrics and Vanilla on reconstruction. The following evaluation scores these frozen endpoints on the existing test cohort.

At the same 10,640-update budget, fresh 1,024/Top-16/cap-0.40 GGF is available for seeds 21 and 22 only, averaging 42.19% Capture, 82.41% AP and 70.90% VE. On these common seeds the larger pilot averages 42.38%, 80.50% and 76.99%. This changes group count, Top-k and cap together, so it does not isolate dictionary width. The missing third reference seed prevents applying the original seed configuration-selection rule. The larger dictionary has four times the core parameters: 75,546,624 versus 18,886,656 at 1,024 groups. Its 384 active coordinates also double the Top-16 model's 192. Published 4,096-group experiments motivate this candidate.

Test evaluation of the six saved models. The evaluation-only run `445e2b6040fc` verifies all six model hashes and TRAIN-selected readouts against the completed pilot before loading test activations. It reuses the declared 1,500-image cohort, normalization, masks, 75 categories and deterministic 32-patch samples. This evaluation performs no gradient updates, readout fitting or backbone extraction. Its scores do not use the separate supervised-comparison holdout evaluated in Appendix C.2.

GGF's seed SDs are 0.10 Capture, 0.14 AP and 0.27 VE

	Capture		Patch AP	
	V	G	V	G
person	15.6±3.4	35.1±0.1	91.4±2.1	96.2±0.2
bicycle	24.8±2.4	49.8±0.2	67.4±5.7	63.4±1.5
car	19.5±2.1	38.9±0.1	74.8±1.4	79.7±0.3
motorcycle	25.7±1.2	48.1±0.2	79.8±5.4	86.9±0.3
airplane	13.1±0.2	45.0±0.7	65.4±22.5	88.2±0.4
bus	21.9±4.8	40.9±0.4	31.1±26.4	83.4±0.3
train	18.7±1.6	44.0±0.6	56.8±13.0	84.3±0.3
truck	12.5±0.7	32.0±0.3	17.2±12.3	43.3±0.6
boat	9.7±2.2	32.5±0.4	60.6±3.2	66.6±0.9
traffic light	26.5±3.6	52.5±0.3	56.3±9.9	60.6±0.5
fire hydrant	20.8±2.7	47.0±0.4	70.5±45.9	98.6±0.1
stop sign	43.4±8.8	70.2±0.5	38.2±46.3	91.7±0.6
parking meter	25.0±3.5	49.7±0.4	78.0±4.8	89.5±0.5
bench	9.3±1.1	34.3±0.8	60.8±5.6	54.4±0.8
bird	13.0±0.8	33.6±0.7	35.7±33.8	79.2±0.2
cat	29.6±0.8	51.2±0.3	91.9±1.6	98.1±0.0
dog	17.7±3.6	39.0±0.4	27.5±9.8	96.4±0.1
horse	30.6±3.2	46.5±0.1	52.8±13.2	96.3±0.1
sheep	24.5±3.7	43.6±0.5	83.1±1.1	93.6±0.3
cow	29.7±3.2	47.7±0.4	64.4±22.2	93.4±0.6
elephant	27.0±10.9	49.8±1.4	94.0±1.2	98.1±0.0
bear	25.8±3.2	39.2±0.3	47.4±24.8	94.5±0.0
zebra	35.9±3.2	62.1±0.4	53.3±48.4	99.0±0.0
giraffe	31.0±3.5	60.1±0.5	42.9±38.2	94.6±0.4
backpack	18.1±2.3	35.4±0.2	35.1±30.3	54.1±1.2
umbrella	22.9±0.8	46.2±0.2	77.0±4.9	82.5±0.4
handbag	12.4±1.9	28.4±0.1	9.2±7.8	31.8±1.1
tie	18.4±3.0	56.2±0.2	48.3±41.8	76.7±0.5
suitcase	18.3±5.9	36.9±0.2	56.7±9.6	66.1±1.1
frisbee	15.4±1.3	39.9±0.7	72.2±21.8	89.9±0.1
snowboard	22.0±2.7	42.4±0.6	20.4±4.0	40.1±0.1
kite	11.1±1.3	36.0±0.5	52.1±14.6	80.9±1.0
baseball glove	13.1±1.6	40.7±1.0	49.4±24.3	89.5±0.3
skateboard	17.9±2.1	41.6±0.3	1.1±1.7	51.1±3.7
surfboard	20.1±0.6	39.7±0.6	65.1±29.6	92.2±1.7
tennis racket	19.6±3.3	61.3±0.1	80.9±12.4	91.8±0.2
bottle	14.4±0.9	43.1±0.2	41.4±4.1	74.8±0.4
wine glass	22.3±1.5	43.4±0.6	35.7±8.3	73.5±0.9

	Capture		Patch AP	
	V	G	V	G
cup	22.0±3.3	41.4±0.5	39.4±6.9	79.0±0.7
fork	30.7±6.6	53.2±0.4	71.6±2.7	80.2±0.3
knife	24.3±4.4	38.1±0.4	5.1±2.9	20.9±0.7
spoon	23.8±5.3	37.1±0.4	22.4±19.4	32.7±3.0
bowl	19.3±1.2	25.4±0.6	15.0±6.9	55.2±0.7
banana	28.1±1.8	48.4±0.4	59.8±21.9	95.0±0.1
apple	20.9±2.8	40.0±0.6	59.0±8.4	69.7±0.4
sandwich	21.9±4.4	35.7±0.7	49.7±23.5	83.0±0.2
orange	24.2±7.5	46.9±0.5	46.8±40.5	86.1±0.2
broccoli	30.8±3.3	56.5±0.4	58.5±49.9	89.5±0.2
carrot	23.5±4.0	36.5±0.3	31.0±5.2	57.1±1.1
hot dog	22.1±5.0	34.7±0.4	25.6±13.3	70.5±1.3
pizza	38.4±3.7	56.1±0.3	73.9±9.4	87.5±1.7
donut	22.4±1.5	40.9±0.3	86.1±1.1	75.9±0.7
cake	13.3±0.5	30.8±1.1	48.9±4.8	80.7±0.4
chair	13.5±2.1	36.3±0.3	47.6±8.7	69.6±0.2
couch	19.9±1.1	36.3±0.1	56.5±2.6	69.8±0.5
potted plant	8.4±1.8	30.9±0.2	28.9±17.7	52.7±1.9
bed	15.2±1.2	31.4±0.3	77.8±4.3	81.8±0.5
dining table	15.5±1.8	34.1±0.4	35.0±3.3	49.5±0.0
toilet	23.5±1.2	57.3±0.3	84.1±9.7	94.0±0.1
tv	20.3±4.2	38.8±0.4	33.0±4.5	59.7±0.3
laptop	25.7±6.9	43.1±0.6	54.2±3.6	81.8±0.1
mouse	22.0±1.3	50.0±1.0	53.2±37.4	90.2±0.6
remote	24.3±4.0	40.4±0.2	32.0±39.4	83.2±1.3
keyboard	30.4±1.7	57.4±0.8	43.5±31.1	74.5±4.2
cell phone	21.3±0.6	33.6±0.3	3.8±4.6	15.1±0.2
microwave	20.2±4.3	40.9±0.8	28.3±19.8	33.4±2.9
oven	21.1±2.9	38.3±0.7	34.4±8.7	45.3±1.0
sink	14.4±2.2	46.0±0.1	55.1±13.4	78.5±0.5
refrigerator	25.0±5.8	41.8±0.6	30.6±15.5	74.9±0.1
book	17.3±4.8	30.5±0.5	39.4±11.1	49.0±0.7
clock	22.8±3.1	48.7±0.7	78.9±6.8	83.1±0.5
vase	17.3±1.5	35.8±0.3	27.0±15.9	59.1±2.4
scissors	21.6±2.1	43.2±0.9	86.2±3.1	89.6±0.5
teddy bear	25.8±2.5	43.8±0.2	84.0±2.5	90.0±0.1
toothbrush	14.0±2.8	43.1±0.4	82.2±1.7	93.8±0.8

Figure S3. From-scratch 4,096-group, 12D, Top-32 comparison on the inspected 1,500-image test cohort. Entries show per-category mean percentages and sample SD, comparing matched Vanilla (V) and GGF (G). Both receive 10,640 updates. GGF uses cap 0.60. Colors encode means on a shared 0–100 scale. Each entry includes sample SD. All 75 categories are shown, including the three whose mean AP decreases. Capture and AP use independently selected training readouts.

points. Vanilla's are 0.84, 1.55 and 0.39. Each seed improves Capture and AP while reducing VE. Category Capture improves in all 75 categories in every run. Category AP improves in 72 of 75 categories after averaging seeds, and in 64 categories in every run. These category counts describe point estimates, not independent statistical replications.

The mean paired gains are 21.40 Capture, 23.10 AP and -3.41 VE points. Separate Student-t 95% intervals over the training starts are [19.13,23.67], [19.52,26.67] and $[-4.10,-2.73]$, respectively. These imprecise seed intervals and the conditional image intervals answer different uncertainty questions. Neither adjusts for multiple comparisons or repeated inspection of this benchmark. All 75 categories have finite scores in all six models. The original native-warm development MSE allowance fails for all GGF endpoints. Reports retain this failed allowance alongside all test scores.

Compared with the older expanded 1,024-group 12D GGF, the fresh pilot has 1.50 points lower Capture, 0.52

Table S8. Values with ± show mean and sample SD. Fresh 4,096-group pilot on the existing test cohort (%). All models use 12D, Top-32 and 10,640 training updates. GGF cap is 0.60.

Seed	Model	Capture	AP	VE
21	GGF	42.88	74.60	77.24
21	Vanilla	21.05	49.97	80.70
22	GGF	42.76	74.87	76.82
22	Vanilla	20.73	52.00	79.94
23	GGF	42.67	74.80	76.75
23	Vanilla	22.31	53.01	80.41
Mean	GGF	42.77 ± 0.10	74.76 ± 0.14	76.94 ± 0.27
Mean	Vanilla	21.37 ± 0.84	51.66 ± 1.55	80.35 ± 0.39

Table S9. GGF-minus-Vanilla gains and paired 95% whole-image intervals on test, in percentage points. Each seed has 1,000 valid draws and zero invalid draws. VE entries are point changes.

Seed	Capture [interval]	AP [interval]	VE
21	21.82 [21.29,22.34]	24.64 [23.09,26.23]	−3.46
22	22.03 [21.47,22.51]	22.86 [21.40,24.52]	−3.12
23	20.35 [19.83,20.84]	21.79 [20.31,23.34]	−3.66

higher AP and 7.43 higher VE. Group count, Top-k, cap, initialization history and training budget all differ, so this is context rather than an isolated width comparison. The test scores do not select a configuration. Full per-category scores, paired intervals, model identities and source hashes are in `evidence/pilot4096_test/`.

C.6. Matched native-baseline comparison

The matched comparison reuses saved GGF models and trains the remaining native controls. Every arm has 1,024 groups, three directions per group, and the same 2,000 training and 400 selection images. All models are scored on the same 1,500 test images, normalized with the original training statistics, across the same 75 categories. Because this cohort was already inspected, the comparison provides a shared evaluation rather than a new untouched confirmation. No test labels enter the new training or readout selection.

Training budgets and reuse. Each warm-up performs 40 epochs, or 7,640 updates, with batches of 2,048 patches. The development-MSE minimum starts continuation. New Grassmannian and Group Lasso warm-ups use the unchanged released classes. Saved Vanilla and mask warm-ups are reused. Every arm has 2,500 plus 500 continuation updates, with Adam restarted between phases. Saved Vanilla supplies the first 2,500 updates, and we train its remaining 500-update phase alongside new continuation runs for Grassmannian, Group Lasso, and mask reconstruction. Mask GGF reuses the 3,000-update ordinary recovery endpoint. New work totals 74,340 gradient updates, with no new guided training.

Continuation uses eight sampled images per update, all 196 patches, learning rate 0.0003, scalar noise standard deviation $0.1\,\mathrm{std}(B)$, gradient clipping at 5, and per-direction decoder normalization. The sampler seeds are $71{,}000 + s$ and $82{,}000 + s$ for training seed s. Batch hashes confirm identical image schedules across arms within each phase. Although the noise rule is shared, initialization consumes different random draws across classes, so their noise arrays need not be identical. Intermediate minimum-MSE checkpoints are retained as diagnostics. Primary comparisons use the final 3,000-update endpoint in every arm, rather than each arm's best test score.

Readouts and activity. Each category's contrast and detector groups are chosen from training data after warm-up and then fixed. Mask reconstruction and GGF share the mask warm-up readouts. Native controls use their own warm-up readouts under the same selection policy. Patch detection AP is scored from raw group norms rather than the learned logistic classifier heads. Released Group Lasso retains its learned hard threshold and mean-activity target of 16. Its measured mean is 17.99 on the test. This is within the declared 20% activity tolerance, but not identical Top-16 sparsity. Table S10 records the differences that remain.

Table S10. Core parameter counts (millions), auxiliary head parameters, and active groups. Group Lasso activity is mean ± sample SD. Top-k activity is fixed. All arms have 3,072 decoder directions. Update matching does not match computation.

Model	Core (M)	Heads	Active
Vanilla	4.722	0	16.00
Grassmannian	2.360	0	16.00
Group Lasso	4.723	0	17.99 ± 0.12
Mask + rec.	4.722	0	16.00
Mask GGF	4.722	150	16.00

Results and uncertainty. Table S11 reports all retained arms in each run. Capture is recomputed from saved per-image contrast errors and target magnitudes. All models share identical image IDs and contrast denominators. GGF Capture, AP, and VE reproduce their previous reports exactly. Artifact hashes and checkpoint-recovery records pass validation.

Paired whole-image bootstraps keep each image's patches

Table S11. Common-test results for every run and arm. All entries are percentages. Reconstruction and object-alignment metrics measure different properties.

Seed	Model	Capture	AP	VE
21	Vanilla	21.178	58.722	65.196
21	Grassmannian	16.437	60.261	64.973
21	Group Lasso	20.764	56.757	64.564
21	Mask + rec.	26.483	68.025	64.934
21	Mask GGF	32.729	71.552	64.789
22	Vanilla	20.393	60.640	65.293
22	Grassmannian	16.892	60.942	64.962
22	Group Lasso	21.510	55.919	64.489
22	Mask + rec.	26.522	69.697	65.003
22	Mask GGF	33.095	72.473	64.852
23	Vanilla	21.241	58.811	65.160
23	Grassmannian	16.223	61.787	65.018
23	Group Lasso	21.388	53.265	64.294
23	Mask + rec.	26.819	69.068	64.907
23	Mask GGF	33.100	71.504	64.766

and objects together. All 200 draws in each comparison retain the full category set. Their interpretation follows Appendix B.2. GGF has positive Capture and AP intervals against every native control in every run.

Table S12. Initialization and guided-loss increments, in percentage points. Init. compares mask reconstruction with Vanilla. Guided compares GGF with mask reconstruction. AP intervals are positive in every row and are included in the accompanying CSV.

Seed	Stage	Capture	95% interval	AP
21	Init.	5.31	[4.90, 5.74]	9.30
21	Guided	6.25	[6.08, 6.40]	3.53
22	Init.	6.13	[5.75, 6.52]	9.06
22	Guided	6.57	[6.40, 6.74]	2.78
23	Init.	5.58	[5.20, 5.91]	10.26
23	Guided	6.28	[6.12, 6.43]	2.44

Mean gains attribute 5.671 Capture points and 9.539 AP points to mask initialization plus reconstruction, followed by 6.367 Capture points and 2.913 AP points from guided losses. These controls separate initialization from object-loss supervision, but do not establish equal total compute or the optimal allocation of training between the stages.

Reconstruction eligibility. The native reference and its 2% development-MSE allowance remain unchanged. All Vanilla endpoints pass. Grassmannian passes in seeds 21 and 23. Group Lasso, mask reconstruction and GGF fail in every run. This is determined on development data, separately from test VE. Similar reconstruction scores therefore do not imply compliance with the declared error limit.

Full per-category outcomes, paired comparisons, protocol, and verification records accompany the paper under `evidence/matched_baselines/`. Large activation and optimizer checkpoints stay in the recorded Drive locations. The compact results archive retains source code and identities.

C.7. The 256-group baseline screen

In the single-seed development screen, Vanilla had the highest native Capture (21.31%, versus 16.84% for Grassmannian and 20.47% for Group Lasso). It also led on Capture and Patch AP after guided training. Synthetic recovery and reconstruction supported this choice (Table S24). The 1,024-group comparisons show metric tradeoffs: Grassmannian has higher Patch AP and Group Lasso slightly higher Capture than Vanilla (Table S11). The supplementary code bundle records the full screening settings and historical three-direction SLiCS pilots. Finite-solver nonconvergence and differing checkpoint rules limit the pilots, which are excluded from the twelve-direction comparison.

D. Supervision and loss-weight studies

These studies separate object supervision, mask initialization and task weighting. The fixed-weight loss ablations select development checkpoints, whereas the matched-component studies use scheduled endpoints.

D.1. Twelve-direction component ablation

The completed component study (run `a2f215ed99c2`) uses 4,096 groups, twelve directions, Top-32 and the declared training starts. Each endpoint receives 7,640 reconstruction warm-up and 3,000 continuation updates. We reuse the three mask warm-ups and six completed Vanilla/GGF endpoints, adding only three mask-initialized reconstruction-only fits and three fixed-guidance fits. No new backbone extraction or unused holdout is required. All twelve endpoints freeze before the new test audit on the existing 1,500 images. Within each run the three mask-initialized arms share their starting weights and fixed training-selected readouts. Capacity, activity, update counts and image/noise streams match. FLOPs differ.

Training-matched fixed weights. Reconstruction keeps weight one. The fixed contrast and detection weights are the arithmetic means of the saved adaptive weights at updates 500, 1,000, 1,500, 2,000, 2,500 and 3,000. They are respectively $(0.36639, 0.24090)$, $(0.35309, 0.23918)$ and $(0.35819, 0.23688)$ for seeds 21, 22 and 23. Only training logs determine these constants, with no development/test tuning. This matches the observed scale of guidance, not the exact average over every update or an optimized fixed-weight baseline. Adaptive GGF retains cap 0.60 and its saved controller schedule.

Sequential gains and uncertainty. Table 2 summarizes the four arms. Table S14 reports paired seed intervals, computed with Student's t and two degrees of freedom. Each of the nine per-seed component comparisons also has 1,000 paired whole-image bootstrap draws. All are valid. Initializa-

Table S13. Values with ± show mean and sample SD. Component test results (%). V: Vanilla, I: mask initialization with reconstruction only, F: added fixed guidance, G: full adaptive GGF. Every arm has the same capacity and update budget.

Seed	Arm	Capture	AP	VE
21	V	21.05	49.97	80.70
21	I	23.61	68.73	79.80
21	F	42.38	74.41	77.54
21	G	42.88	74.60	77.24
22	V	20.73	52.00	79.94
22	I	23.53	68.38	79.33
22	F	42.14	74.76	77.13
22	G	42.76	74.87	76.82
23	V	22.31	53.01	80.41
23	I	23.53	68.53	79.54
23	F	42.23	74.71	77.32
23	G	42.67	74.80	76.75
Mean	V	21.37 ± 0.84	51.66 ± 1.55	80.35 ± 0.39
Mean	I	23.56 ± 0.05	68.55 ± 0.17	79.56 ± 0.23
Mean	F	42.25 ± 0.12	74.63 ± 0.19	77.33 ± 0.20
Mean	G	42.77 ± 0.10	74.76 ± 0.14	76.94 ± 0.27

tion and fixed guidance have positive Capture and AP image intervals in every run. Adaptive Capture intervals are positive in every run, while AP intervals include zero in seeds 22 and 23. Seed intervals assess variation across fitted starts. Image intervals condition on one fitted pair and the inspected cohort. Neither includes researcher-selection uncertainty.

Initialization improves mean category AP in 67 of 75 categories and Capture in 46. Adding fixed guidance improves mean Capture in all 75 and AP in 65. Adaptive weighting improves mean Capture in 71 and AP in 49. These counts use category means across seeds and are not significance tests. Fixed guidance obtains 97.6% of GGF's Capture gain and 99.4% of its AP gain over Vanilla along this sequence, while retaining 0.40 more VE points. The controller adds a smaller Capture benefit. A reliable AP advantage over the tested fixed constants is not established.

Vanilla and initialization-only endpoints pass the original development MSE allowance in all seeds. Both guided arms fail it in every run. The study measures the alignment/reconstruction tradeoff, without showing that reduced reconstruction is necessary or causes the alignment gains. Contrast and detection are jointly enabled here. Their individual effects are examined separately in the three-direction loss ablation below. Complete frozen lineage, training-derived weights, category scores, paired intervals and verification records are retained in `evidence/components_current/`.

The supplementary code bundle contains the three-direction warm-up, continuation, random-start diagnostics and full loss-weight sweeps.

D.2. Group width and adaptive loss weights

This five-arm cap-0.20 study tests group width and loss weighting. Run `e4aeabc90db7` is complete for the declared training starts. It reuses frozen DINOv3 features, the original 75 categories, 2,000 training images, 400 selection images and the inspected 1,500-image test. All 15 endpoints were frozen before test scoring. The comparison measures improvement on this common cohort, not independent confirmation.

Initialization, training and controls. All arms use 1,024 groups and Top-16 gating. Width three supplies 3,072 directions and at most 48 active coefficients. Width four supplies 4,096 directions and at most 64. The corresponding encoder/decoder parameter counts are 4,721,664 and 6,295,552. Guided models add 150 classifier parameters. Vanilla has none. The four-direction control therefore matches dictionary capacity, activity budget and updates, rather than supervision or total computation.

For each category, we accumulate the training foreground outer-product matrix defined in Appendix A.2. We preserve its saved first three float32 directions byte-for-byte, verify agreement with the recomputed leading eigenspace, and obtain a fourth eigen-direction. We project that direction away from the saved three, unit-normalize it, and orient its sign by training projection energy. No development or test masks enter this calculation. The remaining 949 groups are random. The category initialization is shared across seeds. Reserve directions and training order vary.

The original three-direction fixed-weight endpoint is reused, and the three-direction adaptive arm reuses its reconstruction warm-up. The two four-direction masked arms share one new warm-up per seed. Vanilla uses a random four-direction warm-up. Each warm-up runs 40 epochs, or 7,640 updates, with the original Adam/OneCycle recipe. Continuation starts from minimum selection-MSE warm-up weights and makes 2,500 plus 500 updates at learning rate 3×10^{-4}. Both phases evaluate every 250 updates. At the boundary, model Adam resets, but weights and trained heads carry

Table S14. Sequential component gains: mean ± sample SD in percentage points, with paired seed 95% t intervals. These describe this ablation sequence, not independent causal shares.

Change	Capture	Patch AP	Reconstruction VE
Add initialization	2.19 ± 0.85	16.89 ± 1.68	-0.79 ± 0.17
95% interval	$[0.08, 4.31]$	$[12.72, 21.06]$	$[-1.20, -0.38]$
Add fixed guidance	18.69 ± 0.08	6.08 ± 0.36	-2.23 ± 0.03
95% interval	$[18.49, 18.89]$	$[5.18, 6.98]$	$[-2.30, -2.15]$
Use adaptive weights	0.52 ± 0.09	0.13 ± 0.06	-0.40 ± 0.16
95% interval	$[0.29, 0.75]$	$[-0.02, 0.28]$	$[-0.78, -0.01]$

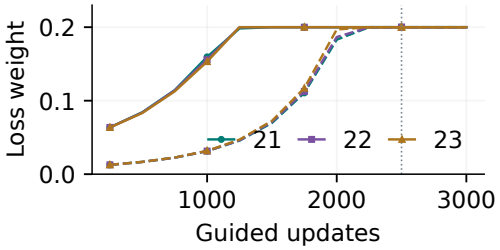


Figure S4. Saved four-direction adaptive-weight trajectories. Lines connect recorded values, not every controller step. Color identifies seed. Solid and dashed lines identify contrast and detection. The dotted vertical line marks the model-Adam reset at 2,500 updates. Controller state is retained.

forward. Adaptive-controller state also carries forward. Final 3,000-update endpoints are used regardless of eligibility, avoiding selection of only successful checkpoints.

Continuation samples eight images with replacement and uses the original noise scale and gradient clipping. Within each phase, sampled-image schedules match across arms. Width-four noise streams also match. Different tensor widths do not have identical noise realizations. Category readouts are selected on training features after warm-up and stay fixed. The two four-direction masked arms share readouts. Other warm-ups can yield different assignments. The new work consists of six warm-ups and twelve continuations, or 81,840 model-gradient updates. No backbone extraction or saved-control retraining is needed.

Observed weight trajectories. The cap is 0.20 in this study. Both object weights reach it in the four-direction runs (Figure S4), motivating the larger-bound screen below.

Complete outcomes and uncertainty. Tables S15 and S16 report all arms. Native eligibility uses each seed's original three-direction native selection-MSE reference and its unchanged 2% allowance, not test VE. All four-direction endpoints in this cap-0.20 study pass. Both three-direction guided arms fail.

Seven declared paired comparisons per seed give 21 comparisons, with 200 whole-image bootstrap draws and no multiplicity correction. Intervals follow Appendix B.2. VE differences are point estimates. Four-direction adaptive GGF improves Capture, AP and VE over the original fixed-weight GGF in every run. Its Capture/AP intervals are positive against original GGF, four-direction fixed-weight GGF and four-direction Vanilla in all seeds. Its VE remains below both four-direction controls. Against three-direction adaptive GGF, AP decreases in seed 22, so width does not uniformly improve detection. The research repository retains all 21 intervals in `evidence/width_balance/paired_comparison.csv`, all endpoint scores, and the category-level summary. The full experimental archive records checkpoint hashes, batch identities, initialization verification and controller histories. No new downstream-classifier evaluation is included.

Against matched four-direction Vanilla, adaptive GGF gains 16.48 Capture and 17.49 Patch AP percentage points on average, with a 0.97-point VE cost.

Cap selection. On the 400-image development set, we compared caps 0.20, 0.40 and 0.60 with matched training updates for seeds 21–23. The 0.40 cap improved Capture and Patch AP at a reconstruction cost. Raising it to 0.60 yielded little further gain, so 0.40 was selected before test scoring. The supplementary code bundle contains the full development grid.

Test evaluation of the selected 0.40 cap. Run `eafd62bd8566` evaluates cap 0.40 for seeds 21, 22 and 23. All saved final endpoints freeze before scoring the same cached 1,500 test images. We reuse the 0.20 GGF, four-direction Vanilla and original GGF control reports and paired audit arrays, with no retraining, test calibration or readout changes. The cap was chosen from development. 0.60, 0.80 and 1.00 are excluded.

Cap 0.40 improves Capture and AP over 0.20 in every run, with mean gains of 1.03 and 0.35 percentage points. Mean VE falls by 0.38 points. Against matched four-direction Vanilla, Capture and AP improve by 17.52 and 17.84 points, while VE is lower by 1.36 points. Against original GGF, all three metrics improve, by 5.86 Capture, 3.35 AP and 0.58 VE points. All nine paired Capture and AP intervals lie above zero. This confirms an object-alignment gain on the shared test cohort, with a reconstruction tradeoff. It does not show improvement in every category or independent generalization. All original GGF controls fail the original native

Table S15. Values with ± show mean and sample SD. Five-arm test means (%). Pass counts concern development MSE.

Arm	Capture	AP	VE	Passes
GGF 3 fixed	32.97 ± 0.21	71.84 ± 0.55	64.80 ± 0.04	0/3
GGF 4 fixed	33.89 ± 0.10	72.38 ± 0.37	66.36 ± 0.03	3/3
GGF 3 adaptive	36.90 ± 0.11	74.37 ± 0.80	64.27 ± 0.05	0/3
GGF 4 adaptive	37.80 ± 0.07	74.84 ± 0.27	65.76 ± 0.02	3/3
Vanilla BSF 4	21.31 ± 0.49	57.35 ± 0.83	66.74 ± 0.05	3/3

Table S16. All five endpoints in all seeds: test percentages.

Seed	Arm	Capture	AP	VE
21	GGF 3 fixed	32.729	71.552	64.789
21	GGF 4 fixed	33.786	72.263	66.317
21	GGF 3 adaptive	36.804	74.278	64.251
21	GGF 4 adaptive	37.727	74.879	65.740
21	Vanilla BSF 4	21.837	57.932	66.761
22	GGF 3 fixed	33.095	72.473	64.852
22	GGF 4 fixed	33.989	72.086	66.370
22	GGF 3 adaptive	36.874	75.213	64.325
22	GGF 4 adaptive	37.868	74.543	65.773
22	Vanilla BSF 4	20.874	57.718	66.770
23	GGF 3 fixed	33.100	71.504	64.766
23	GGF 4 fixed	33.890	72.796	66.379
23	GGF 3 adaptive	37.025	73.628	64.238
23	GGF 4 adaptive	37.795	75.086	65.781
23	Vanilla BSF 4	21.233	56.396	66.679

development-MSE limit. Every four-direction endpoint in this cap-0.40 audit passes.

Each comparison uses 200 paired whole-image draws with shared image IDs and the 75-category denominator (Appendix B.2). The seed means, full category reports, endpoint identities and all nine intervals are in `evidence/cap040_test/`.

E. Capacity and continuation studies

These studies vary group width or active-group count under matched gradient-update budgets. Parameter counts and active-coordinate budgets are also reported because equal updates do not imply equal computation.

The supplementary code bundle records the intermediate four- and five-direction training and continuation studies.

E.1. Matched capacity and continuation studies

What changes and what remains fixed. The starting five-direction GGF and Vanilla checkpoints have 13,640 scheduled updates. The next comparison gives every arm another 3,000: unchanged 5D/Top-16, expanded 6D/Top-16, expanded 8D/Top-16 and 5D/Top-24. All use 1,024 groups and the declared training starts. GGF retains its cap-0.40 controller. Vanilla trains reconstruction only. The final checkpoint supplies scores, not the checkpoint with the highest test result.

Expansion preserves every learned encoder coefficient slot, bias entry and decoder direction in its existing group. New seeded decoder directions are projected away from the learned span and prior additions, then unit normalized. Their encoder coefficients and biases start at zero, preserving the starting gated code and reconstruction up to float32 rounding. Old Adam moments, step counters, trained heads, controller state and reference losses carry forward. New-coordinate moments start at zero and share the existing Adam step counter. All directions may then fine-tune.

The 12D follow-up branches from completed 8D checkpoints at 16,640 updates. One branch adds four coordinates and trains 3,000 updates. The other retains 8D and trains the same additional budget. Matched Vanilla branches follow the same plan, giving twelve new endpoints at 19,640 updates. Training uses eight images sampled with replacement, Adam at 0.0003, noise scale 0.1, gradient clipping at five and decoder unit normalization. Image and noise stream hashes are checked within paired branches. Category Capture and detector readouts stay fixed. Models freeze before scoring the same inspected 1,500-image test cohort. This is a comparison on that cohort, not an unused confirmation set.

At 16,640 updates, 8D exceeds unchanged 5D by 1.29 Capture, 0.15 AP and 2.14 VE points. Top-24 at width five instead lowers Capture by 0.33 points and AP by 0.10 points while raising VE by 3.59 points. This separates the tested effects of within-group capacity and active-group budget. At 19,640 updates, 12D exceeds equally continued 8D by 0.82 Capture, 0.10 AP and 1.60 VE points. Every seed improves in all three point metrics. The 200-draw paired whole-image bootstrap gives positive Capture intervals in every seed. The AP interval is positive only in seed 22. These intervals condition on fitted models and omit seed-level and researcher-selection uncertainty.

Twelve-direction models contain 18,886,656 core parameters versus 12,591,104 at width eight, and expose at most 192 versus 128 active coefficients. GGF adds 150 training-only classifier parameters. Equal updates therefore control training duration but not FLOPs or wall time.

E.2. Width and activity sensitivity

The direction-width and activity screens use seed 21, 2,000 training images, the reused 400 development images, 75 categories and original loss weights. They do not open the final test. Their readouts are training-selected. The separate group-count check below uses mapping-selected readouts.

Total group count. The 1,024-group choice lies halfway on a logarithmic scale between the released 256-group example and the paper's minimum DINOv3 sweep width, 4,096

Table S17. Values with ± show mean and sample SD. Means across runs (%). All four-direction arms pass the original native-reference development-MSE limit.

Model	Capture	Patch AP	VE
GGF 4: cap 0.40	38.83 ± 0.04	75.19 ± 0.25	65.38 ± 0.05
GGF 4: cap 0.20	37.80 ± 0.07	74.84 ± 0.27	65.76 ± 0.02
Vanilla BSF 4	21.31 ± 0.49	57.35 ± 0.83	66.74 ± 0.05
Original GGF 3	32.97 ± 0.21	71.84 ± 0.55	64.80 ± 0.04

Table S18. Cap-0.40 test scores by seed (%).

Seed	Capture	Patch AP	VE
21	38.823	75.217	65.331
22	38.876	74.927	65.388
23	38.796	75.423	65.423

[14, 18]. These anchors motivated the 1,024-group configuration.

A reconstruction-only Vanilla BSF check compares 1,024 and 4,096 groups at width three and Top-16. Both use seed 21, the same 2,000 training images, normalization, Adam/OneCycle recipe, and 40 epochs (7,640 updates). The saved 1,024 control was hash-verified after matching Torch, NumPy, and einops versions. Minimum-selection-MSE checkpoints occur at 7,640 and 7,500 for 1,024 and 4,096 groups. A shared seed does not ensure identical batch/noise streams across widths. Equal updates do not ensure equal computation.

A separate 400-image mapping split selects each model's contrast and detector groups independently. Frozen readouts score all 75 categories on the same 400 confirmation/development images. These images were already inspected. The final test remains unused. This follows our COCO schedule, not the paper's full DINOv3 sweep protocol.

Increasing group count changes Capture by -0.891 percentage points (paired 95% interval $[-1.460, -0.360]$), Patch AP by $+0.193$ points (interval $[-2.808, 2.984]$), and VE by $+0.221$ points. All 100 image resamples retain every category. Capture rises in 30 categories and falls in 45. Mean AP improvement is inconclusive. These conditional intervals do not establish a training-seed population effect. Lower Capture and fourfold parameter cost support 1,024 under this budget, without identifying the best GGF width or longer-training outcome. Records are in `evidence/dictionary_size/` (results JSON and comparison CSV).

Fixed group count. The width/activity screen fixes 1,024 groups: width three gives 3,072 directions for 768-dimensional activations, shared by matched controls. Here b denotes directions per group (main-text r), and k denotes groups selected per patch. At 1,024 groups, we test $b \in \{2, 3, 4\}$ and $k \in \{8, 16, 32\}$, reevaluate the saved $b = 3, k = 16$ control, and give changed models 40 native epochs (7,640 updates) plus 2,500 guided updates with shared image batches. Width changes capacity. Activity changes selected coordinates. Equal updates do not match computation. The group-count check is a separate experiment. Selection maximizes Capture $+0.2$ AP within 2% of each native MSE. No setting meets all cross-setting replacement safeguards. Table S22 compares the $b = 4, k = 16$ endpoint, which fails its native guard at seven audits, with the retained state. Models $b = 3, k = 32$ and $b = 4, k = 32$ retain update zero despite higher endpoint Capture. The CSV retains every selected and final state.

Fixed direction and activity budgets. This screen fixes 3,072 directions, 4,721,664 core parameters, and at most 48 selected coordinates. Models $b = 4, k = 12$, $b = 8, k = 6$, and $b = 16, k = 3$ each receive new native initialization, 40 epochs, and 2,500 guided updates. Wider groups reduce the group count, so these are new dictionaries rather than continuations or rotations of $b = 3, k = 16$. Runtime and the number of independent groups are not matched.

All three layouts pass their own native-MSE guards at every audit, but lose AP and VE against original GGF. Table S23 gives the $b = 16, k = 3$ outcomes. Native AP and VE are already low, so guided learning alone did not cause the deficits. Capture improves in 69 of 75 categories but AP in only 13. Late Capture gains persist, so this run does not establish a Capture plateau.

F. Response geometry and baseline checks

F.1. Coherence-linked group geometry

Figure 2 uses full-category, seed-21 models with 75 native groups each. The selected GGF glove group has three confirmed specific pairs, versus one for each selected SLiCS and supervised Vanilla group and zero for Vanilla. These examples were chosen to illustrate coherence and failure cases before computing their PCA coordinates; they are not representative group samples. All three original pair and whole-group reviews remain frozen. The exhaustive comparison in Table 1 includes every group.

For each displayed group, we decode its native twelve-dimensional codes into 768-dimensional contributions. BSFs retain their signed gated codes; SLiCS retains its independent NNLS projection rather than a joint additive reconstruction. The frozen checkpoints and normalization reproduce every strongest marked patch index, with maximum relative mean-top-four response error below 4.6×10^{-7}. No training or

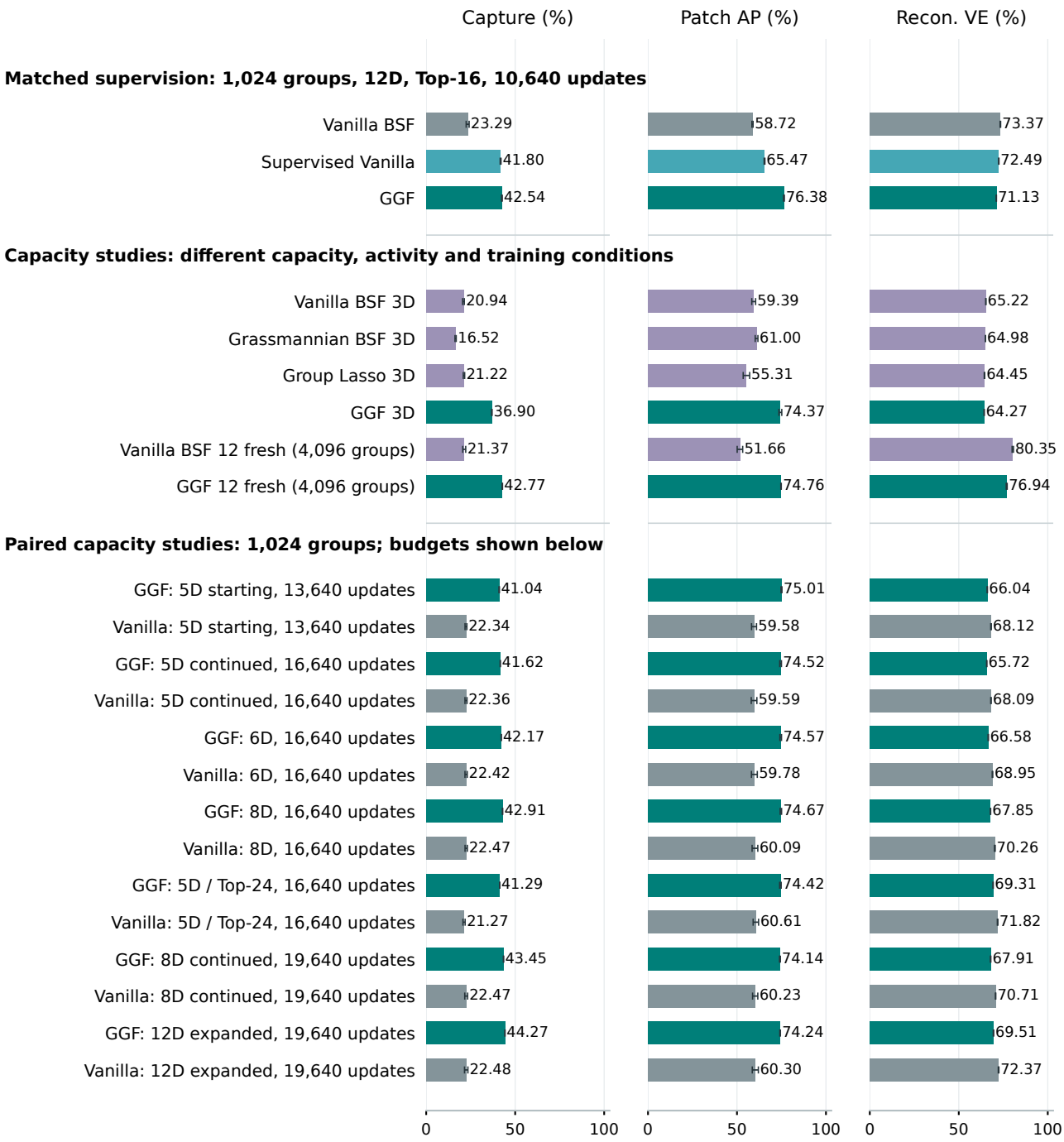


Figure S5. COCO means with sample SD. Top: matched supervision (Appendix C.1). Middle: comparisons at different capacities. Cross-row differences do not isolate width. Fresh 12D uses 4,096 groups, Top-32 and 10,640 updates. Bottom: every paired comparison in Table S19, with its update budget. Top-16 applies except the stated Top-24 control. All bottom rows use 1,024 groups and GGF cap 0.40. GGF caps are 0.20 (3D), 0.40 (matched), and 0.60 (fresh). Three-direction GGF and reconstruction-only rows come from different studies. Group Lasso averages 17.99 active groups. All bars start at zero. All GGF rows use adaptive weights.

annotation changes are involved.

Each group's centered PCA uses its active patches from the same first 64 audit images in the original fixed order, chosen before projection. For covariance eigenvalues λ_j in descending order, the variance retained by the first k principal components is $\sum_{j=1}^{k} \lambda_j / \sum_j \lambda_j$. Here k counts PCA

Table S19. Values with ± show mean and sample SD. Completed capacity studies: means (%). All use 1,024 groups and cap 0.40 for GGF. Top-16 applies except the stated Top-24 arm. Each Vanilla row matches its GGF capacity and update budget. Starting checkpoints are included without retraining.

Configuration	Model	Updates	Capture	Patch AP	VE
5D starting	GGF	13,640	41.04 ± 0.11	75.01 ± 0.26	66.04 ± 0.07
5D starting	Vanilla	13,640	22.34 ± 0.52	59.58 ± 1.40	68.12 ± 0.16
5D continued	GGF	16,640	41.62 ± 0.05	74.52 ± 0.47	65.72 ± 0.04
5D continued	Vanilla	16,640	22.36 ± 0.57	59.59 ± 1.58	68.09 ± 0.13
6D	GGF	16,640	42.17 ± 0.06	74.57 ± 0.45	66.58 ± 0.07
6D	Vanilla	16,640	22.42 ± 0.63	59.78 ± 1.64	68.95 ± 0.16
8D	GGF	16,640	42.91 ± 0.04	74.67 ± 0.39	67.85 ± 0.06
8D	Vanilla	16,640	22.47 ± 0.65	60.09 ± 1.61	70.26 ± 0.20
5D / Top-24	GGF	16,640	41.29 ± 0.03	74.42 ± 0.41	69.31 ± 0.07
5D / Top-24	Vanilla	16,640	21.27 ± 0.61	60.61 ± 1.53	71.82 ± 0.14
8D continued	GGF	19,640	43.45 ± 0.03	74.14 ± 0.23	67.91 ± 0.11
8D continued	Vanilla	19,640	22.47 ± 0.82	60.23 ± 1.47	70.71 ± 0.10
12D expanded	GGF	19,640	44.27 ± 0.03	74.24 ± 0.16	69.51 ± 0.12
12D expanded	Vanilla	19,640	22.48 ± 0.95	60.30 ± 1.71	72.37 ± 0.17

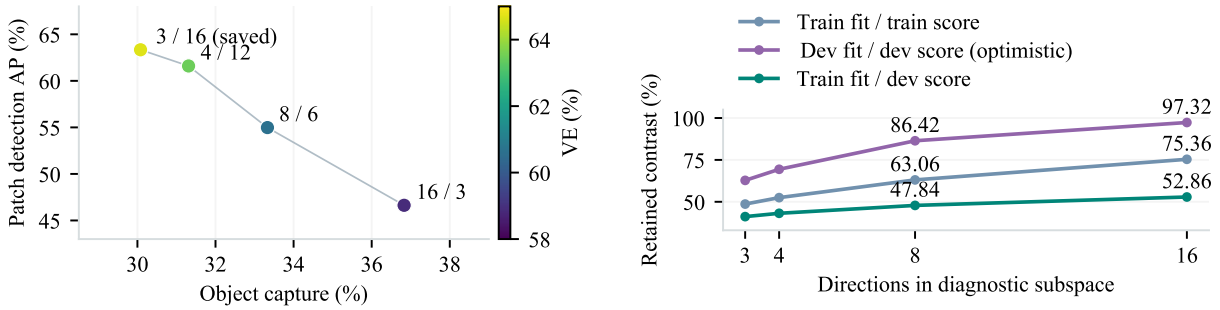


Figure S6. (a) Matched-direction screen, selected development states. Same 3,072 total directions and 48 active coordinates, with one start and reused development images. (b) Category-macro retained contrast energy (percent). The development-fitted column is optimistic and must not be read as population recovery.

Table S20. Follow-up Vanilla group-count check, seed 21, the confirmation/development cohort. Capture, Patch AP and VE are percentages. Parameters are millions.

Metric	$G = 1{,}024$	$G = 4{,}096$
Object contrast Capture	21.32	20.43
Patch detection AP	61.99	62.18
Reconstruction VE	65.21	65.43
Core parameters (M)	4.72	18.89

components, separately from the encoder's group-activity setting. The first three PCs retain 50.7%, 50.2%, 40.6% and 35.3% of reference variance for GGF, SLiCS, supervised Vanilla and Vanilla, respectively. Clouds contain those reference patches; outlined points retain all four original marks from each of the six reviewed images. Arrows locate the strongest mark. Separate bases, signs and scales preclude interpreting cross-panel coordinates as a shared space. No manifold surface is fitted; these projections do not establish intrinsic dimension or topology.

PC1–3 set the RGB channels in both cloud and image overlays. Per-axis limits start at reference percentiles 2 and 98, expand to include all marked points, and map to $[0.04, 0.96]$. Border, arrow and point-outline colors instead annotate visible marked content from the frozen reviews; they are not learned cluster assignments. Let q_{99} be the 99th percentile of that group's active reference norms. Opacity is $0.61 \min(\|z_g\|_2/q_{99}, 1)^{0.65}$. Activity-weighted colors use Gaussian smoothing ($\sigma = 0.35$ patch) and clipped cubic interpolation. Insets are original 96×96-pixel neighborhoods around the strongest 16×16 patch, clamped to the original 224×224 crop. Floor projections are display shadows. Saved arrays, image hashes, patch IDs and transforms reproduce the figure without inferring new geometry or labels.

F.2. Object-response visualizations

Figure S7(a) uses the selected plant/vase photograph COCO 553000 with the fresh seed-23 GGF checkpoint: 4,096 groups, 12 directions per group, Top-32 and cap 0.60. The saved training detector readouts assign potted plant to group 55 and vase to group 71. No additional training is used for this visualization.

We collect gated coefficients on the first 256 images of the fixed training cohort, retaining only patches where the displayed group is active. This gives 13,911 plant-group responses and 5,455 vase-group responses. A fixed uniform sample of 3,000 responses per group is plotted, without

Table S21. The fixed-1,024-group screen reports selected development checkpoints. Width is the number of directions per group, and active groups is the Top-k setting. The active coordinates column gives the number of retained scalar coefficients. Capture, Patch AP and VE are percentages. The first row reuses the saved control, and update zero denotes a retained native start.

Width	Active groups	Groups	Directions	Active coordinates	Selected update	Capture (%)	Patch AP (%)	VE (%)
3	16	1024	3072	48	2500	30.08	63.34	64.66
2	8	1024	2048	16	2500	28.66	60.84	56.46
2	16	1024	2048	32	2500	27.45	60.84	62.88
2	32	1024	2048	64	500	21.14	59.29	69.65
3	8	1024	3072	24	2500	30.52	59.15	57.92
3	32	1024	3072	96	0	15.08	59.46	72.65
4	8	1024	4096	32	2500	32.01	62.03	59.31
4	16	1024	4096	64	750	28.20	64.16	66.35
4	32	1024	4096	128	0	15.58	59.96	74.72

Table S22. Width-four, Top-16 scores. Selection retains update 750.

Model or state	Capture (%)	AP (%)	VE (%)
Selected: 750	28.20	64.16	66.35
Final: 2,500	31.34	65.76	66.27

Table S23. Width-16, Top-3 scores. NR means Capture was not reported.

Model or state	Capture (%)	AP (%)	VE (%)
Native	*NR*	45.54	59.45
Selected: 2,250	36.83	46.63	58.86
Final: 2,500	36.88	46.34	58.92

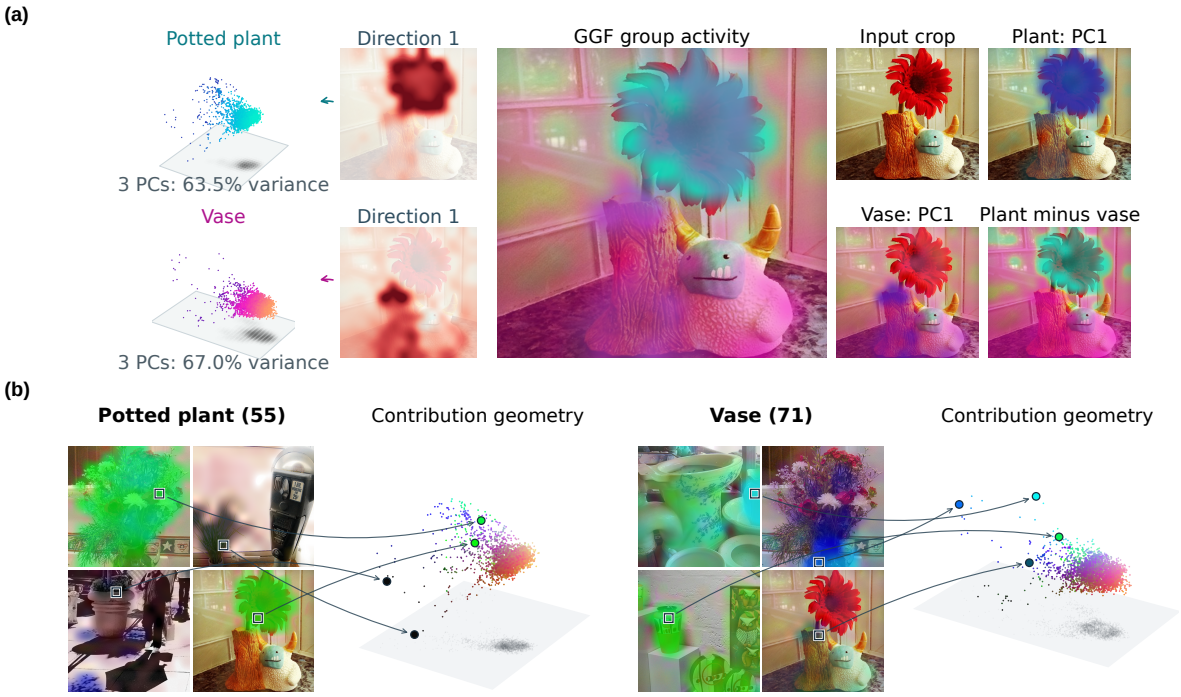


Figure S7. **Object-group responses and geometry.** (a) Plant (55) and vase (71) responses. Cloud/image colors encode PC1, joint colors identify groups, and direction maps show coefficient magnitude. (b) RGB encodes three contribution PCs with matching colors for patches and clouds. Each arrow locates one marked patch. Both panels share the GGF model and training-cloud samples. Overlays use activity-scaled opacity and smooth interpolation.

selecting by masks, cluster shape or test performance. The same saved sample indices are used across display revisions.

For each group, decoding its twelve coefficients produces a contribution in the 768-dimensional activation space. We center these active training contributions and fit PCA, whose orthogonal axes order their variation. The first three axes retain 63.5% of plant-group variance and 67.0% of vase-group variance. The remaining variance is absent from the 3D plot. Neither its shape nor these percentages establish intrinsic concept dimension. Point coordinates are the measured projected responses. Camera framing and a density shadow change their presentation without changing coordinates.

In panel (a), the first principal coordinate determines color within each group: blue–cyan–teal for plant and purple–magenta–orange for vase. The same training-derived color function is applied to cloud points and image patches. Its

limits are the 2nd and 98th percentiles of that group's active training coordinates. Values outside this range saturate for display. The large joint-presence panel instead uses fixed cyan for plant and magenta for vase. These categorical colors identify groups rather than principal coordinates.

Overlay opacity depends on the full twelve-coefficient norm, scaled by the 99th percentile of nonzero activity in the displayed image separately for each group. This makes weaker responses visible but prevents comparing absolute strength by brightness across groups. The difference panel subtracts these normalized group activities. Red direction panels show the absolute first native coefficient using separate within-group scales. They do not depict the first PCA axis. Responses can include surrounding objects and context, so these overlays are not segmentation predictions.

Smooth overlays apply a Gaussian filter with standard deviation $\sigma = 0.35$ patch to scalar responses, then cubic interpolation at patch centers, clipped to the measured range. These display operations change no scores, model outputs or original 14×14 patch values. The source photograph is rendered at high resolution over the same model crop. No image detail is generated. Checkpoint hashes, cohort IDs, projected arrays, sampling indices and display scales accompany the figure.

Panel (b) uses the same saved PCA and training-cloud sample, but maps PC1, PC2 and PC3 to red, green and blue. Each channel is scaled between the 2nd and 98th percentiles of that group's active training coordinates, clipped and mapped to $[0.04, 0.96]$. Exactly the same function colors image patches and cloud points. Four photographs per group illustrate the responses. Three come from high-response images in the fixed training cohort, and the fourth is the existing photograph COCO 553000. These are illustrative examples, rather than a comparison of methods.

Each photograph contains many patch contributions, not a single geometry point. An outlined patch with the largest gated coefficient norm in each image is linked to its exact measured PCA coordinates. The eight highlighted points supplement the fixed cloud sample without changing its coordinates. Let q_{99} be the 99th percentile of active training norms for that group, shared across its four images. Opacity is $0.76 \min(\|z_g\|_2/q_{99}, 1)^{0.65}$. Colors are interpolated after weighting by activity. The source photographs use the same resize and center crop as the model inputs, rendered from their original pixels at higher display resolution. The figure sources retain image IDs, file hashes, patch indices, PCA coordinates and all display scales.

F.3. Synthetic factor recovery with GGF

Manifold Zoo provides a controlled test where individual generating factors are known. We compare a factor-supervised analogue of GGF with the released BSFs. Table S24 reports the common recovery and reconstruction metrics. Figure S8 separates the eight factor families. This reconstructs the benchmark of Fel et al. [14] under the sampling protocol below, rather than reproducing its unpublished arrays or all author settings.

Data and archived controls. Geometry seed 701 defines 128 factors in 128 dimensions: 64 segments and 64 curved factors cycling through circles, disks, spheres, tori, Möbius strips, Swiss rolls and helices. Each observation sums four independently active, distinct factors. Training, selection, calibration and test contain 300,000, 20,000, 20,000 and 100,000 observations. Training-derived scaling gives mean squared norm one without global centering. Summing the stored true components reproduces the observations to 2.51×10^{-8}. This checks the arithmetic ceiling, not learned recovery.

The saved baseline classes [18] match the source hashes used by the new experiment after newline normalization. The original dataset is reused with checksum-verified payload and exact array hashes for all four splits. The nine selected baseline reports are imported unchanged, with no new baseline fits. All models use 256 width-four groups and requested activity four. Baseline seeds 0–2 and two learning rates $(0.0004, 0.003)$ originally gave eighteen fits of 200 epochs (29,200 updates), with batches of 2,048, Adam, OneCycleLR and no added noise. Checkpoints minimize selection MSE. Group Lasso additionally requires activity within 10% of four. Its measured mean activity is 3.808 rather than exactly four.

Factor-supervised GGF. Six new fits use the same seeds, two learning rates and total update budget. Initialization assigns one of the first 128 groups to each factor. The leading four eigenvectors of the uncentered second moment of observed training mixtures conditional on that factor's presence initialize its decoder directions. The encoder begins at the decoder transpose. The remaining 128 groups start randomly. Initialization uses presence labels, not isolated component vectors. Factor-informed initial directions are deterministic and shared across seeds. Reserve initialization and training order vary. Reconstruction warm-up lasts 40 epochs. Guided learning occupies the remaining 160. OneCycleLR spans all 200 epochs, without a reset at the transition.

For a training batch $\mathcal{B}$ of observations indexed by n, let S_n contain the four active factors, let $\mathcal{B}_f = \{n \in \mathcal{B} : f \in S_n\}$, and let $\mathcal{F}_\mathcal{B}$ contain factors represented in the training batch. For the group assigned to factor f, let $z_{nf} \in \mathbb{R}^r$ be its gated coefficient vector and $D_f \in \mathbb{R}^{d \times r}$ its decoder, with $d = 128$ and $r = 4$. These follow the column-vector convention of Equation (2). True component contributions $m_{nf} \in \mathbb{R}^d$ supply a synthetic counterpart of the mask-contrast target.

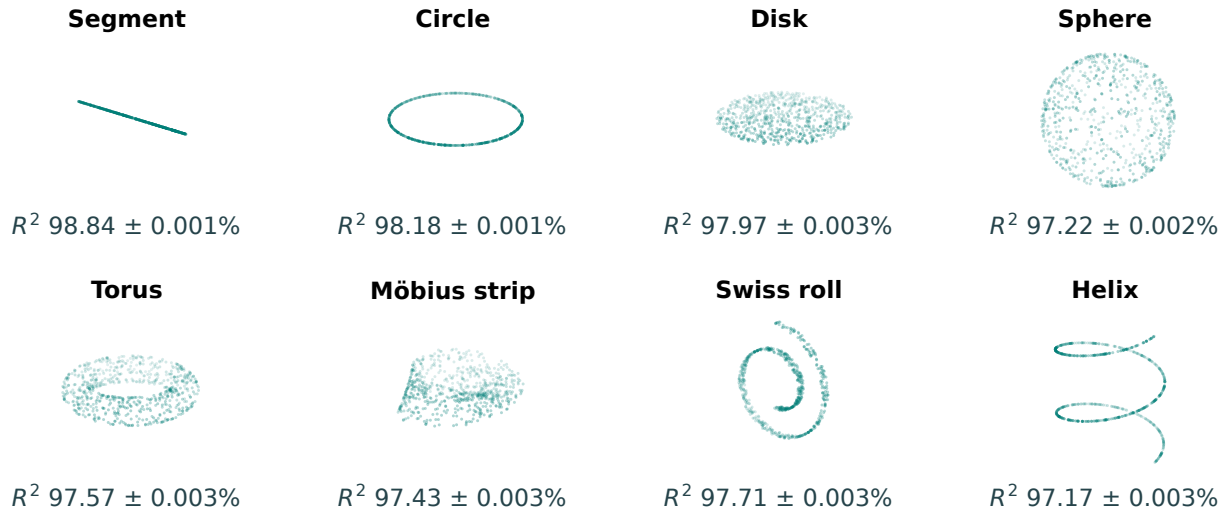


Figure S8. Known factor families and GGF's recovery by geometry. Shapes show samples from the benchmark generating functions of Fel et al. [14], rendered by us and centered and scaled separately. Values show GGF's assigned-group factor R^2, averaged within family, with mean ± sample SD. Decoded predictions appear in Figure S9. There are 64 segments, 10 circles and 9 factors in each remaining family. The benchmark embeds these factors into 128 dimensions and sums four per observation.

Table S24. Mean ± sample SD. Seed synthetic means (%). Test-matched recovery is a secondary diagnostic. Primary recovery locks groups on calibration. GGF final denotes the final endpoint of the selection-chosen trial.

Model	Locked R^2	Test-matched R^2	VE
Vanilla	96.781 ± 0.533	96.781 ± 0.533	97.372 ± 0.043
Grassmannian	86.782 ± 0.776	86.787 ± 0.780	92.049 ± 0.183
Group Lasso	93.295 ± 1.548	93.295 ± 1.548	96.694 ± 0.131
GGF selected	98.231 ± 0.001	98.231 ± 0.001	97.958 ± 0.001
GGF final	98.231 ± 0.000	98.231 ± 0.000	97.958 ± 0.000

The supervised contribution loss is

$$L_{\text{factor}} = \frac{1}{|\mathcal{F}_{\mathcal{B}}|} \sum_{f \in \mathcal{F}_{\mathcal{B}}} \frac{\sum_{n \in \mathcal{B}_f} \|D_f z_{nf} - m_{nf}\|_2^2}{\sum_{n \in \mathcal{B}_f} \|m_{nf}\|_2^2}. \quad \text{(S13)}$$

Each ratio normalizes the factor's mean squared contribution error by its mean component energy, and factors receive equal weight. Balanced factor-presence BCE uses the assigned group's gated coefficient norm, training-only mean/scale measured after warm-up, and a learned positive affine classifier. Its positive and negative pools receive equal weight within each factor. There are 128 heads (256 additional parameters). Reconstruction retains weight one. The same first-order gradient controller adapts both supervised weights every ten guided updates, with exponent 1.5, twenty reference batches and cap 0.60. This uses exact factor contributions and presence, which are stronger targets than COCO masks. It is a synthetic supervision experiment, not the unchanged COCO loss on another dataset.

Each seed's learning rate and checkpoint minimize selection reconstruction MSE before model evaluation on synthetic test. All runs select learning rate 0.0004, at updates 24,820, 24,090 and 27,010 for seeds 0, 1 and 2. Every trial finishes its full budget. Table S24 also reports the selected trial's final endpoint as a separate diagnostic. The GGF core has 263,168 parameters, matching Vanilla's core. Supervision adds heads and gradient computation. Archived BSFs use Torch's cu128 build and new GGF uses cu130, so the numerical environments are not identical.

Recovery and reconstruction. For factor f, let m_{nf} be its true contribution and $\widehat{m}_{nf}$ the chosen group's decoded contribution. With $\overline{m}_f$ the true contribution averaged over observations where the factor is present, recovery is

$$R_f^2 = 1 - \frac{\sum_{n: f \in S_n} \|m_{nf} - \widehat{m}_{nf}\|_2^2}{\sum_{n: f \in S_n} \|m_{nf} - \overline{m}_f\|_2^2}. \quad \text{(S14)}$$

We average over all 128 factors. One denotes exact component recovery. Zero matches the true factor's mean. No fitted readout or rescaling is applied to predictions. For baselines, the primary comparison selects each factor's group by correlation between its presence and group firing on calibration observations, then freezes that assignment for test. GGF's known assigned groups give identical scores to this calibration rule. A paper-style diagnostic selects groups using test

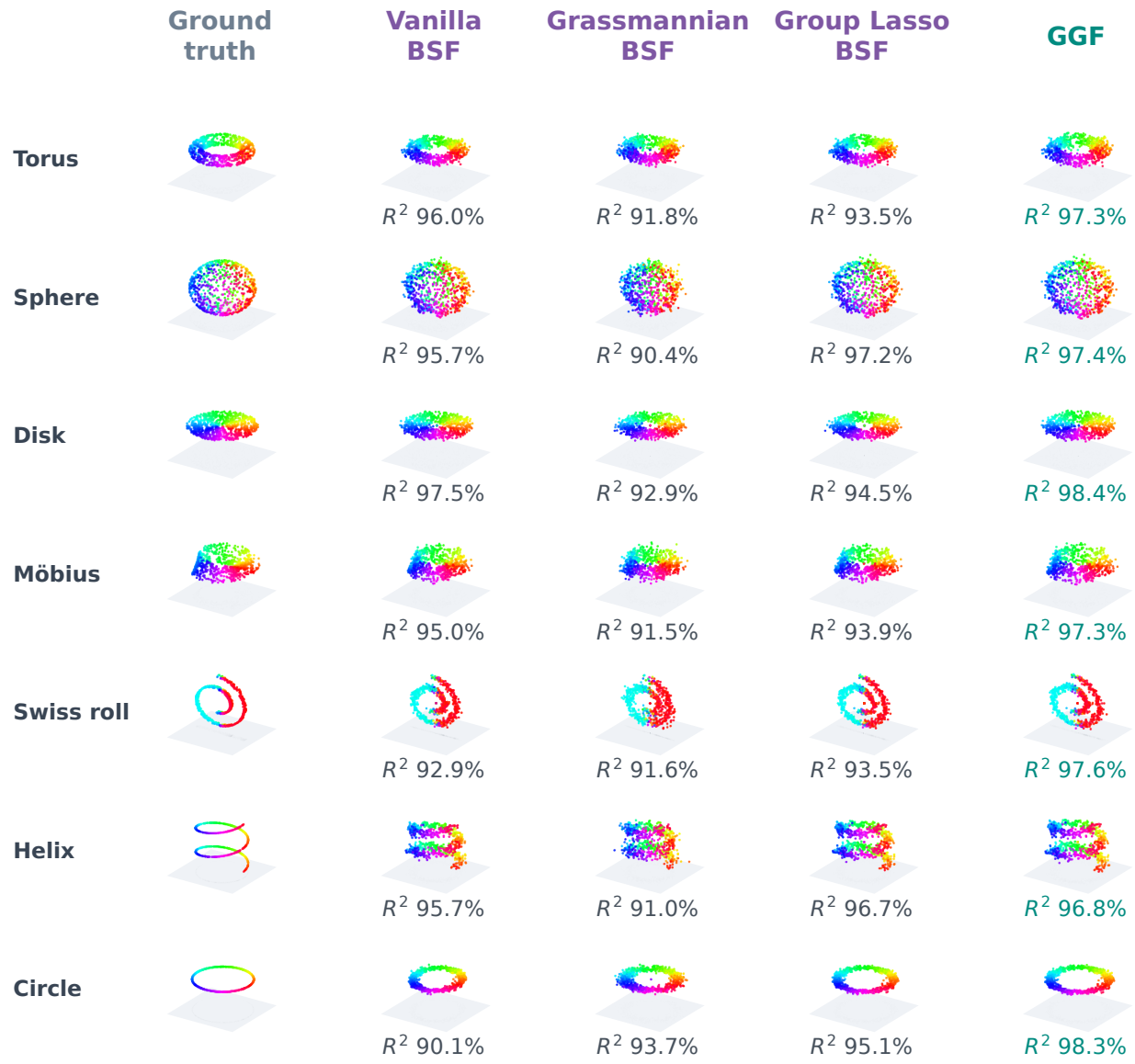


Figure S9. Ground-truth factors and decoded single-group contributions on Manifold Zoo [14], using our saved seed-0 models. Rows show the first indexed factor of each nonlinear family, selected without reference to model scores, and the same 700 present-factor observations in every column. Baseline groups are locked by calibration firing correlation. GGF uses its assigned group. Each row shares the true factor's embedding frame, center, scale and camera. Colors identify corresponding ground-truth points. Gate-zero predictions are retained. These are model outputs on regenerated benchmark observations. The displayed R^2 uses all present-factor test observations and all 128 activation dimensions. Projections can hide off-factor error.

presence masks. Its scores remain separate in Table S24. Reconstruction VE uses deviations from the training mean, unlike COCO's test-mean denominator. Synthetic factor R^2 and COCO contrast Capture measure different quantities.

For Figure S9, we regenerate the fixed benchmark. Seed-0 recovery and VE agree with archived values within 3×10^{-8} in absolute units. Full-dimensional scores accompany common-frame projections.

GGF improves primary recovery by 1.450 percentage points and VE by 0.586 over Vanilla. Both improvements hold in every run. All 128 factors exceed $R_f^2 = 0.90$ in their assigned group on test in each run. This is 100% of factors passing a recovery threshold, not perfect reconstruction: mean recovery is 98.231%. GGF's seed SDs are 0.00057 recovery points and 0.00059 VE points. These describe training-start variation on one fixed synthetic geometry, not uncertainty over generating dictionaries. Mean factor-presence AP is 92.976%. It is not COCO Patch AP, and archived baseline AP is unavailable. The geometry breakdown ranges from 97.17% for helices to 98.84% for seg-

Table S25. Manifold Zoo by seed (%). Recovery uses calibration-locked groups. All GGF assigned-factor test scores exceed 90%. The final row counts those factors, out of 128.

Model or metric	Seed 0	Seed 1	Seed 2
Vanilla recovery	96.20	96.89	97.25
Grassmannian recovery	86.57	87.65	86.13
Group Lasso recovery	94.82	91.72	93.34
GGF recovery	98.231	98.232	98.232
Vanilla VE	97.368	97.417	97.332
Grassmannian VE	92.001	92.252	91.895
Group Lasso VE	96.820	96.557	96.704
GGF VE	97.959	97.958	97.958
GGF factors $R^2 > 0.90$	128	128	128

ments. Neither this conditional synthetic result nor the 93–97% BSF recovery range establishes complete natural-image concept recovery. Full records and source archives are retained in `evidence/zoo_guided/`. The baseline JSON is in `evidence/zoo_baselines/`.

G. Group organization and functional diagnostics

This section brings the shared analysis protocol and its COCO and VOC2012 results together. The metrics distinguish useful individual contributions from stronger claims about semantic coherence.

G.1. Response consistency and dependence on selected groups

These diagnostics retain their original checkpoints, group budgets, readouts and cohorts; they are not pooled with the 75-group study. Figure S11 combines category-response profiles with tests of stability under appearance changes. Detailed protocols, category results, subgroup Capture and selected/random removals follow below; whole-group focus and coordinate-dependent direction agreement remain distinct.

Worst-condition recognition AP is $80.92 \pm 0.33\%$ for GGF, $55.16 \pm 3.37\%$ for Concept-SAE and $80.52 \pm 1.22\%$ for supervised Vanilla. The GGF–Concept-SAE paired 95% interval is positive; GGF's means also exceed Vanilla and SLiCS. Subsequent paragraphs retain every paired interval and the rotation/scale/brightness Capture scores.

G.2. Readouts, subset curves and removal controls

Why these diagnostics. Capture tests reconstructed object contrast; AP tests spatial response ranking. Their derivations and literature connections are given once in Appendix B.2. The subset and removal tests below assess complementary information and fixed-readout dependence.

Capture across group subsets. To test concentration without assuming a known manifold, a greedy rule selects up to four groups on training object contrasts after fitting. At each step it adds the native contribution giving the largest positive training-error reduction. If none improves it, the unchanged prediction is retained for larger budgets. This adapts compact-versus-distributed capture analysis [5] to existing BSF groups. For category c and subset budget $q \in \{1, 2, 4\}$, let $S_{c,q}$ be the selected group list and $C^{(q)}$ the resulting category-macro Capture. These lists freeze before evaluation. Equation (S4) is evaluated with δr_{og_c} replaced by $\sum_{g \in S_{c,q}} \delta r_{og}$, for $q \in \{1, 2, 4\}$. No coefficient/decoder is refitted and encoder Top-k is unchanged. The increment $C^{(4)} - C^{(1)}$ measures additional native-contribution recovery, not a feature-splitting rate: overlap, cancellation and context also affect it. Small increments are informative only alongside substantial one-group Capture. All-group Capture supplies reconstruction context. The first selected group's norm is also scored for AP, ensuring the same group is tested for reconstruction and localization.

Probe removal and its controls. Following probing [1] and feature ablation [16], we fit an evaluation-only linear ridge probe to training DINOv3 patches after freezing the featurizer. It predicts category mask fractions with penalty 10^{-3} and an unpenalized intercept. One probe per dataset is shared across BSF arms. It is separate from GGF's training logistic classifier heads and changes neither featurizer training nor primary Capture/Patch AP.

For category c, let β_c and b_c be the fixed probe's weight vector and intercept, so $f_c(x) = x^\top \beta_c + b_c$. Here g_c is the first training-greedy group. Keeping its code fixed, we subtract $r_{g_c}(x)$ from x, preserving the reconstruction residual. The target effect is

$$\Delta \mathrm{AP}_c = \mathrm{AP}(f_c(x)) - \mathrm{AP}(f_c(x - r_{g_c}(x))). \qquad \text{(S15)}$$

For each category, we separately remove its first training-selected group. This is not one handpicked group or an average over every dictionary group. We macro-average category AP drops within each fitted model, then report the mean and sample SD across fitted models. Four random groups are sampled from the 32 nearest training activity/contribution-RMS neighbors of the selected group. This is approximate matching, not exact energy matching. The same label-blind 32-patch samples, crowd exclusions and positive/negative mask thresholds as the primary AP evaluation are used. A larger drop than random controls indicates that the selected contribution supports the readout, not that information is absent from all remaining groups. The probe is not refitted after removal. Because subtraction is measured through a linear readout, its effects are readout dependence rather than causal changes to the backbone's own computation.

Selectivity is checked separately through mean absolute changes in off-target probe scores, not off-target AP. For each removed group, we average $|r_g(x)^\top \beta_{c'}|$ over $c' \neq c$

and sampled patches, then average categories. A large target AP drop does not imply a small off-target change.

Active coefficient/contribution spectra use a common training cohort. Direction, norm and PCA displays use training-derived bases and fixed image rules. The BSF-family audit covers the primary arms on COCO and VOC. Archived supporting-variant studies remain separate. Concept-SAE retains its native-output evaluation rather than being relabeled as a linear block.

On the matched COCO models, GGF Capture rises from $42.54 \pm 0.21\%$ with one training-selected group to $52.14 \pm 0.30\%$ with four. Vanilla rises from $23.30 \pm 0.85\%$ to $44.93 \pm 0.91\%$. Supervised Vanilla rises from $41.80 \pm 0.24\%$ to $50.81 \pm 0.37\%$. Values are means with sample SD. Both supervised methods place more regional evidence in the first group, while additional groups retain complementary evidence. Removing the selected group reduces target probe AP by 47.17 ± 1.04 points for GGF, 41.10 ± 1.36 for supervised Vanilla and 6.32 ± 1.32 for Vanilla. Matched random removals average 0.01 ± 0.02, 0.02 ± 0.02 and 0.02 ± 0.01 points, respectively. The fixed probe is applied to original activations after subtracting native group contributions. These are readout-based functional diagnostics, not interventions on a backbone output.

G.3. VOC2012 group organization

The model scores and dataset allocation are reported in Appendix C.4. Here we examine how the same fitted models distribute object-associated information across groups.

What the subset curves establish. GGF preserves a Capture advantage at every measured subset size (Figure S10a). Its one-to-four-group gain is 7.00 points, versus 5.36 for supervised Vanilla and 19.13 for Vanilla. The supervised arms therefore capture much more contrast with one group and need less additional contribution than unsupervised Vanilla. However, GGF's larger marginal gain than supervised Vanilla does not establish less fragmentation between those two methods. All-group Capture remains near 80% for every arm, so the first group is useful without containing all object-related information. The Vanilla audit's one-group value differs slightly from its primary score because the greedy plan selects groups under its own post-fit training rule. No held-out image selects a group, coefficient or subset.

What removal establishes. Removing one training-selected group separately for each of the 19 categories reduces target probe AP by 81.93 ± 0.56 points for GGF, 62.14 ± 2.34 for supervised Vanilla and 4.94 ± 1.93 for Vanilla. These are category-macro means with sample SD across fitted models. Matched random removals average below 0.05 points (Table S26). Thus GGF's selected contribution supports the fixed object readout more strongly. This measures readout dependence, not the absence of recoverable information in the remaining groups: no replacement probe is fitted after removal. Larger drops also do not establish greater selectivity. The off-target score-change audit in Table S27 is larger for GGF. Because the frozen probe is linear, removal directly subtracts the contribution's projection onto its weights. It is not an intervention on DINOv3's downstream computation.

Within-group variation and semantic coherence. The training-active coefficient and decoded-contribution spectra measure how many variance-bearing directions a selected group uses, rather than whether those directions share a meaning. Table S27 averages effective ranks over the Capture-selected group for each category, then over training runs. GGF's contribution entropy rank is higher than either control at the same width of twelve. This establishes richer measured within-group variation, not higher semantic coherence or recovery of a known natural-image manifold.

Signed-direction maps, norm overlays and contribution-PCA color maps are generated for fixed annotation-positive person and potted-plant images shared across methods. The image rule is independent of scores. PCA bases and display limits use training data. This follows the contribution-space visualization of Fel et al. [14, Sec. 3.5]: rotating a block's coordinate basis can change individual coefficient maps without changing its decoded contribution. PCA signs and display scales also differ by model, so hue or brightness alone cannot rank coherence. These displays make variation inspectable. The separate controlled-view study below adds fixed transformations and qualitative inspection.

G.4. Controlled appearance changes and qualitative coherence

The preceding subset and removal tests assess information carried by a group. Here we ask whether that group's object association persists across changing appearances, then inspect the meaning of its strongest responses. The models and readouts remain frozen throughout.

Cohort and transformations. The predeclared categories are person, car, dog, chair, potted plant and vase. Annotation coverage and a stable image hash determine the cohort, with at most 32 evaluation instances per category and one instance per base image. The final pool contains 178 evaluation objects. Training objects separately determine PCA bases and display scales. Existing COCO audit images are reused, rather than introduced as a new test set. Each annotated instance is extracted from the pinned crop, resized to a 144-pixel longest side and centered on a 224×224 neutral canvas. We vary in-plane rotation over $\{-30, -15, 0, 15, 30\}$ degrees, scale over $\{0.75, 0.875, 1, 1.125, 1.25\}$ and object

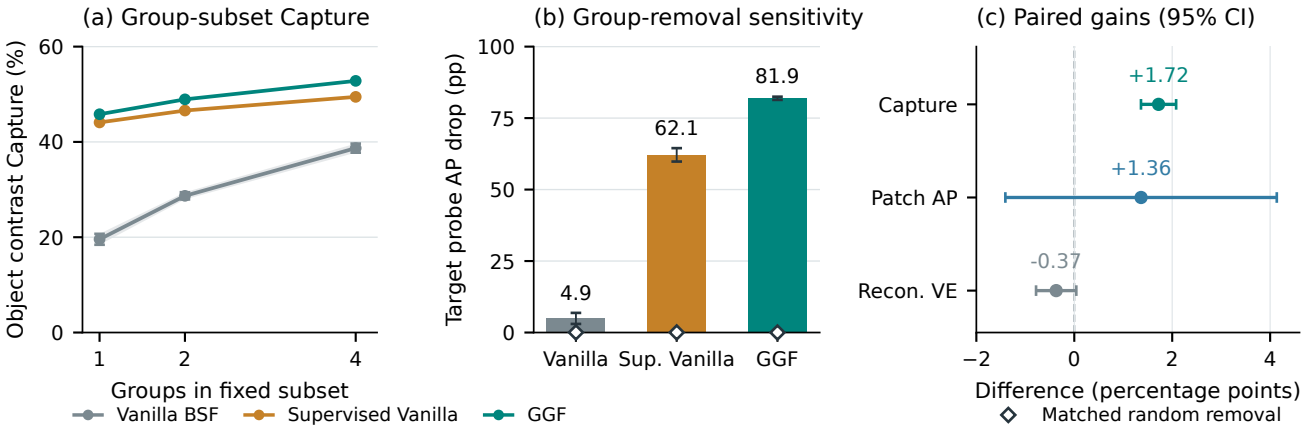


Figure S10. VOC2012 group analyses. (a) Native contributions from training-selected subsets. Lines and bands show mean ± sample SD. The subset size changes, not encoder Top-k. (b) Target probe-AP drop after subtracting one Capture-selected group. Bars show mean ± sample SD and white diamonds matched random removals. (c) GGF minus supervised Vanilla, with paired training-run 95% Student-t intervals. Higher Capture and removal sensitivity support useful individual groups. Neither establishes semantic purity or removal of all complementary information.

Table S26. VOC consolidation audit: mean ± sample SD. Capture (%) uses a post-fit training-greedy plan, separately from the primary readouts. Removal effects are target-category probe-AP drops in percentage points. The random column averages four activity/contribution-matched controls.

Method	One group	Two groups	Four groups	All groups	Selected removal	Random removal
Vanilla BSF	19.57 ± 1.15	28.69 ± 0.75	38.70 ± 0.95	79.80 ± 0.04	4.94 ± 1.93	0.044 ± 0.060
Supervised Vanilla	44.09 ± 0.05	46.58 ± 0.22	49.45 ± 0.21	79.63 ± 0.07	62.14 ± 2.34	0.031 ± 0.050
GGF	45.81 ± 0.10	48.93 ± 0.13	52.81 ± 0.10	79.79 ± 0.07	81.93 ± 0.56	0.006 ± 0.004

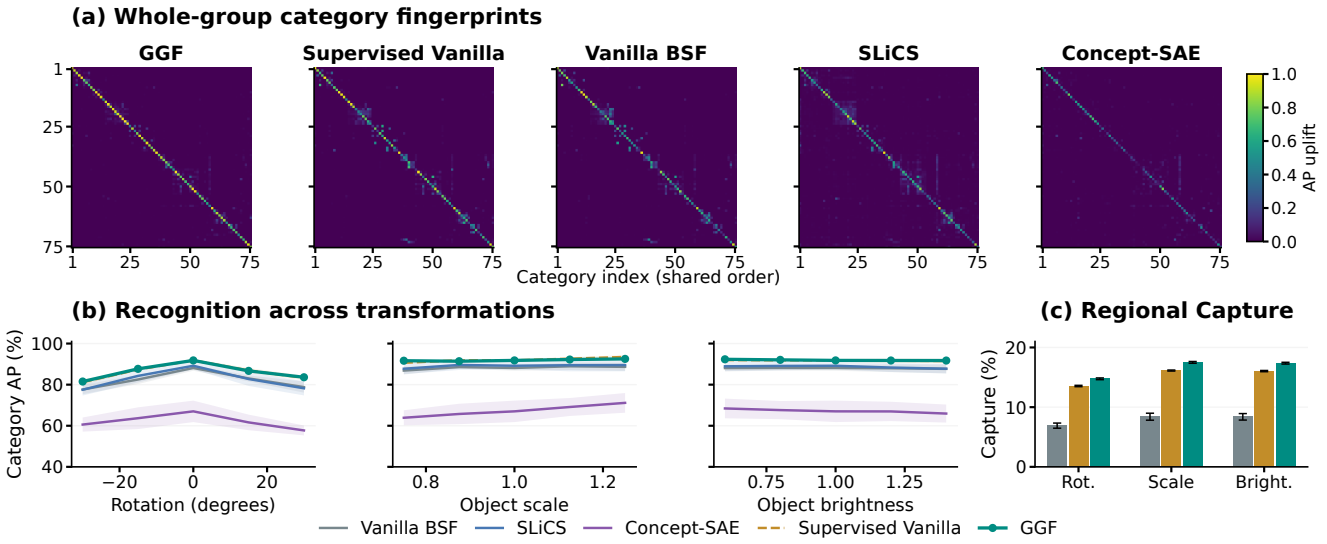


Figure S11. **Object focus and robustness.** Categories shown here were used for guidance. (a) Rows index 75 fixed group or token readouts, and columns test the same categories. Cells show mean above-prevalence AP uplift. A stronger diagonal indicates greater object focus. (b) Fixed readouts on six categories under controlled transformations. (c) Single BSF groups preserve regional contrast. Bands and error bars show mean ± sample SD. Protocols and SD maps appear in Appendices G.4, G.5.

brightness over $\{0.6, 0.8, 1, 1.2, 1.4\}$. Every object must pass annotation-only geometry checks at every level. These are controlled two-dimensional image changes, not three-dimensional pose or physical illumination.

Recognition with a fixed native readout. Each category-associated unit scores every object in the common six-category pool. BSFs use their gated group norms. SLiCS and Concept-SAE retain their native category response interfaces. Training fixes BSF group selection, while the alternatives

Table S27. VOC selected-group variation and collateral readout effects: mean ± sample SD of category means. Entropy rank uses training-active patches. Off-target change is the mean absolute change in other probe scores, multiplied by 10^3, not an AP drop.

Method	Coefficient entropy rank	Contribution entropy rank	Off-target change $\times 10^3$
Vanilla BSF	6.58 ± 0.07	4.99 ± 0.15	0.543 ± 0.050
Supervised Vanilla	5.98 ± 0.12	4.21 ± 0.11	0.849 ± 0.033
GGF	6.40 ± 0.03	6.28 ± 0.03	1.532 ± 0.036

use their category slots. AP ranks instances of the target category above the other five. It is object-category recognition, distinct from Patch AP on natural images. Let $\mathcal{C}$ be these six categories, s index fitted models, a the transformation type and ℓ its level. Write $\mathrm{AP}_{cs}(a, \ell)$ for recognition AP, $A_{cs}^{\min}$ for its worst condition and $\overline{A}_s^{\min}$ for the category mean:

$$A_{cs}^{\min} = \min_{a,\ell} \mathrm{AP}_{cs}(a, \ell), \qquad \overline{A}_s^{\min} = |\mathcal{C}|^{-1} \sum_c A_{cs}^{\min}, \tag{S16}$$

where (a, ℓ) ranges over the fifteen foreground conditions. Taking the minimum within each category before averaging prevents strong categories from hiding another category's weakest condition. The curve in Figure S11(b) instead reports category-macro AP at each level. Both summarize each fitted model before reporting mean and sample SD across models. Table S28 retains every category.

GGF's mean worst-condition AP exceeds Vanilla by 5.09 points, SLiCS by 4.29 and Concept-SAE by 25.76. Paired 95% intervals are $[-1.78, 11.96]$, $[-2.90, 11.47]$ and $[17.47, 34.05]$ points, respectively. Its 0.40-point difference from supervised Vanilla has interval $[-2.32, 3.11]$. The clearest recognition advantage is therefore over Concept-SAE. The archived multiplicity-adjusted result for that contrast is $p = 0.0222$. The other contrasts do not establish a reliable recognition advantage under this protocol.

Capture under the same changes. For the three BSF arms, Equation (S4) uses each category's first frozen training-selected group. Squared errors and target energies pool across objects and transformation levels within each category, followed by equal category averaging. This tests whether an additive group contribution retains regional object information, beyond merely ranking objects correctly. The cutout foreground is compared with its composited background, so these Capture values should not be equated with the natural-image Capture in Figure S2. Table S29 reports every category and foreground transformation. Macro Capture for GGF is $14.75 \pm 0.15\%$, $17.50 \pm 0.15\%$ and $17.37 \pm 0.15\%$ under rotation, scale and brightness. Supervised Vanilla reaches $13.54 \pm 0.11\%$, $16.13 \pm 0.09\%$ and $16.04 \pm 0.10\%$, and Vanilla $6.91 \pm 0.43\%$, $8.40 \pm 0.59\%$ and $8.38 \pm 0.54\%$. SLiCS's independent partial projections and Concept-SAE's native tokens are not added to this BSF-contribution audit.

Qualitative semantic assessment. For every category-associated unit, we inspect the eight strongest distinct original views from the same evaluation pool. The 90 cards cover every method, category and fitted model in this study. Mechanical ranking fixes the examples before inspection. Method identifiers were hidden during the initial assessment, and the authors subsequently verified the images and common-theme explanations. This is a descriptive qualitative assessment, not an independent human agreement study or a statistical coherence test.

The criterion is a recognizable common object, part, shape, material or other meaningful visual property across the strongest responses. A property may span categories. Matching the assigned category and having an interpretable within-concept variation axis are separate questions. GGF and supervised Vanilla consistently show recognizable object themes. The reviewed GGF responses are more consistently coherent than those of unsupervised Vanilla, SLiCS and Concept-SAE, although many of their units also show clear themes. Concept-SAE's chair-associated units include the clearest mixed-response cases. The selected groups and restricted object pool bound these observations.

Reading PCA maps. Following contribution-space PCA [14], each BSF group's first three training-derived principal coordinates set the RGB channels in Figure S7. A group norm indicates where it responds, while PCA color shows the relative location of its contribution in that group's coordinate system. Bases, signs and display limits differ across groups. Compare colors within a group's coordinate system rather than treating them as common semantic labels or an absolute response scale. Figure 2 applies contribution PCA to the matched 75-group coherence examples using a fixed audit reference cohort instead (Appendix F.1). Its border and outline colors annotate marked content separately from the RGB coordinates.

G.5. All-category fingerprints and direction agreement

We test whole-group object focus/reproducibility across all five native interfaces and, separately, within-group direction agreement for the four twelve-direction methods against energy-matched regroupings.

Models, data and fixed readouts. We reuse GGF, supervised Vanilla, unsupervised Vanilla, SLiCS and Concept-SAE checkpoints. No parameters, probes, loss weights or group memberships are updated. The data are all 400 images in the cached COCO development pool, with all 75 categories retained by the original coverage rule. Shared features, normalization, masks and image IDs are identical across methods. We verify checkpoint and feature checksums before inference and checkpoint checksums afterward.

Table S28. Worst-condition object-category AP (%, mean ± sample SD). For each category, the minimum AP across the fifteen foreground conditions is computed within a fitted model before aggregation.

Category	Vanilla	Supervised Vanilla	SLiCS	Concept-SAE	GGF
Person	91.72 ± 0.73	96.37 ± 1.30	92.91 ± 2.85	91.43 ± 4.96	98.00 ± 0.13
Car	79.33 ± 8.60	81.82 ± 3.52	84.37 ± 5.89	55.59 ± 4.31	75.02 ± 3.04
Dog	92.76 ± 1.45	93.20 ± 1.52	95.12 ± 1.42	91.55 ± 1.84	94.92 ± 0.05
Chair	52.95 ± 8.44	58.48 ± 4.66	55.47 ± 11.13	16.63 ± 4.30	56.20 ± 3.37
Potted plant	64.37 ± 6.55	81.94 ± 0.80	83.50 ± 11.03	43.66 ± 9.83	91.67 ± 1.49
Vase	73.83 ± 5.82	71.30 ± 8.46	48.40 ± 11.79	32.07 ± 3.51	69.68 ± 0.90

Table S29. Controlled single-group Capture by category and transformation (%, mean ± sample SD). Squared errors and target energies pool over objects and levels within each category before computing Capture.

Category	Transformation	Vanilla	Supervised Vanilla	GGF
Person	Rotation	7.41 ± 2.03	16.17 ± 0.45	17.86 ± 0.27
Person	Scale	7.61 ± 2.02	16.36 ± 0.44	18.35 ± 0.31
Person	Brightness	7.65 ± 2.00	16.27 ± 0.48	18.25 ± 0.32
Car	Rotation	11.15 ± 1.10	16.60 ± 0.37	15.08 ± 0.29
Car	Scale	14.82 ± 1.42	21.28 ± 0.51	20.04 ± 0.39
Car	Brightness	14.63 ± 1.53	21.01 ± 0.50	19.78 ± 0.39
Dog	Rotation	11.18 ± 0.53	25.37 ± 0.49	27.50 ± 0.59
Dog	Scale	11.63 ± 0.55	26.72 ± 0.43	28.87 ± 0.56
Dog	Brightness	11.68 ± 0.49	26.76 ± 0.41	28.96 ± 0.55
Chair	Rotation	2.59 ± 0.80	6.55 ± 0.39	7.10 ± 0.24
Chair	Scale	4.11 ± 1.14	9.94 ± 0.41	10.64 ± 0.28
Chair	Brightness	4.18 ± 1.11	10.07 ± 0.39	10.79 ± 0.25
Potted plant	Rotation	3.32 ± 0.73	7.96 ± 0.33	11.25 ± 0.13
Potted plant	Scale	3.99 ± 0.90	10.00 ± 0.25	13.46 ± 0.16
Potted plant	Brightness	3.96 ± 0.90	9.64 ± 0.31	12.85 ± 0.08
Vase	Rotation	5.78 ± 0.67	8.62 ± 0.16	9.73 ± 0.03
Vase	Scale	8.25 ± 1.19	12.51 ± 0.11	13.65 ± 0.10
Vase	Brightness	8.19 ± 1.16	12.50 ± 0.08	13.57 ± 0.09

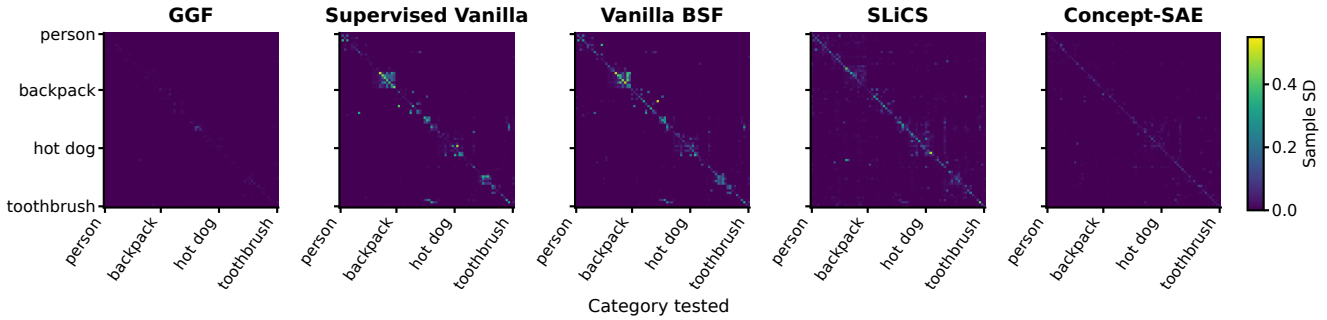


Figure S12. Sample SD of the category-response matrices in Figure S11(a). Each fitted model's matrix averages its two image halves. Rows, columns and category order match Figure S11(a), with a common SD scale across methods.

All 1,024 groups in each BSF, 75 SLiCS dictionaries, and 111 Concept-SAE tokens are profiled. Primary summaries average the 75 existing category-associated readouts: the training-selected contrast group for each BSF category and the corresponding native category dictionary or token for the alternatives. These assignments remain fixed. We neither search for a better group on these images nor discard poorly responding readouts. A group shared by two category assignments contributes to both category summaries. All-unit results remain archived separately from this equal-category comparison.

Two image halves and support requirements. A fixed random seed generates 200/200 image partitions. We retain the first partition with at least three annotation-positive images per category in each half, using masks alone and no model outputs. Each category has at least ten positive images in the complete pool. The halves estimate fingerprints independently, rather than selecting directions in one set and reporting their best responses in the other. Every direction and category enters the defined summaries. These development images were already inspected. This is an exploratory cross-image check within categories used for guidance, rather

than an independent confirmation of transfer to categories excluded from training or discovery of the model's own concepts.

Detection uses all 196 patches per image. A patch is positive at category coverage ≥ 0.5, negative at coverage ≤ 0.1, and otherwise ignored. Crowd coverage above 0.1 is ignored. This full-grid diagnostic differs from the primary benchmark's fixed 32-patch samples. For instance contrasts, known foreground coverage must sum to at least one patch and there must be eight category-negative patches. Foreground pooling uses coverage weights and comparison pooling uses the category-negative mean, as in Equation (S4). There are 1,936 usable object instances. Handbag has only two usable contrast-instance images in one half. Its raw results are retained, while the supported Capture and direction-contribution summaries use the 74 categories with at least three usable contrast images in both halves. Detection and direction-response summaries retain all 75 categories.

What counts as an object-associated fingerprint. Semantic alignment with annotated visual concepts motivates the category-response analysis [3]. Here $\mathcal{C}$ contains all 75 retained categories. For image half $h \in \{A, B\}$, let A^h_{gc} be exact Patch AP for native unit g and category c, including ties, and π^h_c the valid patches' positive prevalence. The unit's response fingerprint is

$$F^h_{gc} = \max\left(0, \frac{A^h_{gc} - \pi^h_c}{1 - \pi^h_c}\right), \tag{S17}$$

Prevalence-aware evaluation motivates adjusting for category frequency [11, Sec. 4.1.1]. Our transformation subtracts positive-patch prevalence and rescales by the remaining distance to perfect AP. It differs from their normalization of precision before averaging. BSFs use gated norms, SLiCS uses its independent per-category NNLS coefficients, and Concept-SAE uses its original mask-times-presence response. Training logistic classifier heads are not used.

We separately record signed single-group Capture C^h_{gc} for every unit and category, extending the single-block recovery perspective [14] to the regional contrasts in Equation (S4). Thus response focus and reconstructed object information have separate fingerprints. For Concept-SAE, Capture uses each native token's actual additive output, including the original nonlinear projector, bias and token aggregator (Appendix C.3). Its 36 free tokens contribute to the all-unit archive. They have no category mask, so their auxiliary spatial response is their decoded contribution norm. Primary comparisons use the original 75 concept-token responses. SLiCS dictionaries remain independent projections and are not summed into a joint reconstruction.

Write $F^h_g = (F^h_{gc})_{c\in\mathcal{C}}$ for a unit's category profile. For a nonnegative profile v, define $n(v) = v/\|v\|_2$, with $n(0) = 0$. Whole-group reproducibility is $n(F^A_g)^\top n(F^B_g)$, and is also computed separately for positive Capture profiles. A fixed category-permutation control measures the similarity remaining when category identities in one half are scrambled. Let $\overline{F} = (F^A + F^B)/2$ be the mean fingerprint matrix. The intended-category response share for readout g_c is

$$Q_c = \frac{\overline{F}_{g_c c}}{\sum_{c'} \overline{F}_{g_c c'}}. \tag{S18}$$

A zero denominator gives zero. Q_c is a share of response-profile weight, not an object-information percentage or a probability. We retain AP and Capture beside it because a narrow profile can be weak. Figure S11(a) shows the unnormalized $\overline{F}$ matrix with one common color scale.

Inspecting individual directions. For direction $j \in \{1, \ldots, r\}$, let D_{gj} be column j of D_g and z_{ipgj} component j of $z_g(x_{ip})$. For object o, define $\delta z_{ogj} = \sum_p w_{op} z_{ipgj}$, using its image i. Its directional contrast contribution is $a_{ogj} = \delta z_{ogj} D_{gj}$; the whole group contributes $m_{og} = \sum_j a_{ogj}$. Let $\mathcal{O}^h_c$ contain eligible objects of category c in half h, and $C^h_{g\setminus j,c}$ be Capture after removing direction j. The signed directional fingerprint is

$$U^h_{gjc} = C^h_{gc} - C^h_{g\setminus j,c}, \tag{S19}$$

where $C^h_{g\setminus j,c}$ uses $m_{og} - a_{ogj}$. Codes, decoder and gate stay fixed during this subtraction. It asks whether that direction helps that group's object reconstruction. For nonorthogonal directions the exact difference is

$$U^h_{gjc} = \frac{\sum_{o\in\mathcal{O}^h_c} \left[2a^\top_{ogj}(\delta x_o - m_{og}) + \|a_{ogj}\|^2_2\right]}{\sum_{o\in\mathcal{O}^h_c} \|\delta x_o\|^2_2}. \tag{S20}$$

This retains interactions among directions instead of treating them as orthogonal or ranking decoder vectors alone. Negative values identify directions whose removal improves that regional reconstruction and are preserved in the raw archive.

A second fingerprint R^h_{gjc} measures response without using the Capture objective. In every eligible image containing category c, we subtract mean absolute coefficient strength on category-negative patches from its category-coverage-weighted foreground mean, then average images equally. Let $\mathcal{I}^h_c$ contain these eligible images in half h. Writing f_{icp} for normalized foreground-coverage weights and $\bar{b}_{icp} = b_{icp}/\sum_{p'} b_{icp'}$ for uniform valid category-negative weights gives

$$R^h_{gjc} = \frac{1}{|\mathcal{I}^h_c|} \sum_{i\in\mathcal{I}^h_c} \sum_p (f_{icp} - \bar{b}_{icp}) |z_{ipgj}|. \tag{S21}$$

Each eligible image has equal weight. This uses all relevant patches rather than choosing visually favorable examples or only the largest responses. We compare positive parts of U and R separately, since a shared helpful category profile and a shared response profile are distinct properties.

Agreement and matched regrouping control. For either direction fingerprint $V \in \{U, R\}$, let $p^h_{gj} = n(\max(V^h_{gj}, 0))$. For direction indices $j, \ell \in \{1, \ldots, r\}$, agreement for a width-$r$ group is

$$W_g = \frac{1}{r(r-1)} \sum_{j \neq \ell} (p^A_{gj})^\top p^B_{g\ell}, \qquad r = 12. \tag{S22}$$

All 132 ordered pairs enter, comparing different directions across separate image halves. Zero positive profiles earn zero instead of being removed. Normalizing each category profile separates the category pattern from a direction's overall amplitude. We also archive the fraction of directions with a positive profile and same-direction reproducibility.

To ask whether agreement exceeds that expected from directions with similar activity, let z_{gj} denote its scalar coefficient and $\mathbb{E}_A$ average over half-A patches. Its decoded energy is $E_{gj} = \mathbb{E}_A[z^2_{gj}]\|D_{gj}\|^2_2$. We sort directions into ten equal-count energy bins and generate 100 fixed random permutations within those bins. Each permutation reassigns direction identities to the existing width-twelve slots and is applied to both image halves. Thus widths and slot energy bins are preserved, as is each direction's cross-half fingerprint. The reported grouping lift is W_g minus the mean agreement of these random regroupings. This is a control on the fingerprint statistic, not a newly trained or physically exchanged model. The control matches approximate decoded energy rather than every activation statistic. The permutations provide a descriptive reference rather than a calibrated hypothesis-test distribution.

Native interfaces and coordinate dependence. GGF, both Vanilla controls and SLiCS expose twelve additive directions per group. Concept-SAE remains in every whole-token comparison without artificial scalar coordinates. Direction agreement characterizes the learned basis. For BSFs, a coordinated orthogonal rotation of the encoder and decoder preserves group norms and decoded contributions [14, Sec. 3.5], while individual-direction fingerprints can change. Whole-group focus and reproducibility are therefore more direct measures of object-associated coherence. This symmetry does not extend to arbitrary rotations of SLiCS's nonnegative codes.

Aggregation, uncertainty and verification. For each model, we average scores over the fixed category readouts, giving categories equal weight. We then report the mean and sample SD across training runs. Random regrouping repetitions and directions are not treated as independent model replicates. Paired seed differences and all category results remain available in the evidence archive. We verified the removal formula against explicit subtraction and exact AP against the standard tied-score implementation. Across all saved models, 180 checks against the archived native interfaces agree to within 8.0×10^{-6} AP. This confirms interface fidelity rather than creating a new performance comparison.

Whole-group results. GGF's intended-category response share and response-profile reproducibility exceed each comparator in every seed (Table S30). Figure S12 shows variation in the response matrices across seeds. Capture-profile reproducibility is similar for GGF and SLiCS, and supervised Vanilla is slightly higher. GGF therefore has stronger whole-group object focus, with a particularly clear response-profile advantage over the supervised alternatives, rather than an advantage on every fingerprint property.

Direction-level results and their interpretation. Both the contribution and response fingerprints give GGF higher grouping lift than SLiCS in every seed (Figure S13, Table S31). The Vanilla controls have higher agreement than GGF under both definitions. These results distinguish GGF's stronger whole-group object focus from the closer alignment of individual native direction profiles in the controls. Alongside the qualitative object themes, the comparison supports greater object-associated coherence over SLiCS and Concept-SAE at the whole-unit level, with additional internal-direction evidence against SLiCS. Object labels do not describe every shared shape, part or texture, so this operational definition concerns object-associated organization.

G.6. Matched 75-group coherence and concentration

Purpose of the group budget. SLiCS uses one 12-direction dictionary per category, while the BSF methods allow a choice of group count. A common 75-group setting for the 75-category evaluation therefore matches SLiCS's native count and makes exhaustive inspection feasible. Vanilla, supervised Vanilla and GGF retain 75 twelve-direction groups. This does not assume that each learned group represents one named category; parameter counts, activity and native inference still differ. Concept-SAE 111 remains exploratory, and native category-split SLiCS 37/38 is an unmatched seed 21 reference. The 1,024- and 4,096-group experiments retain their original protocols and are never pooled with this study.

Literature basis and choice of review tasks. ACE treats meaningfulness and coherence of visual examples as separate requirements and tests coherence by asking people to identify the conceptually different image among six [17, Secs. 2,4]. CRAFT uses crop-level intruder detection, with the intruder and other examples drawn from the same object class [13, Sec. 4.2]. These studies motivate inspecting shared content beyond a broad category label. We apply that principle to fixed strongest-response patches. Three disjoint pairs use all six views without reusing an image and ask whether

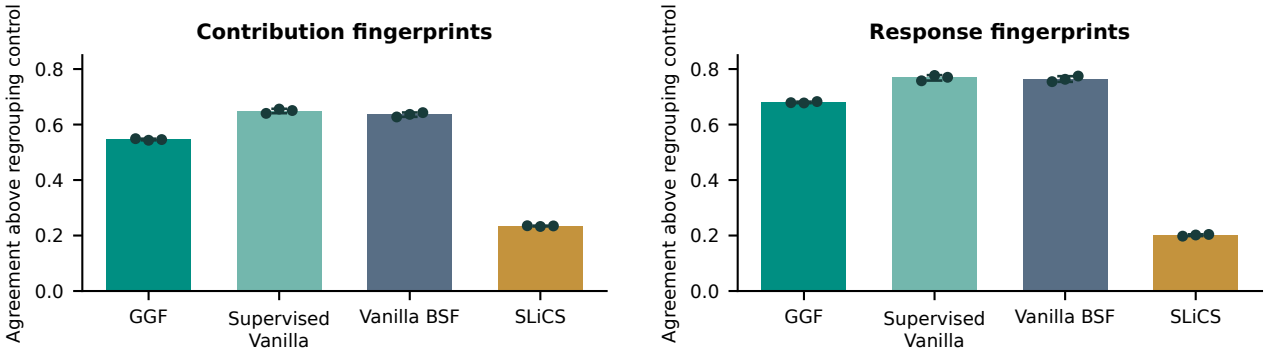


Figure S13. Category profiles show agreement among group directions in their contributions (left) and responses (right). Bars report lift over energy-matched random regrouping, with mean ± sample SD across fitted models and dots for individual runs. All twelve directions and eligible categories are included.

Table S30. Whole-group fingerprints: mean ± sample SD. Response and Capture reproducibility are cosine similarities. Category share is the intended category's fraction of positive response profile weight. Capture covers 74 supported categories, and the other summaries cover the 75 fixed category readouts. These full-grid development diagnostics are separate from the primary benchmark scores.

Method	Capture (%)	Patch AP (%)	Response reproducibility	Capture reproducibility	Category share (%)
GGF	41.6 ± 0.2	80.6 ± 0.3	0.9594 ± 0.0013	0.9685 ± 0.0001	77.1 ± 0.4
Supervised Vanilla	40.5 ± 0.2	62.7 ± 1.4	0.9265 ± 0.0110	0.9728 ± 0.0030	63.9 ± 1.6
Vanilla BSF	22.1 ± 0.9	52.4 ± 2.3	0.9091 ± 0.0220	0.9638 ± 0.0036	59.8 ± 2.4
SLiCS	25.0 ± 0.7	52.9 ± 1.7	0.8893 ± 0.0166	0.9680 ± 0.0004	46.5 ± 1.3
Concept-SAE	1.1 ± 0.1	32.4 ± 0.4	0.7818 ± 0.0017	0.5577 ± 0.0080	55.4 ± 0.6

Table S31. Direction grouping lift: mean ± sample SD. Contribution fingerprints compare signed Capture-removal effects through their positive parts. Response fingerprints compare foreground/background coefficient differences.

Method	Contribution	Response
GGF	0.5461 ± 0.0029	0.6798 ± 0.0027
Supervised Vanilla	0.6488 ± 0.0079	0.7683 ± 0.0098
Vanilla BSF	0.6356 ± 0.0078	0.7641 ± 0.0101
SLiCS	0.2342 ± 0.0014	0.2012 ± 0.0031

the marked regions share a specific meaning. A separate whole-set task checks that one theme spans the group: three individually coherent pairs could otherwise depict three different concepts. Requiring five of six matches permits one exception while demanding a common theme across most views. This is our explicit operational cutoff; the six-image ACE task uses an inserted intruder and does not validate this cutoff. Specificity is recorded separately so that a generic description does not automatically establish a discriminating meaning.

Automated assessment of strongest-response examples also has a close precedent. ELUDe compares grids of nine top-activating images using three vision-language judges and majority consensus [2, App. B.1]. Its judgments are preferences between two units' grids. Our absolute judgments concern marked pairs or one six-view set, and use three fresh contexts of the same reviewer model. The cited evaluation designs motivate our tasks without making their scores interchangeable.

Fits, category allocation and leakage checks. Training orders 21–27 share initialization and source image pools. BSFs use width 12 and Top-32; new fits use 10,640 updates. Two complementary category folds guide on 37 or 38 categories. The 75-group capacity remains fixed as guidance is reduced. Guide-only annotations determine initialization, targets, losses, controllers and model selection; excluded categories do not supply these signals. Only guided development outcomes in order 21 select supervised settings, frozen for orders 22–27. Excluded objects can still appear in reconstruction images. The task is withheld supervision categories, not unseen visual exposure. Frozen native units are calibrated after fitting; calibration may use the excluded-category labels, so the test is not zero-shot recognition. Shared Vanilla is one native bank per run, not two independent models. Original weight, native-response, crop and card hashes were checked.

Coherence scores. For group g, fixed pair $p \in \{1,2,3\}$ and reviewer context $r \in \{1,2,3\}$, let y^e_{gpr} indicate a yes vote on endpoint $e \in \{\text{semantic}, \text{visual}\}$. With $\mathcal{G}$ the complete planned group bank, the majority indicator and pair score are

$$m^e_{gp} = \mathbf{1}\left[\sum_{r=1}^{3} y^e_{gpr} \geq 2\right], \quad S_e = \frac{1}{3|\mathcal{G}|}\sum_{g\in\mathcal{G}}\sum_{p=1}^{3} m^e_{gp}. \tag{S23}$$

Let a_{gr} indicate a clear whole-group judgment and b_{gr} a specific or broad-but-discriminating specificity judgment.

Table S32. GPT coherence scores by seed (%). Column headings 21–27 are training seeds. Sup. Vanilla denotes supervised Vanilla BSF. Category-split scores include all native groups.

Specific-pair meaning

Method	21	22	23	24	25	26	27
Category-split guidance							
GGF	81.78	78.00	79.33	81.33	77.56	79.56	81.56
Sup. Vanilla	68.89	77.33	72.67	70.67	70.00	75.11	74.67
Vanilla BSF	54.67	63.11	61.33	58.67	58.22	68.44	58.67
All category labels							
GGF	93.33	93.78	95.11	92.89	90.22	93.78	94.67
SLiCS	80.89	83.56	84.44	87.11	80.89	81.33	85.78
Sup. Vanilla	63.11	65.33	67.56	70.22	72.00	66.67	70.22
Vanilla BSF	54.67	57.33	54.22	61.78	63.56	66.22	61.33

Pairwise visual structure

Method	21	22	23	24	25	26	27
Category-split guidance							
GGF	81.33	75.78	77.56	70.67	71.11	71.11	70.67
Sup. Vanilla	69.33	75.56	72.89	61.78	63.11	66.00	67.33
Vanilla BSF	57.78	60.89	61.33	55.11	52.00	59.11	54.22
All category labels							
GGF	90.22	93.78	89.78	82.67	80.89	83.11	81.33
SLiCS	86.22	82.22	84.44	78.67	76.44	74.22	80.89
Sup. Vanilla	62.67	64.00	67.56	62.67	62.22	60.00	60.89
Vanilla BSF	53.33	55.11	56.44	51.56	54.22	56.44	56.00

Clear whole-group theme

Method	21	22	23	24	25	26	27
Category-split guidance							
GGF	86.67	82.00	83.33	84.67	80.00	82.67	80.00
Sup. Vanilla	75.33	78.67	82.67	72.67	71.33	75.33	72.00
Vanilla BSF	65.33	69.33	74.67	57.33	57.33	66.67	60.00
All category labels							
GGF	100.00	100.00	100.00	96.00	100.00	100.00	100.00
SLiCS	97.33	97.33	100.00	96.00	97.33	96.00	96.00
Sup. Vanilla	77.33	72.00	69.33	72.00	72.00	60.00	72.00
Vanilla BSF	61.33	72.00	68.00	70.67	66.67	61.33	62.67

Clear theme with discriminating specificity

Method	21	22	23	24	25	26	27
Category-split guidance							
GGF	86.00	81.33	82.67	84.67	78.67	82.00	79.33
Sup. Vanilla	74.67	77.33	81.33	72.67	71.33	75.33	72.00
Vanilla BSF	65.33	69.33	74.67	57.33	57.33	66.67	57.33
All category labels							
GGF	100.00	100.00	100.00	96.00	100.00	100.00	98.67
SLiCS	97.33	97.33	100.00	96.00	96.00	94.67	96.00
Sup. Vanilla	77.33	70.67	66.67	72.00	70.67	60.00	70.67
Vanilla BSF	60.00	69.33	64.00	69.33	62.67	58.67	61.33

Table S33. GPT whole-group outcomes and specificity, pooled counts across seven training orders. Disagreements count groups with non-identical reviewer votes on the panel's measure. Unknown votes are individual insufficient-evidence votes.

Theme outcome

Setting	Method	Clear	Partial	No theme	Insufficient	Unresolved	Disagreements	Unknown votes
Split	GGF	869	177	4	0	0	37	0
Split	Supervised Vanilla	792	245	11	0	2	75	1
Split	Vanilla BSF	338	176	11	0	0	57	0
All	GGF	522	3	0	0	0	8	0
All	SLiCS	510	15	0	0	0	25	0
All	Supervised Vanilla	371	143	11	0	0	47	0
All	Vanilla BSF	347	166	11	0	1	57	0

Theme specificity

Setting	Method	Specific	Broad, discriminating	Generic	Unclear	Unresolved	Disagreements
Split	GGF	697	343	8	0	2	230
Split	Supervised Vanilla	718	321	7	0	4	260
Split	Vanilla BSF	310	203	5	0	7	156
All	GGF	432	92	1	0	0	95
All	SLiCS	378	145	2	0	0	104
All	Supervised Vanilla	302	213	7	0	3	141
All	Vanilla BSF	276	230	15	0	4	145

Define $A_g = \mathbf{1}[\sum_r a_{gr} \geq 2]$ and $B_g = \mathbf{1}[\sum_r b_{gr} \geq 2]$. The two whole-group rates are

$$S_{\text{whole}} = \frac{\sum_g A_g}{|\mathcal{G}|}, \qquad S_{\text{discr}} = \frac{\sum_g A_g B_g}{|\mathcal{G}|}. \tag{S24}$$

Table S34. All category labels, negative quantile 0.95: mean ± sample SD.

Method	Groups	Coverage	False pos.	Precision	Pos. overlap	Neg. overlap
GGF	1	90.62 ± 0.10	5.05 ± 0.07	33.79 ± 0.27	—	—
GGF	2	93.15 ± 0.23	8.87 ± 0.05	23.90 ± 0.10	51.13 ± 0.65	17.72 ± 0.45
GGF	4	94.70 ± 0.26	15.34 ± 0.14	16.51 ± 0.11	33.62 ± 0.43	15.04 ± 0.35
GGF	8	96.50 ± 0.09	26.78 ± 0.20	11.11 ± 0.07	18.09 ± 0.33	9.75 ± 0.23
SLiCS	1	79.60 ± 0.79	4.66 ± 0.11	32.76 ± 0.61	—	—
SLiCS	2	84.48 ± 0.94	7.91 ± 0.29	24.37 ± 0.50	54.76 ± 3.07	23.76 ± 1.52
SLiCS	4	87.80 ± 0.58	13.51 ± 0.53	17.34 ± 0.40	37.06 ± 1.83	17.79 ± 1.49
SLiCS	8	91.32 ± 0.76	23.70 ± 0.87	11.82 ± 0.27	20.57 ± 1.17	11.70 ± 0.76
Supervised Vanilla	1	62.59 ± 1.83	4.97 ± 0.17	26.22 ± 1.16	—	—
Supervised Vanilla	2	70.89 ± 2.03	9.36 ± 0.26	18.66 ± 0.72	41.87 ± 4.58	8.09 ± 1.43
Supervised Vanilla	4	78.21 ± 1.67	17.32 ± 0.46	12.83 ± 0.39	25.78 ± 1.85	6.75 ± 0.76
Supervised Vanilla	8	84.82 ± 1.20	30.87 ± 0.73	8.76 ± 0.15	14.22 ± 0.51	5.30 ± 0.29
Vanilla BSF	1	55.11 ± 1.77	4.97 ± 0.12	23.82 ± 1.17	—	—
Vanilla BSF	2	63.94 ± 1.87	9.46 ± 0.21	17.09 ± 0.76	37.41 ± 2.42	7.82 ± 1.21
Vanilla BSF	4	72.05 ± 2.06	17.41 ± 0.41	11.95 ± 0.40	25.87 ± 1.93	6.61 ± 0.55
Vanilla BSF	8	79.98 ± 1.65	31.12 ± 0.51	8.23 ± 0.15	14.45 ± 0.87	5.11 ± 0.26

Table S35. Categories withheld from guidance, negative quantile 0.95: mean ± sample SD.

Method	Groups	Coverage	False pos.	Precision	Pos. overlap	Neg. overlap
GGF	1	66.13 ± 1.87	5.09 ± 0.14	26.93 ± 0.91	—	—
GGF	2	74.42 ± 1.14	9.35 ± 0.15	19.33 ± 0.35	45.63 ± 1.46	11.55 ± 0.79
GGF	4	82.63 ± 1.00	17.04 ± 0.23	13.67 ± 0.19	27.10 ± 1.17	9.01 ± 0.54
GGF	8	89.02 ± 0.47	29.97 ± 0.34	9.36 ± 0.06	14.58 ± 0.60	6.64 ± 0.21
Supervised Vanilla	1	59.20 ± 1.76	5.27 ± 0.23	24.21 ± 0.84	—	—
Supervised Vanilla	2	68.28 ± 1.77	9.67 ± 0.25	17.89 ± 0.56	41.63 ± 2.50	9.46 ± 1.53
Supervised Vanilla	4	76.58 ± 1.45	17.54 ± 0.45	12.65 ± 0.22	27.93 ± 1.49	7.55 ± 0.90
Supervised Vanilla	8	83.42 ± 0.98	31.07 ± 0.61	8.65 ± 0.14	14.53 ± 0.87	5.60 ± 0.40
Vanilla BSF	1	57.15 ± 1.99	5.22 ± 0.10	23.62 ± 0.93	—	—
Vanilla BSF	2	65.33 ± 2.24	9.75 ± 0.22	16.99 ± 0.71	40.19 ± 1.17	8.00 ± 1.54
Vanilla BSF	4	73.46 ± 1.88	17.82 ± 0.45	11.91 ± 0.47	27.27 ± 1.59	6.68 ± 0.83
Vanilla BSF	8	80.93 ± 1.94	31.51 ± 0.63	8.19 ± 0.28	15.06 ± 0.84	5.25 ± 0.35

Table S36. Guided categories in category-split fits, negative quantile 0.95: mean ± sample SD.

Method	Groups	Coverage	False pos.	Precision	Pos. overlap	Neg. overlap
GGF	1	90.07 ± 0.20	4.71 ± 0.03	34.86 ± 0.12	—	—
GGF	2	92.22 ± 0.17	8.93 ± 0.08	23.50 ± 0.16	41.03 ± 1.26	14.51 ± 0.54
GGF	4	94.38 ± 0.16	16.29 ± 0.24	15.71 ± 0.17	24.45 ± 1.06	10.69 ± 0.68
GGF	8	96.17 ± 0.16	29.38 ± 0.32	10.08 ± 0.07	13.15 ± 0.59	7.03 ± 0.33
Supervised Vanilla	1	81.28 ± 1.82	4.77 ± 0.15	32.66 ± 0.49	—	—
Supervised Vanilla	2	85.74 ± 1.29	8.86 ± 0.23	22.56 ± 0.40	54.27 ± 2.43	9.72 ± 1.12
Supervised Vanilla	4	90.00 ± 0.96	16.97 ± 0.16	14.67 ± 0.23	29.89 ± 0.99	7.57 ± 0.52
Supervised Vanilla	8	93.27 ± 0.64	30.50 ± 0.29	9.58 ± 0.14	14.97 ± 0.74	5.68 ± 0.36
Vanilla BSF	1	57.15 ± 1.99	5.22 ± 0.10	23.62 ± 0.93	—	—
Vanilla BSF	2	65.33 ± 2.24	9.75 ± 0.22	16.99 ± 0.71	40.19 ± 1.17	8.00 ± 1.54
Vanilla BSF	4	73.46 ± 1.88	17.82 ± 0.45	11.91 ± 0.47	27.27 ± 1.59	6.68 ± 0.83
Vanilla BSF	8	80.93 ± 1.94	31.51 ± 0.63	8.19 ± 0.28	15.06 ± 0.84	5.25 ± 0.35

A clear judgment requires one theme matching at least five of six HIGH views. A vote that is unclear, insufficient or unresolved is not counted as yes or clear, and never reduces these planned denominators. Outcome and specificity distributions count groups; disagreement counts groups with non-identical votes, and unknown counts raw insufficient-evidence votes.

All-group evidence and blinded review. Pair and whole-group reviews use GPT-6.1 Sol (`gpt-6.1-sol`) with high reasoning effort [26]. Archived execution metadata verifies the inherited model settings for all actual reviewer identities, including the reused seed 21 reviews. Fresh contexts are review replicates, not independent models or human raters.

Each group's six strongest audit images are selected by the signed mean of its four largest native patch responses over 196 positions, without clipping. Deterministic image ties and hash presentation fix the six views and three disjoint pairs. Up to four positive marked patches and

Table S37. All category labels, negative quantile 0.90: mean ± sample SD.

Method	Groups	Coverage	False pos.	Precision	Pos. overlap	Neg. overlap
GGF	1	93.48 ± 0.46	9.96 ± 0.15	21.93 ± 0.21	—	—
GGF	2	96.55 ± 0.11	16.68 ± 0.10	15.46 ± 0.09	65.71 ± 1.24	22.87 ± 0.46
GGF	4	97.60 ± 0.11	27.02 ± 0.22	10.88 ± 0.06	47.81 ± 0.66	19.82 ± 0.19
GGF	8	98.88 ± 0.15	42.69 ± 0.33	7.71 ± 0.06	31.64 ± 0.32	15.94 ± 0.25
SLiCS	1	85.98 ± 1.23	9.25 ± 0.31	21.87 ± 0.62	—	—
SLiCS	2	91.26 ± 0.71	14.88 ± 0.51	16.35 ± 0.45	68.71 ± 2.60	28.46 ± 1.68
SLiCS	4	94.22 ± 0.45	23.62 ± 0.86	11.94 ± 0.26	55.92 ± 1.43	24.72 ± 1.30
SLiCS	8	96.38 ± 0.45	37.75 ± 1.49	8.43 ± 0.21	37.89 ± 1.37	19.31 ± 1.05
Supervised Vanilla	1	75.30 ± 2.62	9.67 ± 0.32	18.96 ± 1.06	—	—
Supervised Vanilla	2	83.54 ± 1.97	17.55 ± 0.32	13.29 ± 0.50	57.41 ± 4.39	12.05 ± 1.78
Supervised Vanilla	4	89.87 ± 1.30	30.39 ± 0.56	9.32 ± 0.21	41.38 ± 1.81	11.60 ± 0.91
Supervised Vanilla	8	94.63 ± 0.84	49.87 ± 0.66	6.47 ± 0.12	26.61 ± 1.15	9.75 ± 0.54
Vanilla BSF	1	68.65 ± 2.15	9.69 ± 0.24	17.57 ± 0.53	—	—
Vanilla BSF	2	77.03 ± 1.53	17.35 ± 0.34	12.64 ± 0.22	51.98 ± 1.29	13.14 ± 0.96
Vanilla BSF	4	85.27 ± 0.97	30.18 ± 0.36	8.95 ± 0.07	40.52 ± 1.34	11.54 ± 0.45
Vanilla BSF	8	91.76 ± 1.16	49.82 ± 0.55	6.28 ± 0.09	26.83 ± 1.15	9.46 ± 0.52

Table S38. Categories withheld from guidance, negative quantile 0.90: mean ± sample SD.

Method	Groups	Coverage	False pos.	Precision	Pos. overlap	Neg. overlap
GGF	1	79.17 ± 1.27	9.73 ± 0.12	19.45 ± 0.34	—	—
GGF	2	87.74 ± 0.55	17.52 ± 0.28	13.86 ± 0.15	62.92 ± 1.76	15.58 ± 0.98
GGF	4	93.50 ± 0.38	29.35 ± 0.22	9.96 ± 0.08	46.71 ± 0.89	14.92 ± 0.28
GGF	8	97.05 ± 0.43	47.35 ± 0.49	7.04 ± 0.07	29.33 ± 0.52	12.28 ± 0.34
Supervised Vanilla	1	71.73 ± 1.16	9.95 ± 0.14	17.88 ± 0.37	—	—
Supervised Vanilla	2	81.45 ± 0.98	18.05 ± 0.23	12.81 ± 0.25	55.38 ± 2.88	13.02 ± 1.16
Supervised Vanilla	4	88.62 ± 0.86	30.83 ± 0.43	9.13 ± 0.14	42.45 ± 1.63	11.86 ± 0.82
Supervised Vanilla	8	93.88 ± 0.58	50.22 ± 0.37	6.44 ± 0.08	27.41 ± 1.27	9.84 ± 0.44
Vanilla BSF	1	69.45 ± 1.53	9.67 ± 0.15	17.78 ± 0.37	—	—
Vanilla BSF	2	78.75 ± 1.02	17.44 ± 0.35	12.79 ± 0.29	55.23 ± 1.85	13.19 ± 1.56
Vanilla BSF	4	85.90 ± 0.66	30.10 ± 0.74	9.03 ± 0.24	42.53 ± 1.17	11.50 ± 1.25
Vanilla BSF	8	92.20 ± 0.60	50.03 ± 1.00	6.27 ± 0.16	27.51 ± 1.16	9.33 ± 0.63

Table S39. Guided categories in category-split fits, negative quantile 0.90: mean ± sample SD.

Method	Groups	Coverage	False pos.	Precision	Pos. overlap	Neg. overlap
GGF	1	93.68 ± 0.17	9.64 ± 0.08	22.36 ± 0.14	—	—
GGF	2	96.67 ± 0.21	17.05 ± 0.18	15.24 ± 0.11	56.75 ± 0.88	20.49 ± 0.54
GGF	4	97.57 ± 0.26	28.12 ± 0.20	10.63 ± 0.05	41.44 ± 1.53	17.87 ± 0.55
GGF	8	98.77 ± 0.16	46.10 ± 0.43	7.21 ± 0.05	25.26 ± 0.95	13.21 ± 0.35
Supervised Vanilla	1	88.06 ± 1.60	9.57 ± 0.25	21.62 ± 0.44	—	—
Supervised Vanilla	2	92.63 ± 0.62	17.06 ± 0.24	14.82 ± 0.22	68.58 ± 2.28	13.69 ± 1.29
Supervised Vanilla	4	95.68 ± 0.38	29.67 ± 0.24	10.00 ± 0.13	44.51 ± 1.18	12.26 ± 0.56
Supervised Vanilla	8	97.87 ± 0.27	49.15 ± 0.60	6.75 ± 0.09	26.71 ± 1.18	10.03 ± 0.40
Vanilla BSF	1	69.45 ± 1.53	9.67 ± 0.15	17.78 ± 0.37	—	—
Vanilla BSF	2	78.75 ± 1.02	17.44 ± 0.35	12.79 ± 0.29	55.23 ± 1.85	13.19 ± 1.56
Vanilla BSF	4	85.90 ± 0.66	30.10 ± 0.74	9.03 ± 0.24	42.53 ± 1.17	11.50 ± 1.25
Vanilla BSF	8	92.20 ± 0.60	50.03 ± 1.00	6.27 ± 0.16	27.51 ± 1.16	9.33 ± 0.63

their original-resolution zoom strips are inspected. Category/method names, native unit identities, numerical scores and other reviews are hidden. The semantic endpoint requires a specific shared object, part, action, relation, material or region at both marks. Generic edges or a broad umbrella category are insufficient. Specific visual structure has its own yes/no/unclear vote; cross-category geometry and textures count. Scores divide majority-yes pairs by all three planned pairs per group, with unclear outcomes retained. A group is not required to match its intended training category to be coherent.

A supplemental, separately blinded whole-group review asks for one theme across all six HIGH responses: clear at 5/6 or 6/6, partial at 3/6 or 4/6, and no clear theme at 0–2/6. Specificity is separately specific, broad but discriminating, generic/nondiscriminating or unclear. Majority-clear plus at least two discriminating-specificity votes is secondary. LOW views are absent, so HIGH–LOW contrast is not as-

sessed. Unknown evidence and unresolved votes remain in the denominator. The whole-group request followed the pair results; it is not an independent confirmatory allocation, and these are AI reviewers working in separate fresh contexts, rather than human raters.

There are 675 matched groups per run and 4,725 across all seven orders. Matched pair review covers 14,175 fixed pairs per context. Including 75 native reference groups from order 21 gives 14,400 pairs per context and 43,200 two-endpoint assessments across three contexts. Whole review covers 4,725 matched groups and yields 14,175 ratings. The order 21 judgments and provenance are reused after evidence-identity audits. Orders 22–23 add 339 packets for each review task and orders 24–27 add 675 for each. Review annotations are finalized before aggregation and retained with their source hashes. Independent recounts reproduce majorities, denominators, unique group IDs and source hashes, and verify that original seed 21 scores and completed numeric/concentration results remain unchanged.

Presence-based concentration and false positives. All 1,900 crops receive blinded category-presence labels: 400 calibration and 1,500 audit crops. Calibration uses the original confirmation cohort, which is separate from training, development and audit images. A fixed 40-crop second AI review covers 3,000 category cells; canonical labels remain unchanged after QA. Negative response quantiles 0.95 (primary) and 0.90 (sensitivity) define per-group thresholds, with strict exceedance. Calibration balanced accuracy ranks all 75 candidate groups for each category. Subsets of 1/2/4/8 ranked groups freeze before audit scoring and use an OR detector. Coverage is the fraction of known target-present crops detected; false positives are the fraction of known target-absent crops detected. Unknown presence labels are not imputed negative. At least five positive and ten negative crops are required in both splits. Truck and hot dog lack calibration support; all 75 targets stay in support tables, with rates over 73 supported targets, never zero imputation. No group is selected because it won a coherence review. Positive/negative overlap, rank-wise unique detection and incremental coverage are retained in the accompanying statistics. The one-to-eight-group curve concerns detection concentration around these predefined targets, not an interchangeable reconstruction-Capture score or a proof of fragmentation of every discovered concept.

The two rates are conventional true-positive and false-positive rates [12]. Restricting the number of selected native groups makes them a test of response concentration: how much target coverage is accessible through a small subset, at its measured false-positive cost. The 95th-percentile rule sets a common calibration operating point; it does not force the audit false-positive rate to equal 5%. Budgets 1, 2, 4, 8 examine progressively larger, nested subsets. Unlike a standard threshold-swept ROC curve, these curves vary subset size while retaining calibration-fixed thresholds.

Concentration and overlap formulas. Let s_{ig} be group g's crop-level response for image i: the mean of its four largest signed native patch responses across all 196 positions, without clipping. For category c, let $\mathcal{P}_c$ and $\mathcal{N}_c$ be crops with known positive and negative presence. Calibration sets group threshold τ_{gc} to negative-response quantile q. Let TPR_{gc} and FPR_{gc} be that threshold's calibration true- and false-positive rates. Balanced accuracy $\mathrm{BA}_{gc} = (\mathrm{TPR}_{gc} + 1 - \mathrm{FPR}_{gc})/2$ ranks groups. For subset budget $k \in \{1, 2, 4, 8\}$, separate from encoder Top-k, let $K_{c,k}$ be its frozen top-ranked subset. The OR prediction is

$$\widehat{y}_{ic}^{(k)} = \mathbf{1}\left[\max_{g \in K_{c,k}} (s_{ig} - \tau_{gc}) > 0\right]. \tag{S25}$$

Let TP_c and FP_c count these detections in $\mathcal{P}_c$ and $\mathcal{N}_c$, respectively. Coverage Cov_c, false-positive rate FPR_c and precision Prec_c are

$$\mathrm{Cov}_c = \frac{TP_c}{|\mathcal{P}_c|}, \quad \mathrm{FPR}_c = \frac{FP_c}{|\mathcal{N}_c|}, \quad \mathrm{Prec}_c = \frac{TP_c}{TP_c + FP_c}. \tag{S26}$$

Precision is undefined when there are no positive predictions. Unknown presence labels enter neither set. Each rate is averaged equally over categories with the required support; undefined values are retained as missing, not imputed zero.

For sign $\sigma \in \{+, -\}$, let D_{gc}^{σ} be the crops in $\mathcal{P}_c$ or $\mathcal{N}_c$ detected by group g alone. With $\mathcal{E}_{c,k}^{\sigma}$ the selected-group pairs having nonempty union, mean prediction overlap is

$$J_{c,k}^{\sigma} = \frac{1}{|\mathcal{E}_{c,k}^{\sigma}|} \sum_{(g,h) \in \mathcal{E}_{c,k}^{\sigma}} \frac{|D_{gc}^{\sigma} \cap D_{hc}^{\sigma}|}{|D_{gc}^{\sigma} \cup D_{hc}^{\sigma}|}. \tag{S27}$$

Zero-union pairs are excluded and counted separately. Overlap is undefined for a one-group subset. Incremental coverage at rank j is the fraction detected by that group and no preceding selected group; unique coverage is the fraction detected by that group and none of the other selected groups. Both use the same positive or negative denominators as Equation (S26).

Aggregation and uncertainty. All concentration tables use 0–100 scales for coverage, false positives, precision and Jaccard overlap, with 73 supported of 75 planned targets per run. One-group overlap is undefined and never zero-imputed.

The matched 75-group summaries use seven run-level scores (seeds 21–27). Sample SD uses the $n-1$ denominator at the training-order level. Coherence combines equal-size 75-group folds within each run; shared Vanilla counts once. Presence, Patch AP and Capture use equal-category macro averages after fold aggregation. Reconstruction is averaged

Table S40. Full-grid category-mask Patch AP and regional Capture (%) by training seed, with mean ± sample SD over seeds 21–27. Each metric selects its group on calibration data and scores it on audit data. The split settings pool categories across complementary folds. The same Vanilla checkpoint contributes to both target strata.

Full-grid Patch AP

Method	Mean ± SD	21	22	23	24	25	26	27
All category labels								
GGF	74.38 ± 0.30	73.75	74.45	74.60	74.43	74.31	74.43	74.70
SLiCS	49.93 ± 1.02	49.43	49.14	51.67	50.47	48.61	50.40	49.81
Supervised Vanilla	31.12 ± 2.40	33.25	28.70	31.97	30.83	34.86	29.93	28.30
Vanilla BSF	20.96 ± 1.13	19.61	20.37	21.47	23.16	21.08	20.64	20.38
Category split, guided targets								
GGF	69.72 ± 0.29	69.92	69.87	69.91	69.46	69.77	69.92	69.17
Supervised Vanilla	55.65 ± 2.14	51.21	56.53	57.12	56.68	55.24	55.28	57.52
Vanilla BSF	22.35 ± 1.43	20.83	20.79	22.64	24.79	23.29	22.52	21.59
Category split, withheld targets								
GGF	22.92 ± 1.17	23.03	20.47	23.58	23.06	23.03	23.05	24.22
Supervised Vanilla	22.19 ± 0.69	22.15	22.52	21.94	21.31	22.32	23.43	21.64
Vanilla BSF	22.35 ± 1.43	20.83	20.79	22.64	24.79	23.29	22.52	21.59

Regional Capture

Method	Mean ± SD	21	22	23	24	25	26	27
All category labels								
GGF	40.96 ± 0.11	40.99	40.86	40.99	41.16	40.85	40.90	41.01
SLiCS	26.05 ± 0.60	26.61	24.98	25.97	26.48	25.69	25.97	26.64
Supervised Vanilla	28.73 ± 0.62	29.33	27.72	29.35	28.88	28.80	28.99	28.04
Vanilla BSF	9.08 ± 0.56	9.64	8.57	8.90	10.03	8.53	8.99	8.91
Category split, guided targets								
GGF	43.55 ± 0.15	43.37	43.65	43.35	43.75	43.56	43.53	43.61
Supervised Vanilla	38.60 ± 0.42	39.12	38.38	38.24	38.07	39.17	38.57	38.66
Vanilla BSF	8.52 ± 0.46	8.88	8.24	8.09	9.37	8.14	8.50	8.43
Category split, withheld targets								
GGF	20.34 ± 0.10	20.26	20.17	20.32	20.40	20.37	20.50	20.35
Supervised Vanilla	17.71 ± 0.58	17.48	16.92	17.62	18.11	17.25	18.69	17.88
Vanilla BSF	8.52 ± 0.46	8.88	8.24	8.09	9.37	8.14	8.50	8.43

Table S41. Native full-bank reconstruction on the audit set. MSE and VE are mean ± sample SD across seeds 21–27. VE is reported for the full bank in each setting. The SLiCS concept projections do not supply a comparable additive full-bank reconstruction.

Setting	Method	MSE	VE
All labels	GGF	0.1762 ± 0.0032	0.8237 ± 0.0032
All labels	SLiCS	N/A	N/A
All labels	Supervised Vanilla	0.1428 ± 0.0020	0.8571 ± 0.0020
All labels	Vanilla BSF	0.1417 ± 0.0021	0.8582 ± 0.0022
Category split	GGF	0.1379 ± 0.0003	0.8620 ± 0.0003
Category split	Supervised Vanilla	0.1382 ± 0.0009	0.8617 ± 0.0009
Category split	Vanilla BSF	0.1293 ± 0.0004	0.8706 ± 0.0004

across fitted banks for each method and run. Only then are means and SD computed across runs. Full-grid category-mask AP in this study differs from the 32-patch-per-image benchmark; they are reported separately. All 75 AP categories remain supported. The archive retains every per-run value, per-bank/fold distribution, specificity outcome, unknown, disagreement and source fingerprint. Shared initialization and image pools preclude interpreting these SDs as independent-model population uncertainty. No new superiority test or confidence claim is inferred. Independent human validation, visual target alignment and a genuinely new-image test remain outstanding.

Cross-model coherence check. We repeated the blinded evaluation with Gemini 3.5 Flash-Lite [19] (API ID `google/gemini-3.5-flash-lite`) via OpenRouter

and Google AI Studio Flex, with minimal reasoning effort and temperature zero. Gemini judged the same 4,725 matched groups from orders 21–27. Each group received one fresh call for the three fixed pairs and a separate call for the whole-group task. Both calls received the frozen PNG and lossless crops of its numbered views and zoom strips. The same rubric applied, with prompts that explicitly excluded the review overlays as evidence and requested concise pair explanations. Method identities and GPT judgments were hidden. Pilot calls informed the prompt choices and are excluded from the results.

Of the 9,450 replies, 16 pair-task and 131 whole-group replies failed rubric validation. These remain missing, with planned denominators of 14,175 pairs and 4,725 whole groups. GGF leads all aggregate endpoints even if every missing baseline judgment counts as positive. It leads specific-pair coherence in every order under both guidance settings. Gemini's absolute coherence scores are lower than the GPT-majority scores. Among comparable judgments, exact agreement with the GPT majority is 77.4% for specific-pair coherence (10,932/14,127), 75.1% for visual structure (10,613/14,127), 78.0% for whole-group coherence (3,582/4,594), and 54.2% for specificity (2,490/4,594). The method rankings therefore agree in aggregate across both models, while absolute rates and specificity depend on the reviewer. Both models use the same image evidence, and this comparison uses one Gemini judgment per task and the majority of three GPT contexts.

H. Computational resources and measurement scope

H.1. Fitting costs across datasets

Fitting costs and saved-model execution answer separate questions. Tables S43–S45 and Figure S14 retain each hardware, dataset and native group-count protocol rather than pooling them.

Resource summaries. Let Θ_{learned} contain learned tensors θ, excluding frozen DINOv3 and evaluation probes. With numel counting tensor entries, parameters total $\sum_{\theta\in\Theta_{\text{learned}}} \mathrm{numel}(\theta)$. Allocated and reserved memory are separate recorded maxima, including resumed fitting invocations. For method m, let $\overline{T}_m$ be its unrounded mean time within one protocol block, R_m its time ratio to GGF, and H_{SV} GGF's percentage time overhead over supervised Vanilla (SV):

$$R_m = \frac{\overline{T}_m}{\overline{T}_{\text{GGF}}}, \quad H_{\text{SV}} = 100\left(\frac{\overline{T}_{\text{GGF}}}{\overline{T}_{\text{SV}}} - 1\right). \tag{S28}$$

For F_{call} registered operations in a call with N_{images} images, operations per image are $F_{\text{call}}/N_{\text{images}}$. Means and sample SD follow Equation (S8), with the declared training runs or independent calls as units.

Training schedules and timings. T4 VOC BSFs use 1,024 groups, 7,640 warm-up and 3,000 continuation updates, and eight images per update. Each Vanilla arm is charged the shared random warm-up; supervised Vanilla includes its selected loss pair. Concept-SAE sums its three native stages, with 2,090 updates and effective batch 64 in eight-image microbatches. The measured loops give GGF 3.4% overhead over supervised Vanilla and a $1.66\times$ speedup over Concept-SAE. Concept-SAE's COCO loops average 76.23 minutes; corresponding BSF loop coverage is incomplete.

The category-matched SLiCS studies each use 10,640 updates. Their GGF loop speedups are $14.67\times$ on COCO/T4 with 75 groups and $18.33\times$ on VOC/A100 with 19 groups. The VOC fit-invocation ratio is instead $9.26\times$: it includes initialization and development checks, while excluding final evaluation, resource profiling and result packaging. All underlying times and their sample SD remain in the tables.

Timer boundaries and memory. BSF loops include losses, backpropagation, model/controller updates and synchronization after batch/noise preparation. SLiCS includes its NNLS solve and dictionary update; Concept-SAE includes microbatch handling. Loop timers exclude initialization, feature preparation, development checks, checkpoint transport and search.

Faster loops need not use less memory. In the VOC 19-group study, GGF's 350,476 learned parameters include an untied encoder, decoder, classifier heads and controller; SLiCS has 175,104 dictionary parameters. GGF has larger allocated/reserved fitting peaks there. In the T4 1,024-group comparison, allocated memory favors GGF over Concept-SAE, while reserved memory does not. Tables retain every peak. COCO 75-group GPU peaks were not recorded. Loading a completed checkpoint measures reuse, not its original fitting cost.

H.2. Execution profiles and operation coverage

We profile three independent cold calls on disposable copies of the seed-21 VOC models on A100-SXM4-40GB, with TF32 off. Copies reset between calls and source checkpoints stay intact. Timers span input preparation through synchronized return and exclude construction, loading, validation and checkpoint I/O.

Registered FLOPs and coverage. PyTorch's registered formulas count a fused multiply-add as two operations and cover executed matrix/convolution operations, including backward for updates. Elementwise operations, reductions, indexing, eigendecompositions and external SciPy/BLAS

Table S42. Cross-model check with Gemini 3.5 Flash-Lite. Scores are mean ± sample SD across seven training orders (%). Whole + discr. requires a clear theme rated specific or broad but discriminating. Rejections count missing pair-task and whole-group judgments, respectively. Shared Vanilla is counted once in the category-split setting.

Method	Specific pairs	Visual pairs	Whole clear	Whole + discr.	Rejections
All category labels					
GGF	79.37 ± 2.86	69.59 ± 3.41	92.38 ± 2.14	92.38 ± 2.14	0/1
SLiCS	69.84 ± 1.65	62.79 ± 3.63	86.48 ± 5.68	86.10 ± 6.10	2/8
Supervised Vanilla	51.62 ± 3.36	47.17 ± 2.77	58.29 ± 2.74	57.90 ± 2.97	0/18
Vanilla BSF	43.43 ± 4.83	41.33 ± 4.22	49.33 ± 3.77	48.57 ± 4.34	3/27
Category-split guidance					
GGF	64.13 ± 1.24	58.63 ± 1.75	72.00 ± 2.52	71.81 ± 2.66	3/25
Supervised Vanilla	59.40 ± 2.98	54.89 ± 2.53	65.62 ± 5.50	65.43 ± 5.60	3/27
Vanilla BSF	47.17 ± 5.06	45.02 ± 5.06	51.62 ± 3.33	51.05 ± 3.75	5/25

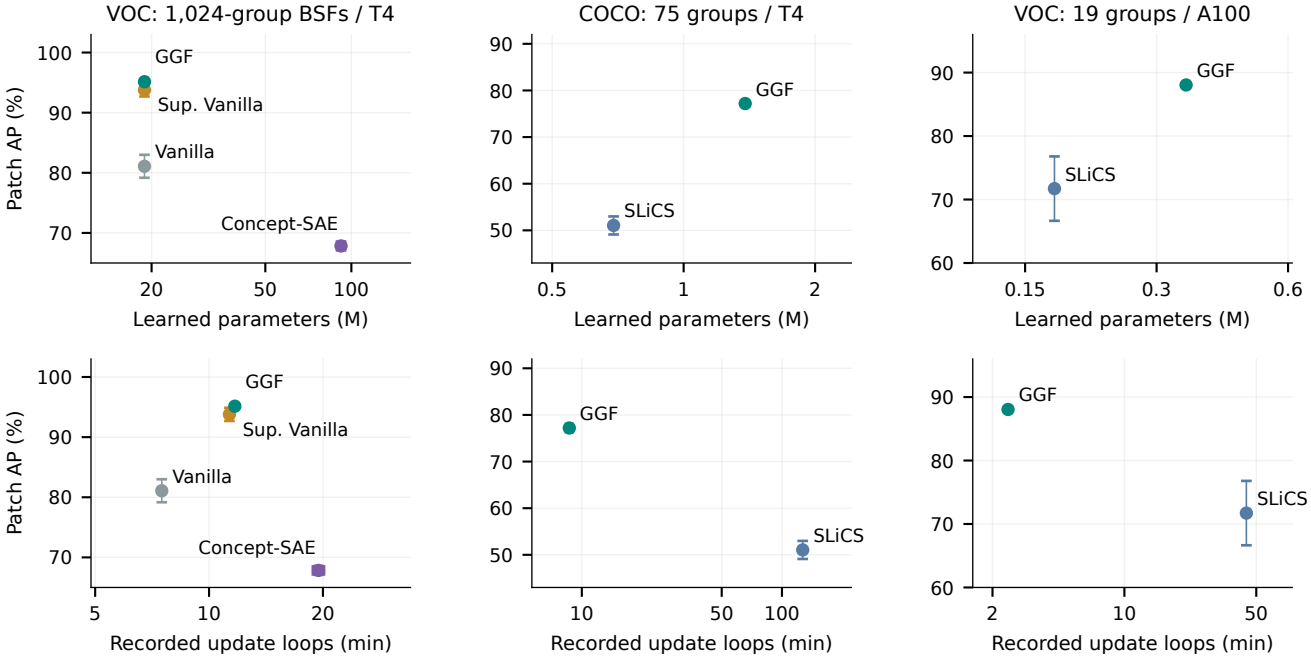


Figure S14. Performance and fitting resources in three separate comparisons. Columns retain the VOC 1,024-group/T4, COCO 75-group/T4 and VOC 19-group/A100 protocols. Top: learned parameters versus Patch AP. Bottom: recorded update-loop minutes versus Patch AP. Error bars show sample SD where measured. Higher points have better AP, while points farther left use fewer parameters or less loop time. Timers exclude validation, checkpoint transport and search. Devices and native training schedules are not pooled.

Table S43. VOC fitting resources under two protocols. Update loops: minutes, mean ± sample SD. GPU peaks: maximum allocated/reserved GiB across fits. Devices and group counts differ between blocks (Appendix H).

Method	Params (M)	Update min	GPU peaks
T4: 1,024 groups for BSF arms			
Vanilla BSF	18.89	7.50 ± 0.02	0.74/0.85
Supervised Vanilla	18.89	11.32 ± 0.03	0.87/2.10
GGF	18.89	11.70 ± 0.04	1.04/2.21
Concept-SAE	92.00	19.47 ± 0.66	1.32/1.48
A100: 19-group SLiCS comparison			
GGF	0.350	2.42 ± 0.03	0.074/0.102
SLiCS	0.175	44.31 ± 0.08	0.055/0.074

Table S44. COCO 75-group T4 fitting resources. Loop time is mean ± sample SD in minutes. GPU peaks were not recorded.

Method	Parameters	Update min
GGF	1,383,452	8.66 ± 0.33
SLiCS	691,200	126.96 ± 1.63

Table S45. VOC A100 fit-invocation time in minutes (mean ± sample SD). This includes initialization and development checks but excludes final evaluation.

Method	Update loops	Fit invocation
GGF	2.42 ± 0.03	6.40 ± 0.36
SLiCS	44.31 ± 0.08	59.26 ± 0.90

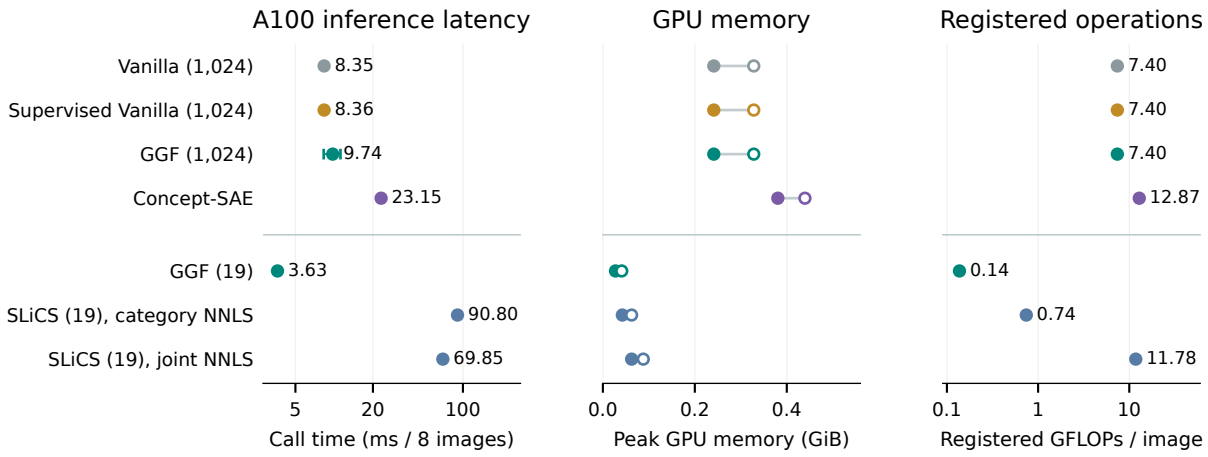


Figure S15. A100 execution profiles on the same eight VOC training images. Left: inference-call time, with sample SD over three independent calls. Center: maximum allocated (filled) and reserved (open) GPU memory. Right: registered GFLOPs per image from one instrumented call. Labels in parentheses give BSF/SLiCS group counts. SLiCS category and joint NNLS serve object scoring and full reconstruction, respectively. The separator distinguishes the 1,024-group/native-token study from the 19-group study. Latency and operation counts use logarithmic axes. These calls exclude the frozen DINOv3 backbone and retain each method's native outputs.

Table S46. VOC A100 execution profiles excluding the DINOv3 backbone. Times are mean ± sample SD over three calls. GPU peaks list allocated and reserved memory. GFLOPs use one instrumented call and count each fused multiply-add as two operations.

Phase	Images	Call (ms)	Peak GPU (GiB)	GFLOPs per image
GGF (19)				
Inference	8	3.63 ± 0.11	0.027/0.041	0.137
Ordinary update	8	43.39 ± 1.22	0.057/0.068	0.415
Controller update	8	57.89 ± 0.08	0.058/0.070	0.663
SLiCS (19)				
Category NNLS	8	90.80 ± 0.50	0.042/0.062	0.742
Joint NNLS	8	69.85 ± 0.68	0.063/0.088	11.779
Training update	8	244.30 ± 11.07	0.054/0.070	1.614
GGF (1,024)				
Inference	8	9.74 ± 1.45	0.241/0.328	7.399
Ordinary update	8	60.41 ± 0.84	0.590/0.748	25.907
Controller update	8	85.77 ± 0.05	0.624/0.854	40.710
Vanilla (1,024)				
Inference	8	8.35 ± 0.03	0.241/0.328	7.399
Training update	8	17.80 ± 0.09	0.453/0.594	18.497
Supervised Vanilla (1,024)				
Inference	8	8.36 ± 0.10	0.241/0.328	7.399
Training update	8	59.07 ± 0.16	0.590/0.748	25.907
Concept-SAE				
Inference	8	23.15 ± 0.97	0.381/0.439	12.869
Stage 1 update	64	198.00 ± 36.34	0.577/0.643	8.820
Stage 2 update	64	152.37 ± 20.27	0.576/0.643	4.500
Stage 3 update	64	383.13 ± 9.52	1.319/1.477	21.405

CPU work remain uncounted. Thus Table S46 reports partial GFLOPs per image, with coverage limits in `FLOP_COVERAGE.json`. That file, the per-phase operator inventories and hardware records are in `evidence/voc_slics_final/`.

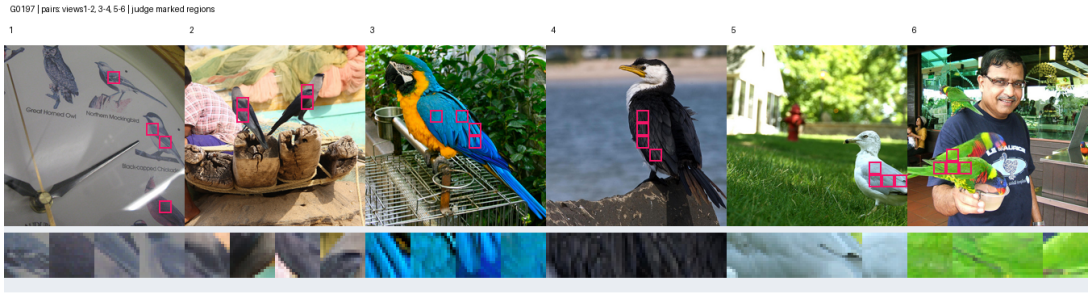
Execution measurements. Inference uses the same eight training images. BSFs execute encoder, gate and decoder; Concept-SAE retains its native output. SLiCS uses independent category NNLS for object scores and joint NNLS for full reconstruction. GGF's measured inference speedups are $2.38\times$ over Concept-SAE at 1,024 groups, and $25.02\times/19.25\times$ over SLiCS category/joint NNLS at 19 groups. The table gives all means, SDs and peak allocated/reserved memory; the peaks are maxima, not means.

Training uses batches of eight for BSFs and SLiCS, and effective batch 64 in eight-image microbatches for Concept-SAE. BSFs restore available optimizer state and Concept-SAE starts fresh Adam moments for each stage update. We time controller updates separately because they run every ten steps. These cold calls complement the whole-fit timings.

I. Qualitative comparisons

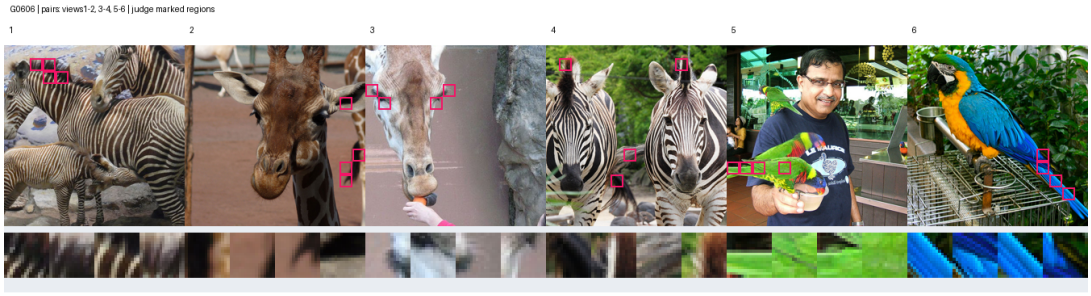
(a) GGF; group 14; G0197 Specific pairs: 3/3; visual pairs: 3/3.

Overlapping feathers along a bird wing or body. Theme matches: 6/6, 6/6, 6/6 (clear).



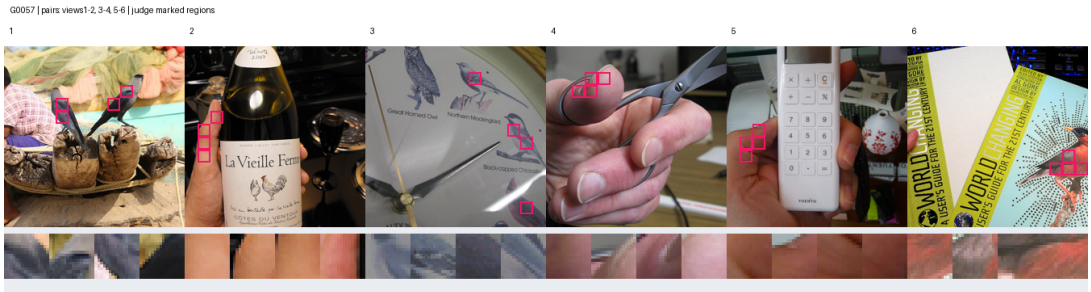
(b) SLiCS; group 14; G0606 Specific pairs: 1/3; visual pairs: 1/3.

Animal hair or feather coverings, with fine strands or layered edges. Theme matches: 6/6, 4/6, 6/6 (clear).



(c) Supervised Vanilla BSF; group 35; G0057 Specific pairs: 1/3; visual pairs: 1/3.

Human fingers wrapped around or gripping a handheld object. Theme matches: 3/6, 3/6, 6/6 (partial).



(d) Vanilla BSF; group 35; G0136 Specific pairs: 1/3; visual pairs: 1/3.

Birds or bird depictions, especially their heads and plumage. Theme matches: 4/6, 3/6, 4/6 (partial).



Figure S16. **Matched bird readouts.** GGF consistently marks feather surfaces. SLiCS combines hair and feathers under a broader animal-covering theme; the Vanilla variants mix bird-related regions with fingers and other surfaces. GGF has three confirmed specific pairs, versus one for each comparator. Its visual-pair score is also higher. SLiCS still has a clear whole-group majority, while both Vanilla variants are partial. For each method, the group is its frozen category-mask-AP calibration readout; each sheet shows that group's six strongest audit images. Pink boxes and zoom strips retain all original marked patches. Specific pairs and theme matches follow Figure 2. Displayed themes quote reviewer 1; counts use each reviewer's own theme. These seed-21, 75-group cases were selected after review for GGF's specific-pair advantage over every comparator; the all-group comparison is in Table 1.

(a) GGF; group 51; G0716 Specific pairs: 3/3; visual pairs: 2/3.

Doughnut surfaces with sugar, glaze, or toppings. Theme matches: 6/6, 6/6, 6/6 (clear).



(b) SLiCS; group 51; G0697 Specific pairs: 1/3; visual pairs: 1/3.

Browned, bread-based savory foods, with topping-covered or filled surfaces. Theme matches: 6/6, 6/6, 6/6 (clear).



(c) Supervised Vanilla BSF; group 7; G0468 Specific pairs: 1/3; visual pairs: 1/3.

Open sky, including blue and cloudy portions. Theme matches: 4/6, 4/6, 4/6 (partial).



(d) Vanilla BSF; group 6; G0364 Specific pairs: 1/3; visual pairs: 1/3.

Dark chocolate-coated doughnut surfaces with an irregular granular texture. Theme matches: 3/6, 3/6, 3/6 (partial).

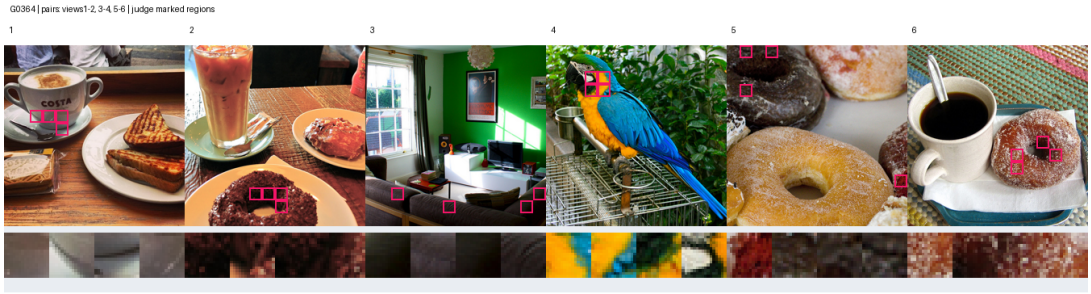


Figure S17. **Matched doughnut readouts.** GGF repeatedly marks doughnut surfaces, whereas SLiCS groups other browned or topped foods, supervised Vanilla BSF mixes doughnuts with sky, and Vanilla BSF mixes them with other regions. GGF has three confirmed specific pairs, versus one for each comparator, and a higher visual-pair score. SLiCS's broader food theme is clear; both Vanilla variants have partial whole-group outcomes. For each method, the group is its frozen category-mask-AP calibration readout; each sheet shows that group's six strongest audit images. Pink boxes and zoom strips retain all original marked patches. Specific pairs and theme matches follow Figure 2. Displayed themes quote reviewer 1; counts use each reviewer's own theme. These seed-21, 75-group cases were selected after review for GGF's specific-pair advantage over every comparator; the all-group comparison is in Table 1.